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Beyond the barrel: Europe's hidden energy vulnerability

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Authors

Yacine Rouimi
Senior Economist

Mark Wall
Chief Economist

Recent supply chain stress, highlighted by European PMI data, has renewed focus on the continent's external vulnerabilities. In this note, we argue that Europe's exposure is twofold:

— A direct, tangible dependency on the Gulf for specific commodities like oil products (e.g., jet fuel, diesel), oil derivatives, chemicals and metals, representing a measurable cost.

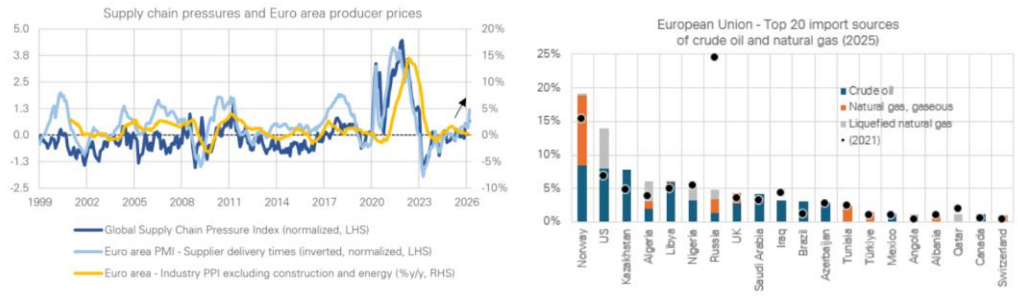
— A more elusive and structural indirect risk from "hidden" energy embedded within industrial imports, particularly from Asia.

This indirect exposure is qualitatively different, comprised of refined products and electricity (not just crude oil) and threatens shortages of specific, non-fungible industrial inputs like plastics, chemicals and metals. This upstream vulnerability is far harder to substitute than direct commodity imports and, as we've argued previously, is a prime example of a non-linear risk that can amplify stagflation and complicate monetary policy.

Note: 'Direct' exposure hereafter refers to Europe's reliance on imported goods from a specific region, for instance energy from the Middle-East. 'Indirect' exposure refers only to the "embodied" content of an import – the energy consumed along the value chain to produce that good before export to Europe.

A ceasefire's reprieve, a system's vulnerability. The fragility of the agreed two-week ceasefire in the Middle East was underscored this weekend by the failure of US-Iran talks. While any pause in hostilities offers a welcome reprieve for global supply chains, this development highlights that underlying risks remain in place.

Figure 1: Supply chain pressure is increasing rapidly. EU direct energy exposure to GCC States not as pronounced as to Russia in 2022



Source: Deutsche Bank Research, NY FED, S&P Global, Eurostat (datasets nrg_ti_gas and nrg_ti_oil) for the right-hand chart. GCC countries refers to a sum of Bahrain, Iran, Kuwait, Qatar, Saudi Arabia and the UAE. 2025 trade share covers January to September.

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Recent data has already provided a clear warning signal of this vulnerability, with the March PMI releases for Europe showing a notable lengthening of supplier delivery times and a fresh acceleration in input costs (Figure 1, left).

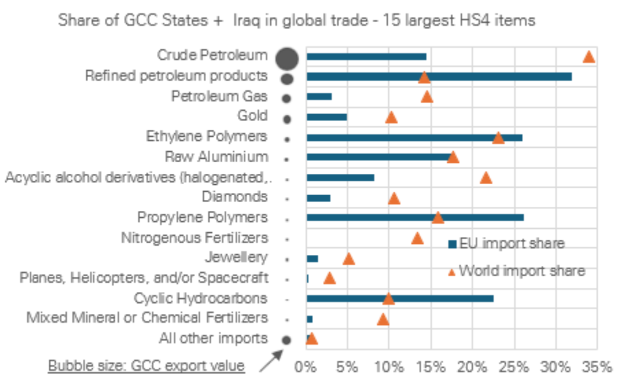
While this offers a glimpse of what could happen if the conflict escalates again, the transmission mechanism to Europe is more complex than a simple repeat of 2022. There are two distinct channels of vulnerability when it comes to this energy shock: a direct, measurable impact on the cost of specific imports – largely energy –, and a more subtle, structural and indirect risk embedded within global value chains, potentially impacting the vulnerability of various inputs.

Channel 1: A tangible, product-level exposure. The direct risk channel has two layers. The first is the overall cost of energy. As we quantified in a recent note, the euro area's energy import bill should increase by around 0.8% of GDP, based on current market pricing (Brent at 98 USD/bbl and gas at 47 EUR/MWh).

The second risk is a more specific dependency on the region central to the crisis in terms of physical supply. This is most visible not at the continent-wide level – where suppliers like Norway, the US, and Kazakhstan now dominate Europe’s energy supply (Figure 1, right) – but at the product-specific level, where Gulf countries can provide upwards of 15% of EU imports. A key example is refined products, where Europe sources a notable share from the region (Figure 2). While Europe's significant domestic refining capacity (supplying 81% of the domestic market in refined petroleum in 2022) provides a crucial buffer, the import dependency remains.

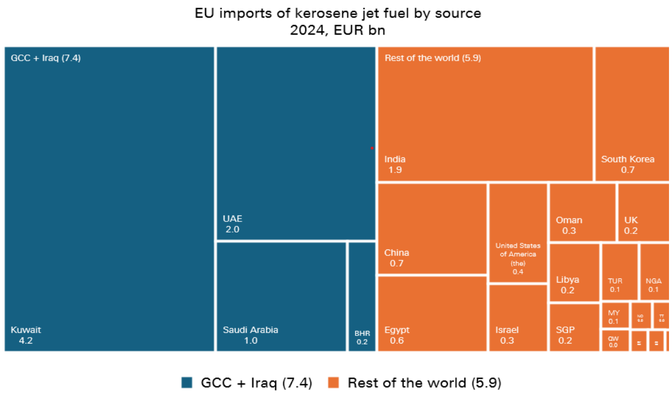
For certain types of refined petroleum products, such as kerosene, the share of GCC countries as a source for EU imports reaches 56% (Figure 3). On this point specifically, our in-house specialists think that the air transport sector has 45-55 days of effective cover, higher than the normal 30-35 days thanks to the release of reserves by the International Energy Agency. Uncertainty prevails, however, as the ceasefire has done little to ease the Strait of Hormuz situation.

Figure 2: The EU’s reliance on the Gulf is meaningful in specific categories, although generally lower than the rest of the world



Source: Deutsche Bank Research, BACI

Figure 3: An example of high exposure is kerosene, with an import share of 56%

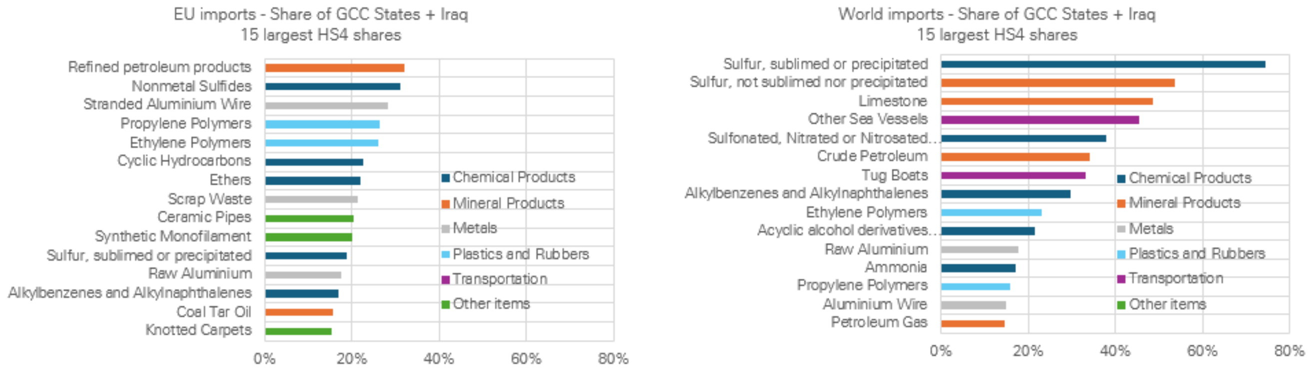


Source: Deutsche Bank Research, Eurostat

Furthermore, granular data reveals pockets of high dependency for specific petrochemicals, plastics, and metals (Figure 4, left). These concentrated exposures represent a measurable cost to Europe's import bill and a vulnerability for the industries that rely on them. In short, while Europe's product-level exposure may be lower than in other parts of the world, it is real and cannot be dismissed. The second channel of risk, however, is qualitatively different and arguably more complex.

Channel 2: A hidden, structural risk in value chains. While Europe's visible exposure is specific, the GCC and Iraq region's role in supply chains more globally creates a broader bottleneck risk. The region's global importance extends beyond crude oil, and it holds a commanding position in key industrial inputs, particularly in mineral and chemical products (Figure 4, right).

Figure 4: The granular set of products where the Gulf makes a large fraction of EU and world imports



Source: Deutsche Bank Research, BACI

For instance, the region accounts for over 70% of global exports of certain forms of sulfur and 48% for limestone. This is complemented by significant global shares in chemical feedstocks like sulfonated/nitrated derivatives (38%) and plastics like ethylene polymers (23%).

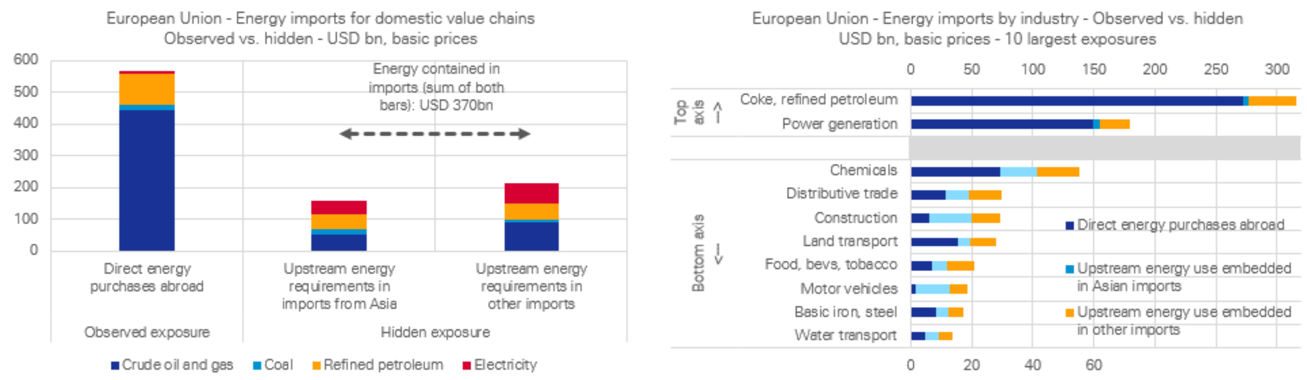
This global bottleneck risk is one indirect threat to Europe. Another transmission channel is through second-round effects via Asia, whose industries are more reliant on these Gulf supplies. Recent headlines (see here, here and here) already flag this, with reports of Asian producers cutting output due to soaring feedstock costs.

We quantify this indirect channel by calculating the energy "embodied" in Europe's imports – that is, the energy consumed all along the global value chain to produce imported goods. Based on OECD Input-Output tables for 2022 – the last year available –, we find that in addition to Europe's USD 567bn (3.8% of EU GDP) in direct energy imports, the goods imported into Europe for use in supply chains required an additional USD 370bn (2.5% of GDP) of indirect energy to be consumed in their production (Figure 5, left). A significant share of this – USD 170bn (1.1% of GDP) – was embedded in goods and services from Asia alone. These measures capture total energy use across production stages, including energy used within the energy sector itself, and is therefore not 100% comparable to observed import flows.

Crucially, this hidden exposure has a different risk profile. While direct imports are dominated by crude oil, the indirect energy footprint has a larger share of refined products and electricity. This shifts the focus of risk away from just oil wells and towards the stability of foreign refining capacity and power grids – critical, yet less monitored, factors for European industrial resilience.

Pinpointing upstream vulnerabilities. What inputs could create disruptions? The impact of this indirect exposure is not uniform across the European economy. Beyond direct energy consumers (refined petroleum and power generation) sectors such as chemicals, manufacturing, and construction carry a significant indirect energy footprint tied to their Asian supply chains (Figure 5, right).

Figure 5: Energy dependency – What is observed and what hides in imports. The exposure of many manufacturing sub-segments is largely indirect



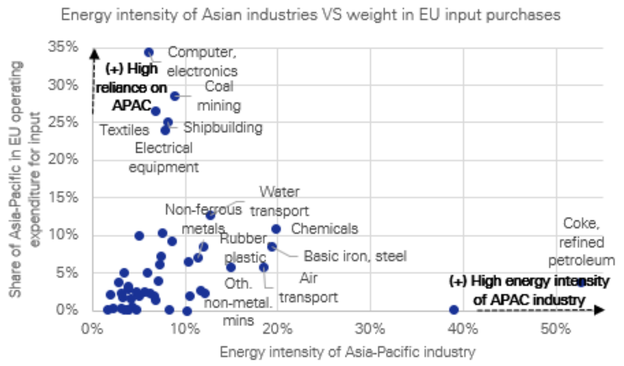
Source: Deutsche Bank Research, OECD Inter-Country Input-Output Table (2022). Analysis covers imported energy only. The OECD's $567bn figure differs from Eurostat's ~$700bn primarily due to valuation (OECD uses 'basic prices', Eurostat uses 'CIF') and scope (OECD tracks by exporting industry, not purely by product). The $567bn measure also only accounts for energy as an input (intermediate demand).

The transmission risk is most acute for inputs that combine two characteristics: high sourcing from Asia and high energy intensity. This makes basic chemicals and industrial materials – like plastics and steel –primary vectors through which cost and availability shocks can propagate into European industries (Figure 6).

Having measured total energy requirements along value chains, we now isolate the pure energy cost absorbed by each economy. This strips out non-energy components of the energy sector's output (intermediate inputs as well as margins, taxes and labour costs), providing a cleaner measure of underlying energy exposure. This shifts the focus from production requirements to income absorption/value added contributions from energy as an input.

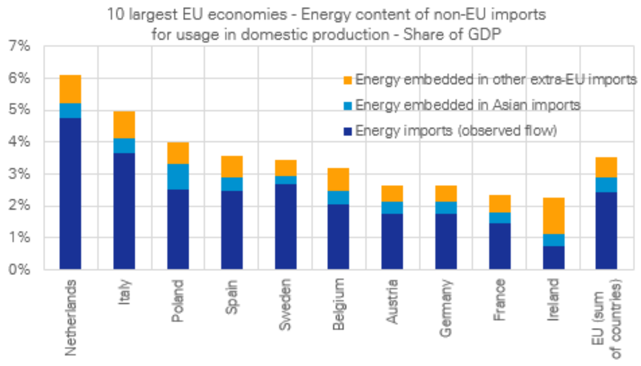
This country analysis reveals a remarkably widespread "hidden" burden. Factoring in this absorbed energy cost adds an amount equivalent to between 0.8% and 1.5% of GDP to the import bill for most major European economies. This effectively increases their total energy dependency by a third to a half, compounding already high exposure in countries like Italy, Spain, and Poland (Figure 7).

Figure 6: Which inputs from Asia could be at risk in terms of supply?



Source: Deutsche Bank Research, OECD ICIO

Figure 7: Energy absorption shifting higher when hidden energy is captured



Source: Deutsche Bank Research, OECD ICIO

Conclusion: A structural vulnerability. The key takeaway is that Europe's exposure in terms of supply is twofold. The direct part (trade exposure to Gulf countries on crude oil, gas, oil-derivatives, chemicals and metals is tangible and should not be overlooked. The indirect part is more elusive, structural and difficult to manage: energy that arrives embedded within industrial inputs – with a precise fit in those chains – is much harder to substitute than fungible commodities like oil or natural gas, for which Europe's sourcing has already shifted to other suppliers.

This creates the potential for shortages of specific, critical components, a prime example of the non-linear risks we highlighted in a recent piece. The nature of such a structural vulnerability means it is often less responsive to broad macroeconomic tools and requires a more targeted level of monitoring. The ultimate message is that even countries with limited direct dependence to the Gulf face a meaningful – and arguably less manageable – upstream vulnerability through their reliance on global industrial supply chains.

The lingering risk of shortages amplifies the stagflation risks already prevalent from the energy shock (Figure 6), and complicates the conduct of monetary policy. Our view is that the risk of persistent inflation will likely drive the ECB to hike rates to 2.50%, the upper end of the range of neutral rates. This would signal the ECB's commitment to price stability without overburdening growth when there are downside risks.

Appendix 1

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How the war has disrupted markets and economies

April 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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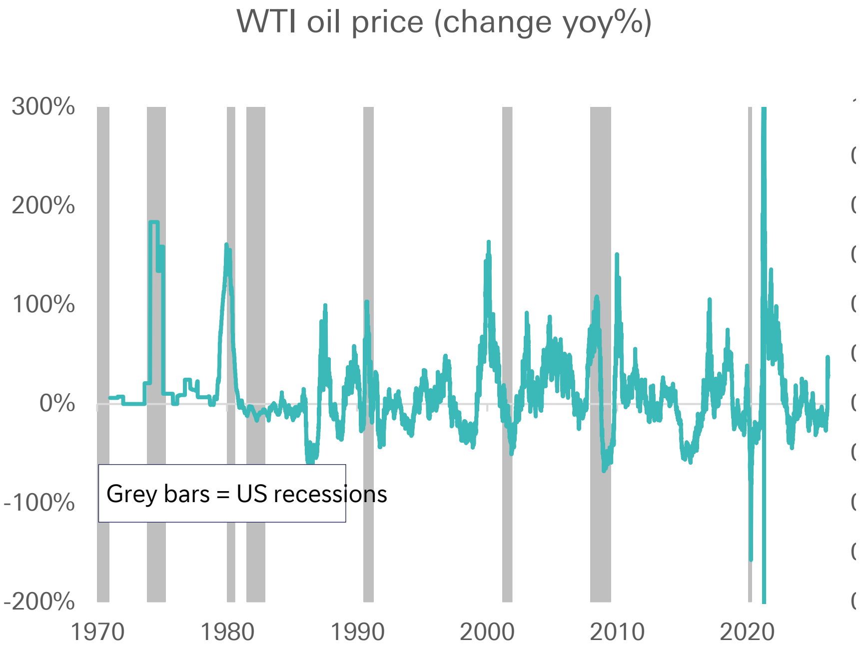
Summary: it may be too early to call the end of the conflict with Iran despite the encouraging signs. Thus, we are watching for how resilient the disruption to the world's economies and markets will be for the rest of 2026.

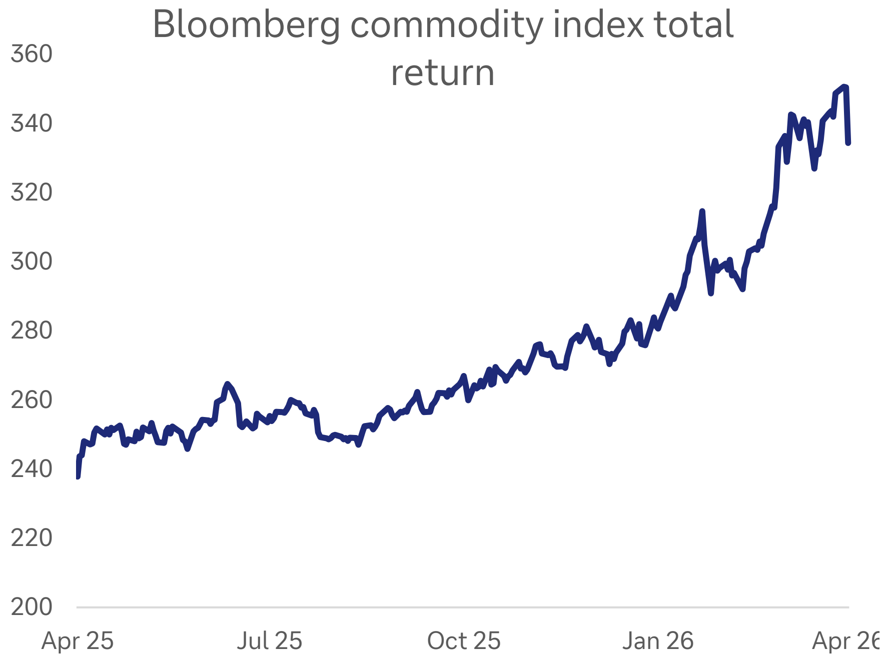
A reset? Or the return to a dangerous status quo?

- Following the announcement of the two-week conditional ceasefire between the US and Iran overnight, attention is now turning to exactly how oil tankers might begin to transit the Strait of Hormuz.
- There are inconsistencies in the logistics with Iran claiming the US has accepted its control over the strait and that transit will only be allowed under the "supervision" of its armed forces and subject to "technical limitations".
- There is also a real fear that the broader situation may return to the status quo of US sanctions and fruitless talks while a reinvigorated Iran restarts its nuclear programme. That will be a key discussion point ahead of likely US-Iran negotiations on Friday in Pakistan.
- On a day-by-day basis, there are deep uncertainties as to how long the ceasefire can hold.
- Upcoming negotiations will be difficult given Iran has claimed that the US will offer sanctions relief and security guarantees, things that may not eventuate in part or full. The most important demand in Iran's ten-point plan, its right to enrich uranium, also crosses a US red line.

It wasn't just oil – other commodities have surged and stayed high

The spike in oil prices (yoy) was less pronounced than some prior episodes that coincided with US recessions

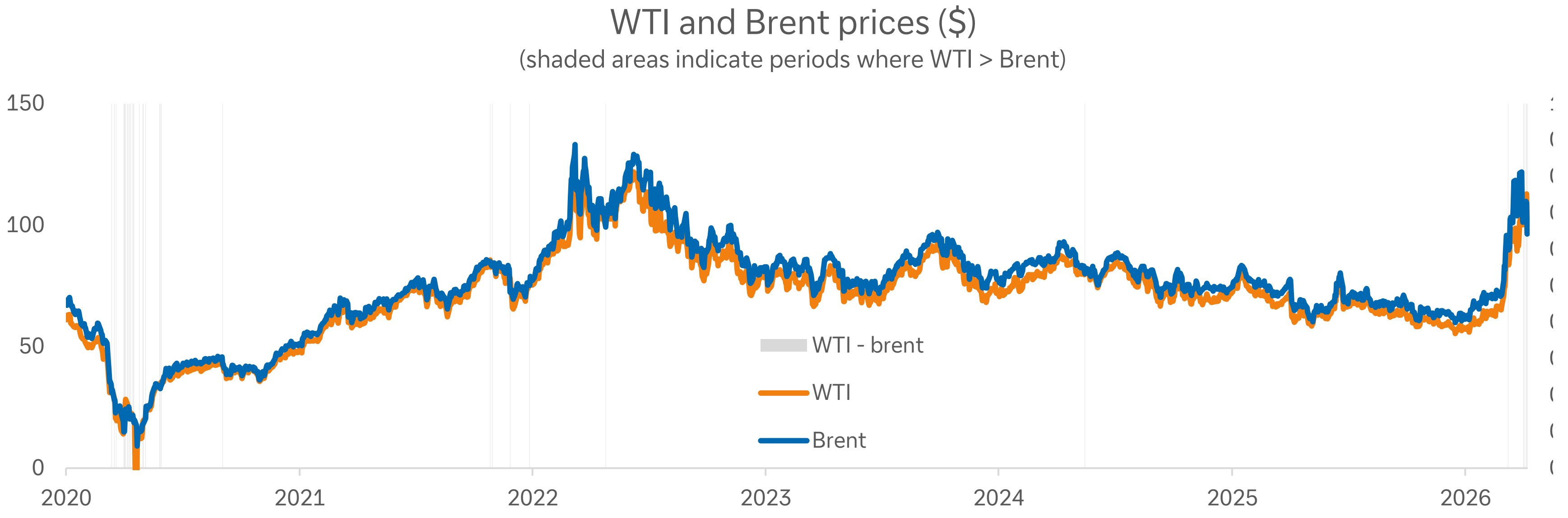




Source: Bloomberg Finance LP, Deutsche Bank.

WTI oil prices jumped above brent – that is extremely unusual this decade

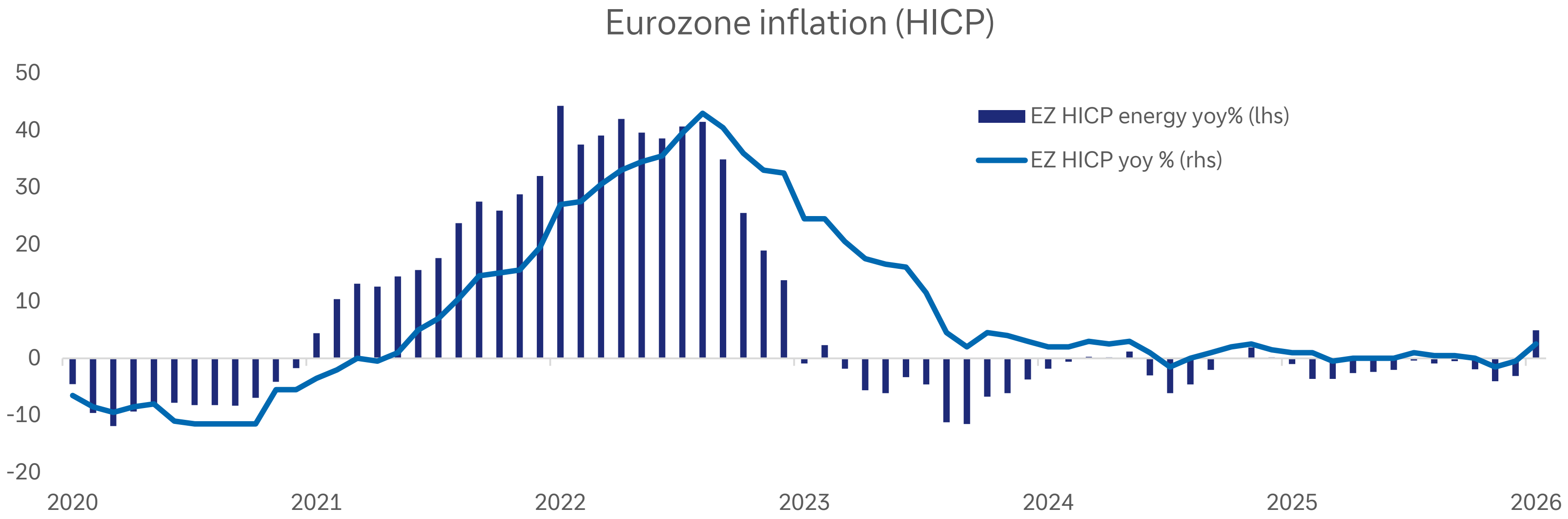
WTI's one-month delivery schedule (compared with two months for brent) was a key reason why traders moved to WTI contracts



Source: Bloomberg Finance LP, Deutsche Bank.

European inflation has climbed, however, the energy shock is so far nothing like that around the time Russia invaded Ukraine.

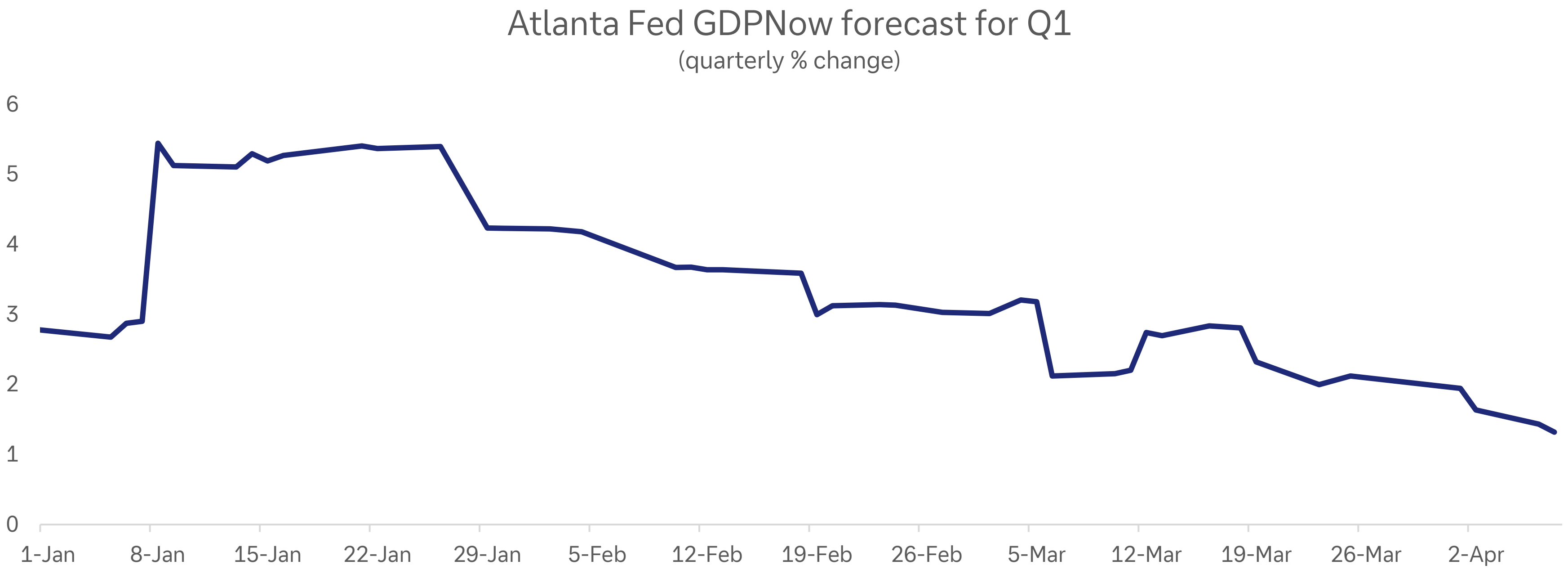
Once April's figures come in, they seem on track to show a continuing upwards trend in energy inflation. But Europe's dependence on the Middle East does not compare with its dependence on Russia five years ago.



Source: Bloomberg Finance LP, Deutsche Bank.

The war has exacerbated the falling trend for US growth forecasts

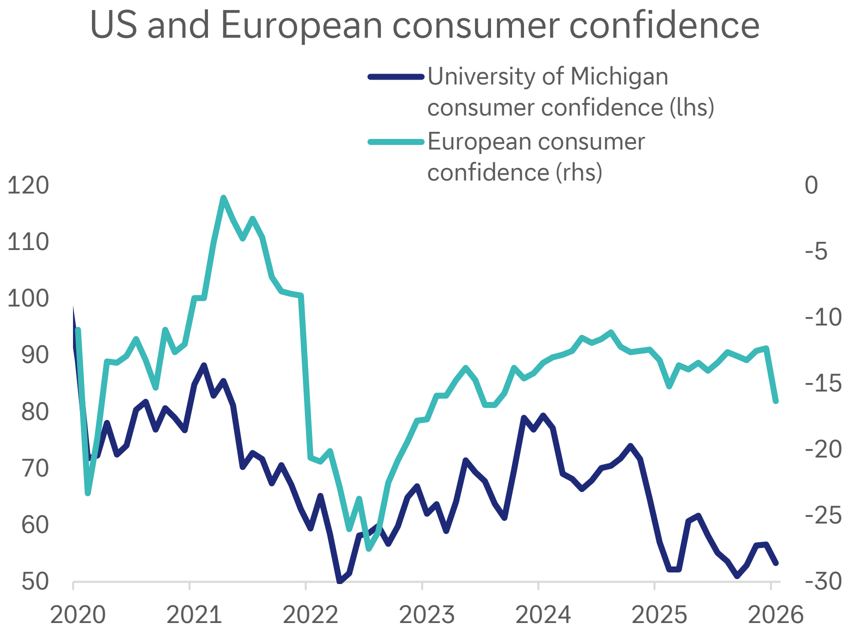
Economic growth expectations were falling before the war in Iran started. However, are more data points are released this month some will certainly put downwards pressure on growth estimates.

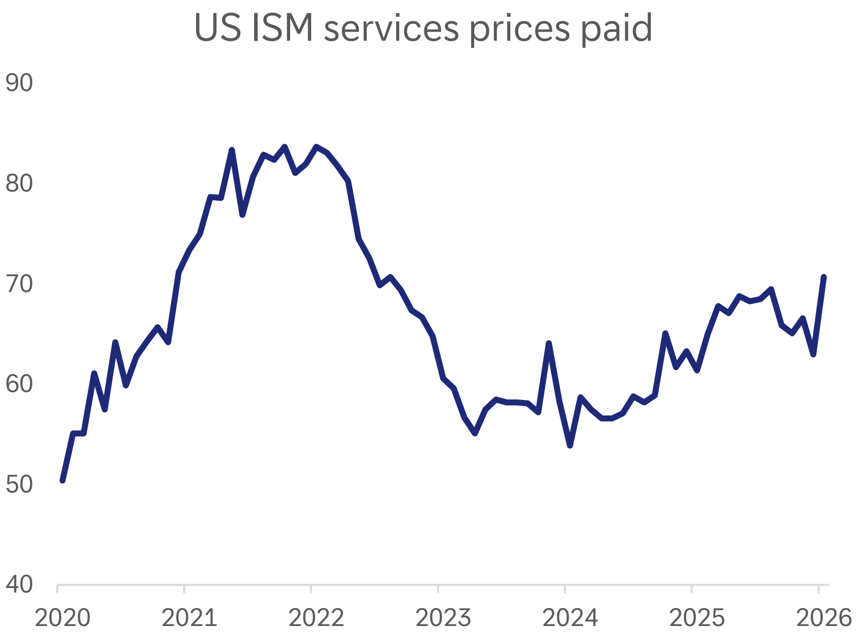


Source: Bloomberg Finance LP, Deutsche Bank.

The drop in consumer confidence might be expected. How long it lasts will be influenced by the resulting inflation.

Consumer confidence is, at least partially, dependent on whether the uptick in input cost inflation remains robust. That will also be influenced by how inflation filters through to consumers. Some corporates may decide to absorb the costs themselves.

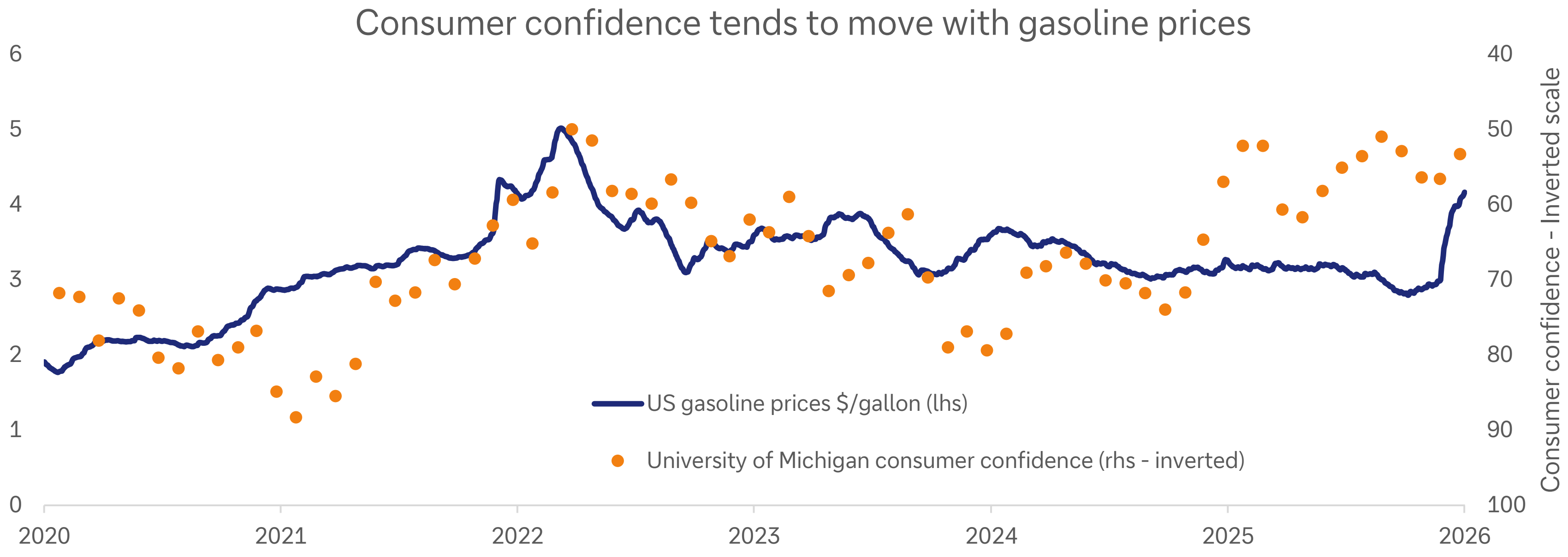




Source: Bloomberg Finance LP, Deutsche Bank.

Petrol prices

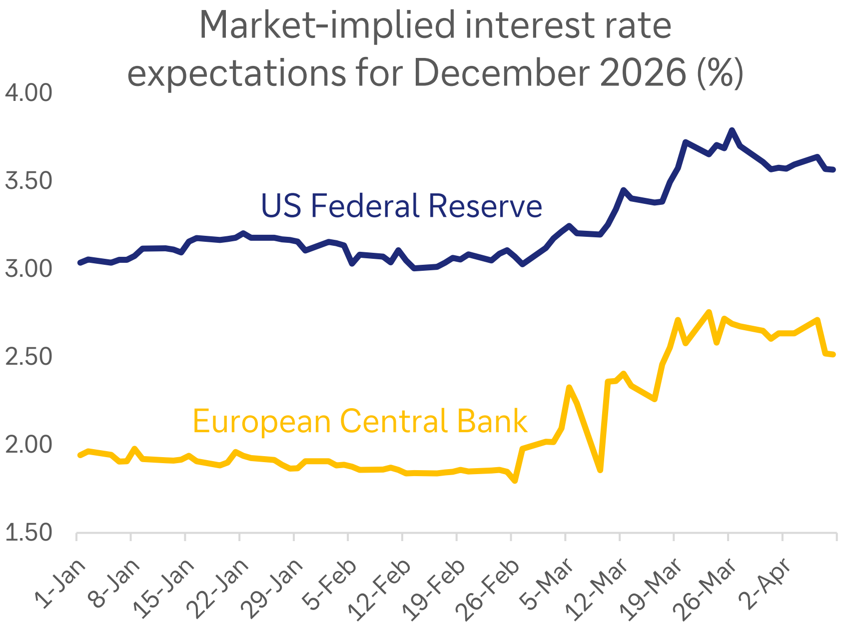
It is an inescapable reality that people care about the price they pay at the pump. That has a disproportionate impact on their overall confidence in the economy and feeds through to their voting intentions.

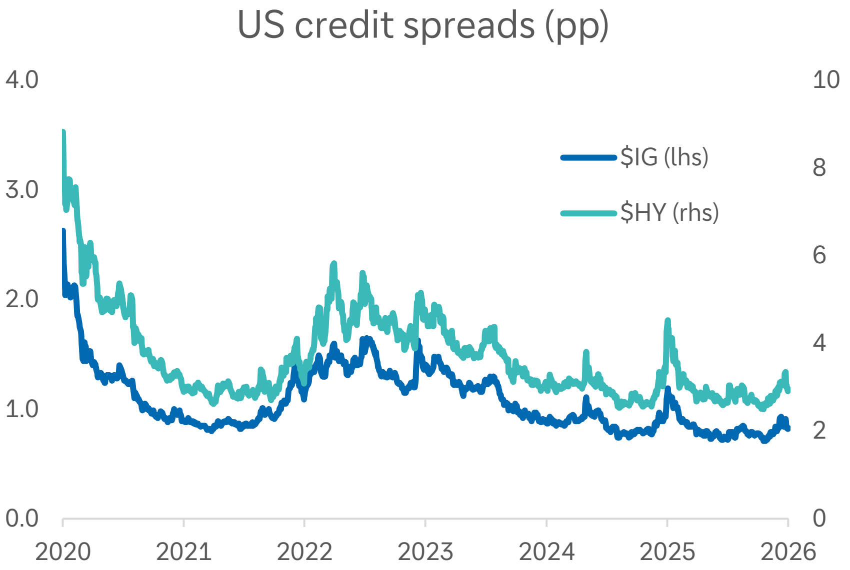


Source: Bloomberg Finance LP, Deutsche Bank.

Interest rate expectations have jumped

Credit spreads ticked upwards as the war commenced, but they weren't too concerned. Even as private credit and software worries continued to plague the market, and especially high-yield, investors seemed relatively sanguine about credit quality. The large amount of liquidity in the market certainly helps



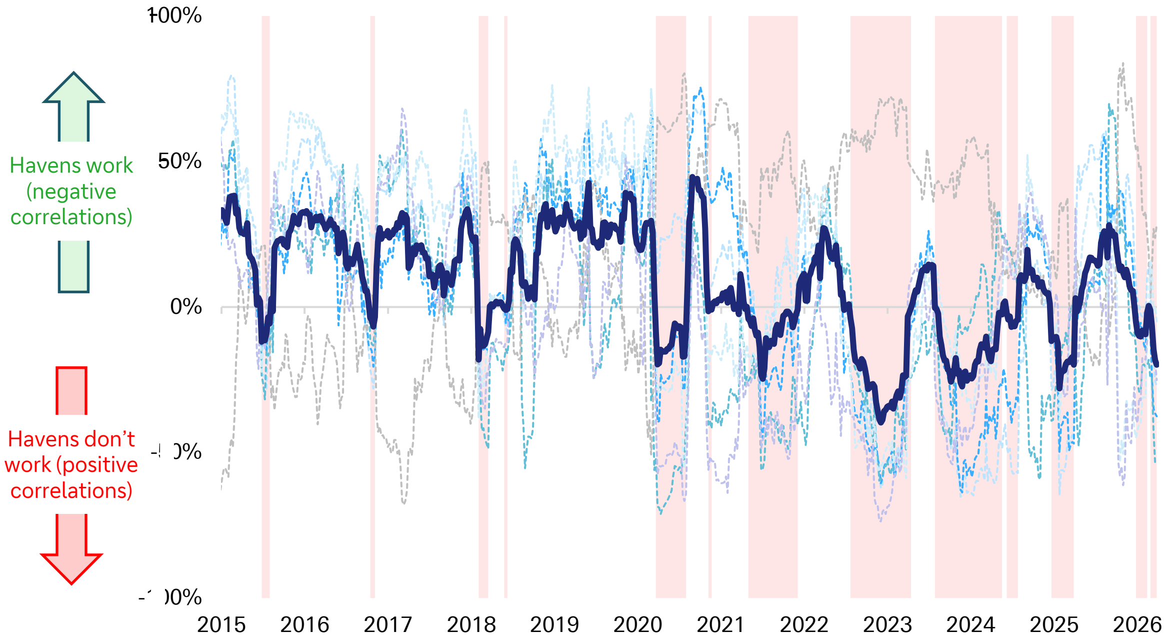


Source: Bloomberg Finance LP, Deutsche Bank.

Haven assets were little help during the Iran war – a 2020s trend

Our 'haven indicator' shows that traditional havens have been largely absent in the 2020s

Our haven indicator (20-wk rolling, avg across pairwise correlations with S&P 500)



Source: Bloomberg Finance LP, Deutsche Bank.

How our haven indicator works

We take six traditional haven assets:

1. Gold
2. US dollar
3. Japanese yen
4. Swiss franc
5. 10yr treasury
6. 10yr bund

We then analyse their individual correlation with the S&P 500 (our proxy for 'risk') over time. Finally, we take an average of the six correlations.

The red shading indicates periods where the average haven **did not work** (ie. positively correlated with the S&P 500). This includes:

1. Iran war in 2026
2. US tariff crisis in 2025
3. Peak fear about interest rate rises in 2022
4. Initial covid turmoil in 2020

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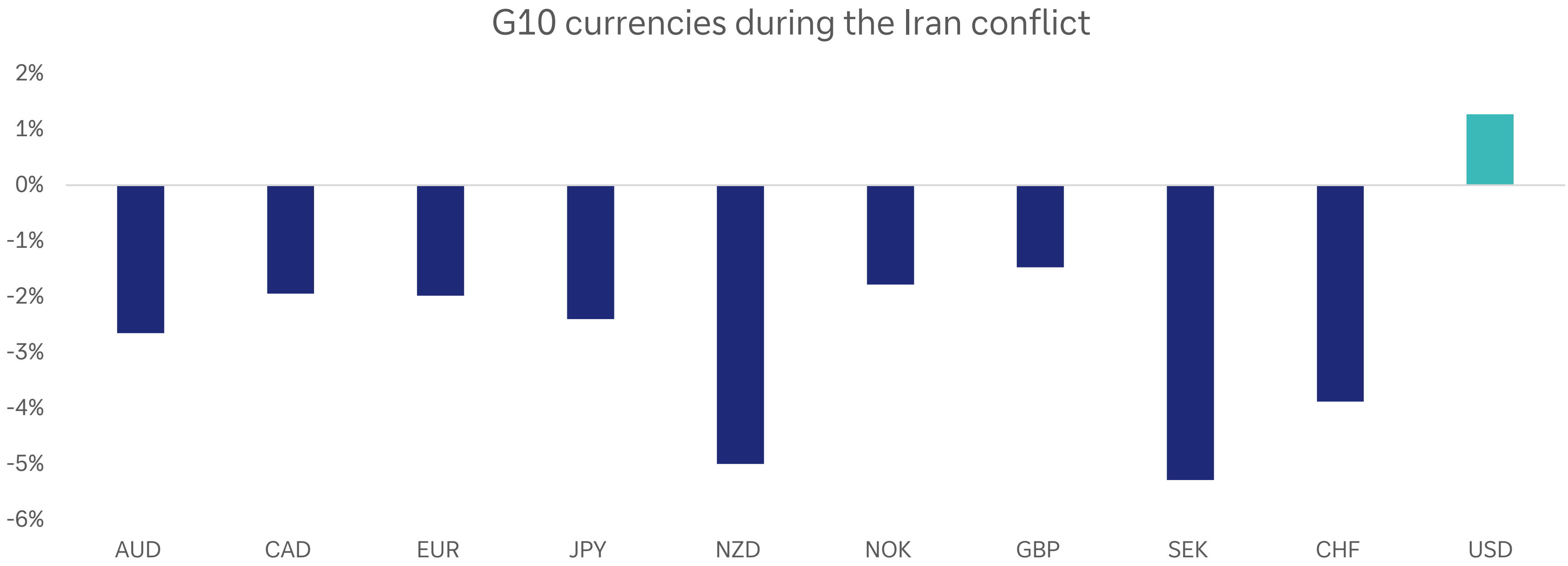
The US dollar was an exception

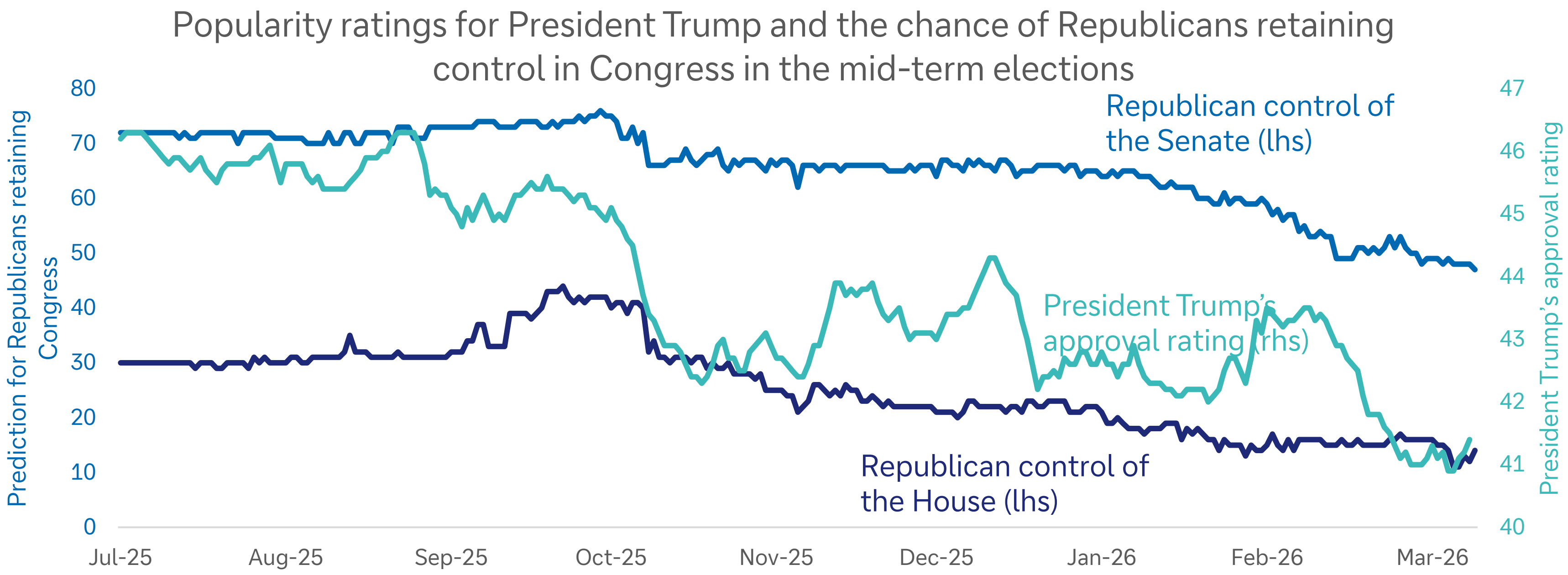
There was a sharp dichotomy between the risk-off period during the Iran war and the risk-off period during the tariff crisis in 2025. This time, investors were content with the safety of the US dollar. That was no doubt influenced by the fact that a kinetic conflict presented a different type of risk-off profile to the economic challenges last year.



Source: Bloomberg Finance LP, Deutsche Bank.

The political cost

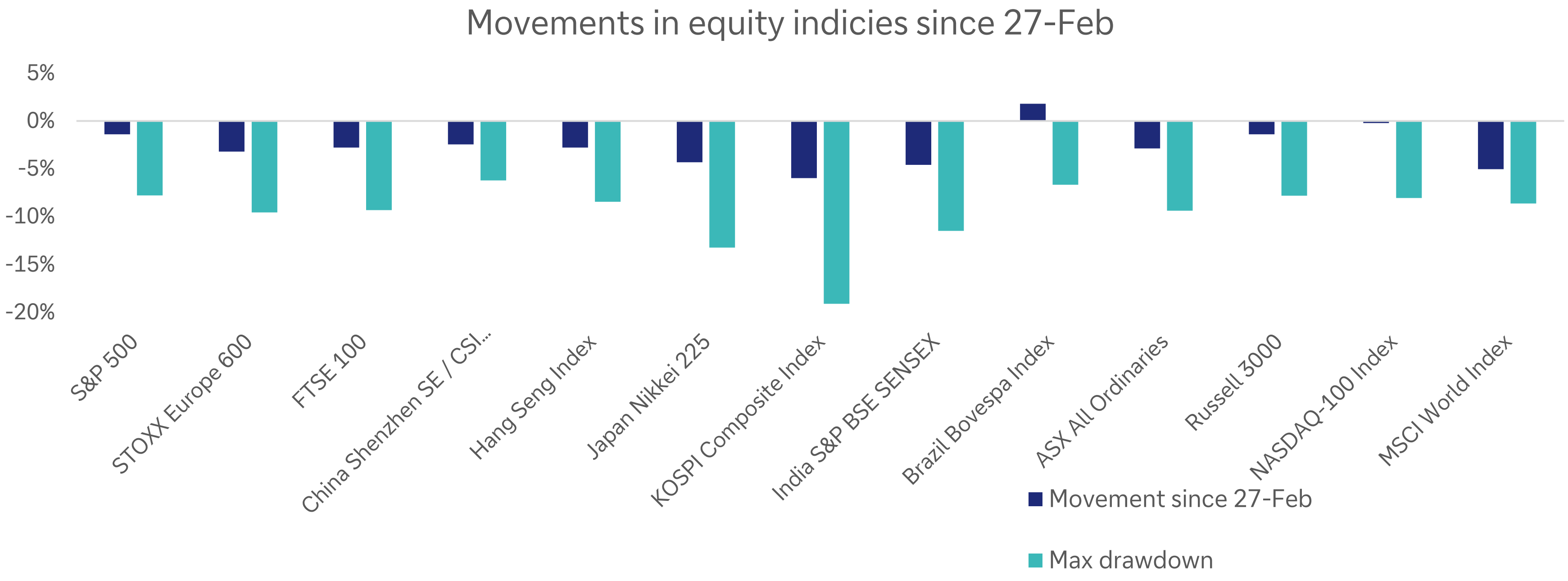
President Trump's approval rating, and the estimate of the Republicans holding the Senate, have fallen during the last five weeks.



Source: Bloomberg Finance LP, Deutsche Bank.

Global equities – most were relatively stubborn considering the severity of the closure of the Strait of Hormuz

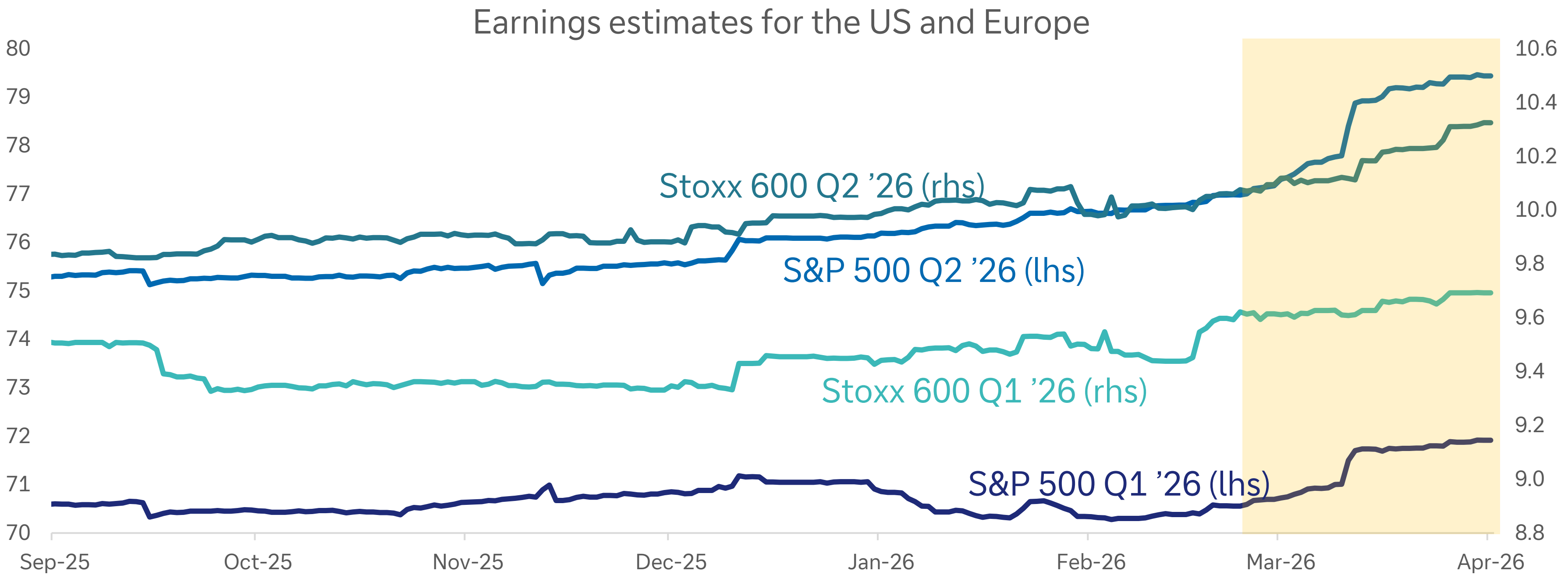
Many investors have learned the lessons of several geopolitical shocks in the 2020s and were expecting a rapid rebound. It may be too early to be quite so optimistic.



Source: Bloomberg Finance LP, Deutsche Bank.

Earnings estimates have stayed strong despite the energy shock

It appears many investors have been reluctant to price in too much downside into corporate earnings until they actually see the financial impact

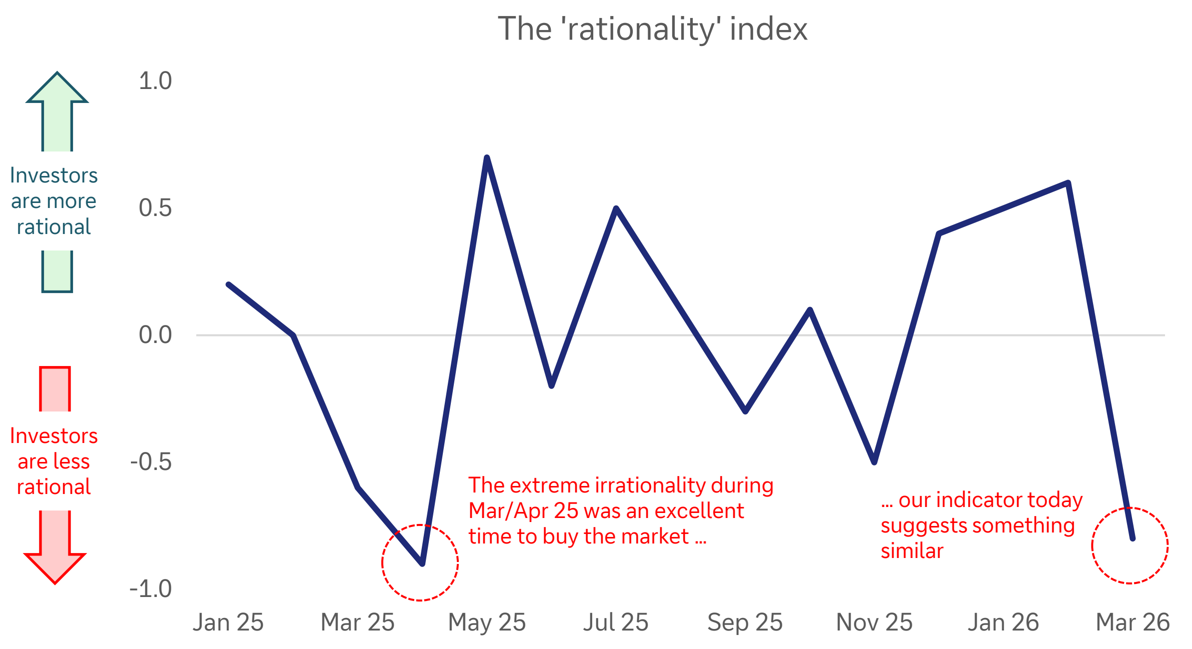


Source: Bloomberg Finance LP, Deutsche Bank.

Are investors rational? During the thick of the crisis, our indicator said "No"

Investors can be good at analysing markets during normal times and terrible during periods of uncertainty

Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions



Source: dbLumina, Deutsche Bank.

What does it mean to be 'rational'

March 2026 Extreme Irrationality (Panic/Fear):
The escalation of the Iran conflict led to a spike in oil prices and a broad market sell-off, characterized as a "stagflationary shock". This overwhelming fear, with investors largely selling off risk assets, indicates a highly emotional and irrational response, despite brief moments of optimism.

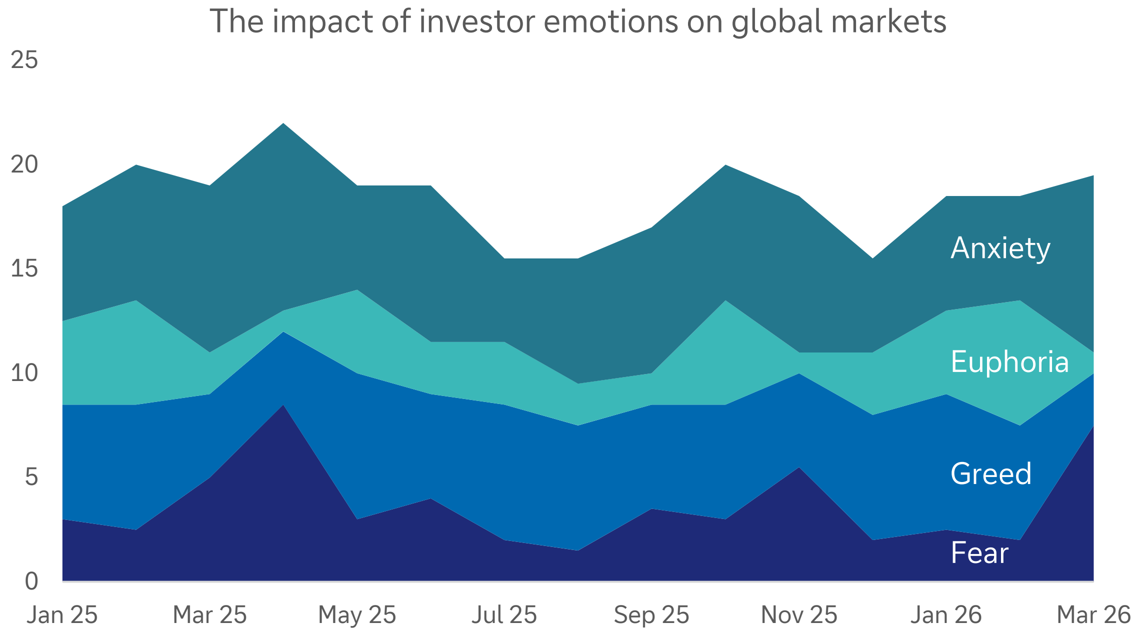
Fear Indicators:

Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted, strong emotional language in descriptions (e.g., "roiled," "plunge," "blistering sell-off," "panic selling,").

The impact of four market-moving emotions: fear, greed, euphoria and anxiety

dbLumina tells us that anxiety and greed can go hand-in-hand even as fear and euphoria wax and wane.

Anxiety and fear ruled Q1



Source: dbLumina, Deutsche Bank.

Status-quo bias is strong. That is creating more market volatility

Investors have been reluctant this decade to consider the upside and downside before the event. So, when something 'bad' happens, market drops can be more aggressive than expected.

Some examples:

1. Demography and fiscal largesse is slow moving and predictable
2. Rush to price in the downsides of tech disruption
3. Fear of considering sovereign risk
4. Fear of underperforming peers is greater than the desire to beat them
5. Unwillingness to price in the upside of iterative technology improvements
6. Narrow vision of what globalisation is and the fear that it has peaked

Appendix 1

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AI, Iran and the Gulf 101

Why the Gulf needs AI – and vice versa

April 2026

Adrian Cox

With deep dedication.

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IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Iran war threatens to derail Gulf ambitions to become third AI region after US, China
How will it affect AI – both inside and outside the region?

Artificial Intelligence has been a driver of growth in the Gulf as the region diversifies from oil and gas income. It has leaned into its natural strengths, such as abundant energy for data centres, and mitigated the challenges, such as how to cool them in the desert heat.

The five-week war centered on Iran may derail those ambitions. Whether or not the current ceasefire holds, it has threatened the region's economy, supply lines and facilities, and could yet squeeze investment not only in the region but also abroad.

Iran's Islamic Revolutionary Guard has reportedly hit several Amazon Web Services (AWS) data centres in the UAE and Bahrain. It has also threatened to attack the giant 1GW planned Stargate (UAE-US AI Campus) on the outskirts of Abu Dhabi. For their part, the US and Israel have targeted Iranian data centres.

In this short report, we examine why AI is so important to the Gulf, why the region is so well-suited for it, what progress it has made so far, why this matters to the rest of the world, with a focus on its sovereign wealth funds, and what the long-term implications of the war may be.

Number of data centres by country

Tehran Times
@TehranTimes79
X.com

#BREAKING
Spokesman of Iran's Khatam al-Anbiya Headquarters:

Nothing is hidden from our sight.

All ICT companies in the region will be considered legitimate targets for us.

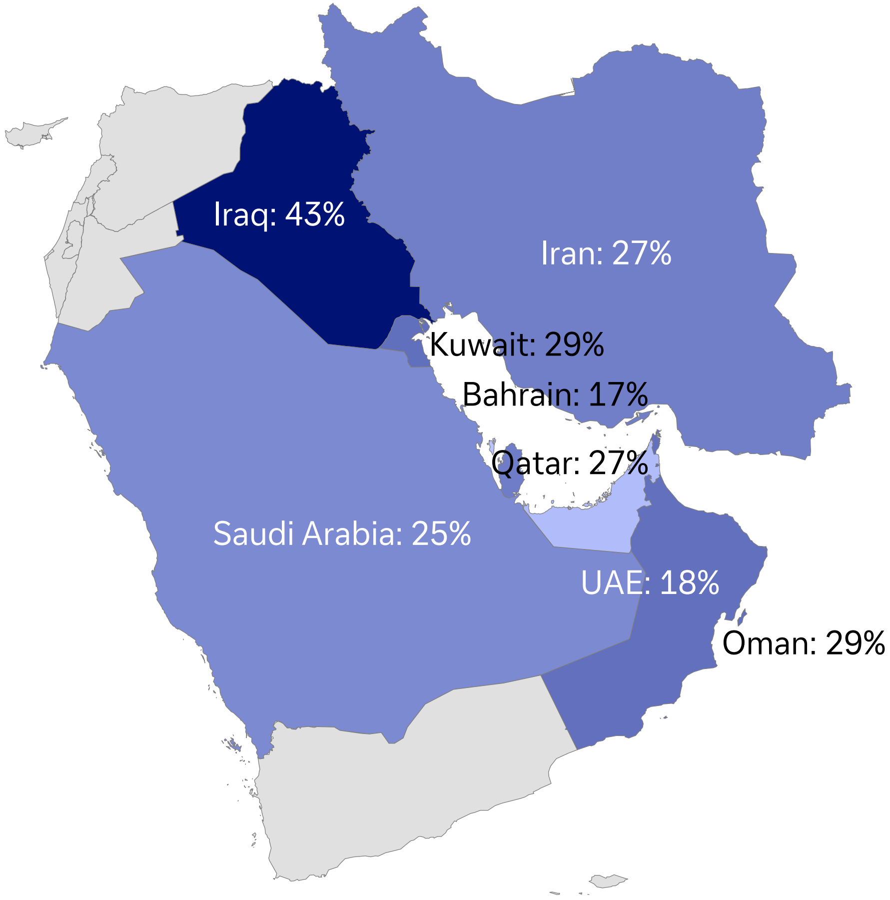
07:02 · 03/04/2026 · **115K** Views

84 675 2.4K 200

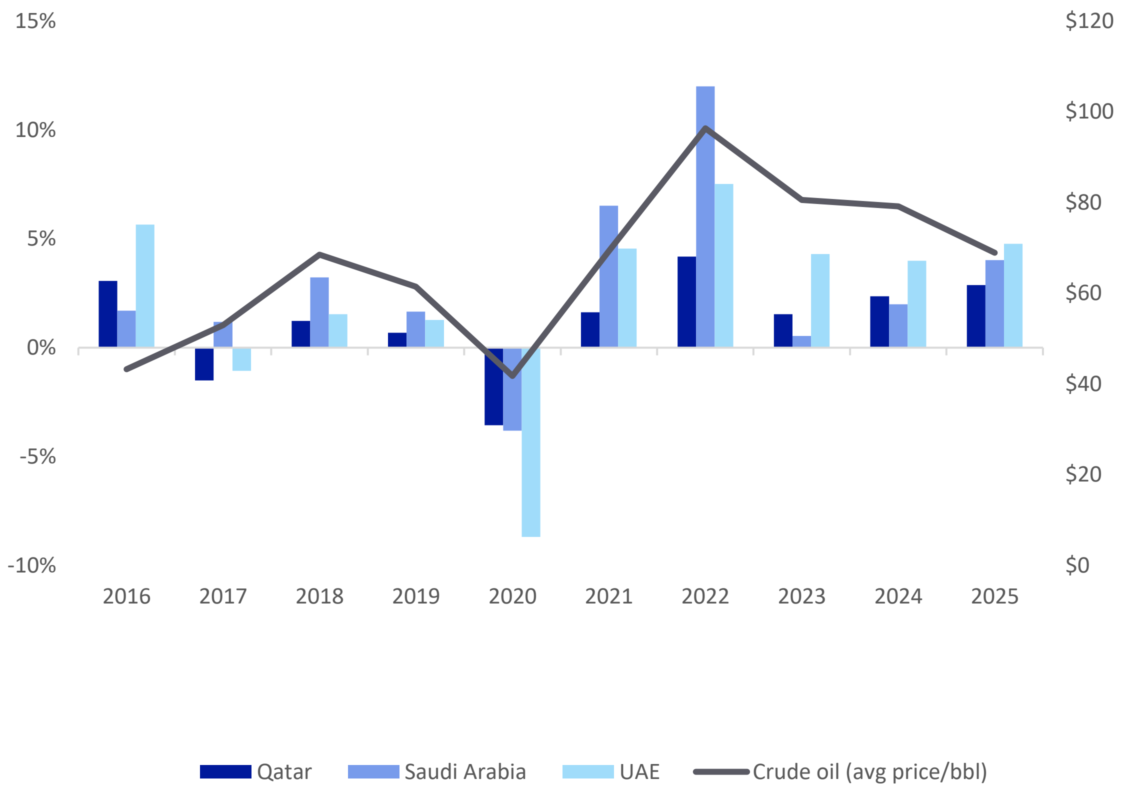
Source: Data Center Map, X, Deutsche Bank Research

1. Why does AI matter so much to the Gulf?
The region is seeking to diversify from its dependence on oil and gas

Oil and natural gas economic profits (% of GDP)



Average oil price ($/bbl) and real GDP growth (%)



Source: Our World in Data, World Bank, Haver Analytics, Deutsche Bank Research

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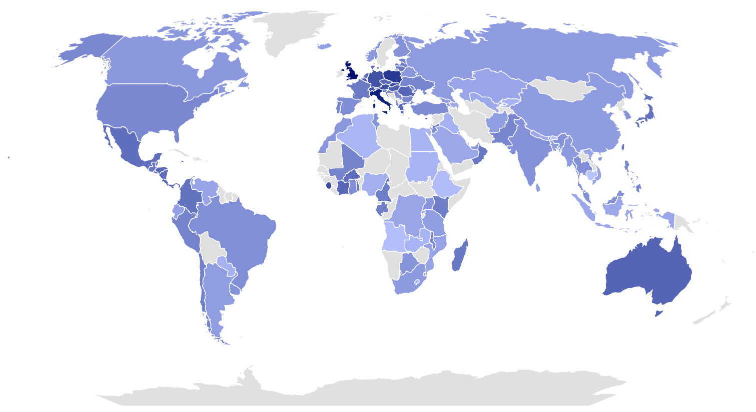
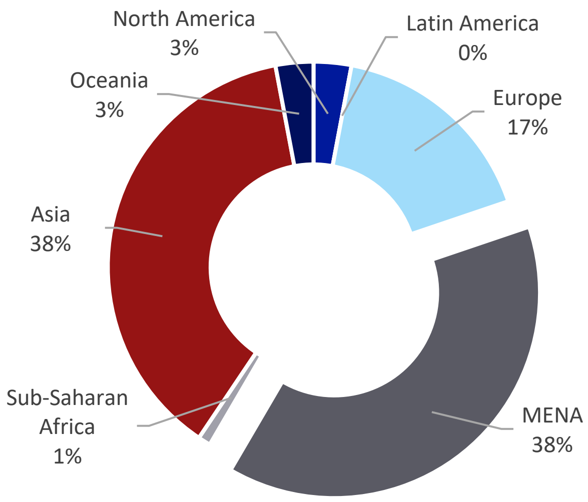
2. Why is the Gulf so well suited for AI?

i. Cheap energy; ii. Abundant, focused sovereign wealth; iii. High citizen use and trust

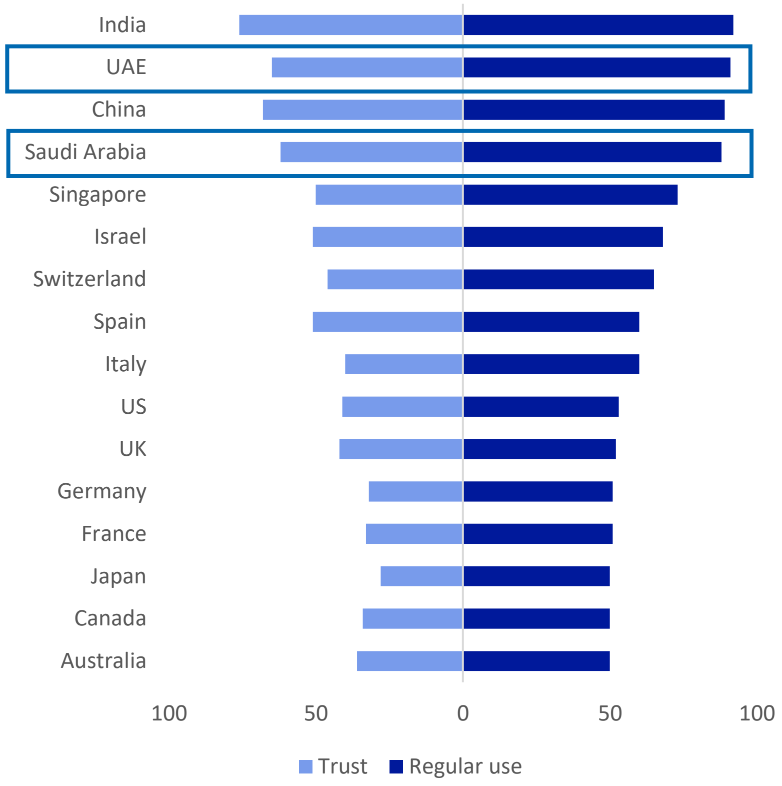
Business electricity rates world, 2023-2026 average (darker shade is highest at $0.445/kWh)



Sovereign Wealth Funds USD 15.8tn, by region



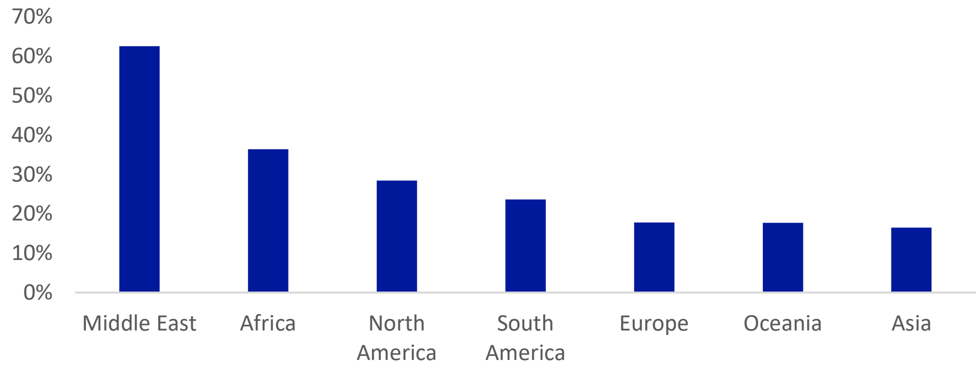
Regular use and trust in AI systems across selected countries (%)



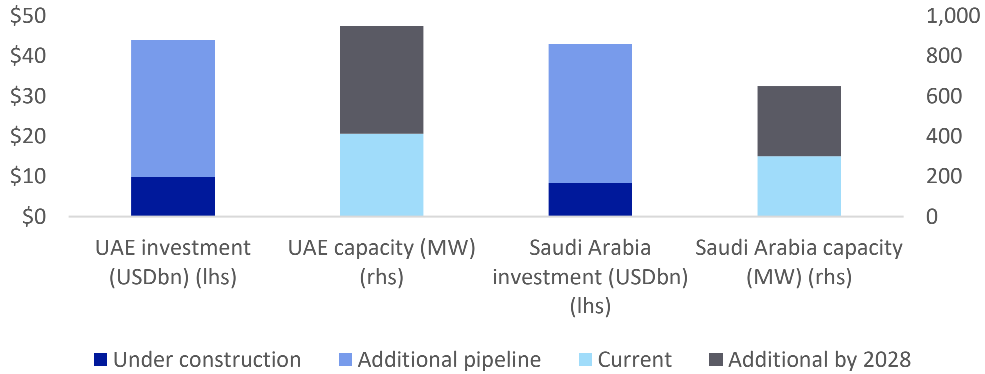
Source: globalpetrolprices.com, Global SWF, Gillespie, N., Lockey, S., Ward, T., Macdade, A., & Hassed, G. (2025). Trust, attitudes and use of artificial intelligence: A global study 2025. The University of Melbourne and KPMG, Deutsche Bank Research

3. What progress has it made so far?
i. Data centres: Saudi Arabia and UAE rank in top four locations due to favourable environment

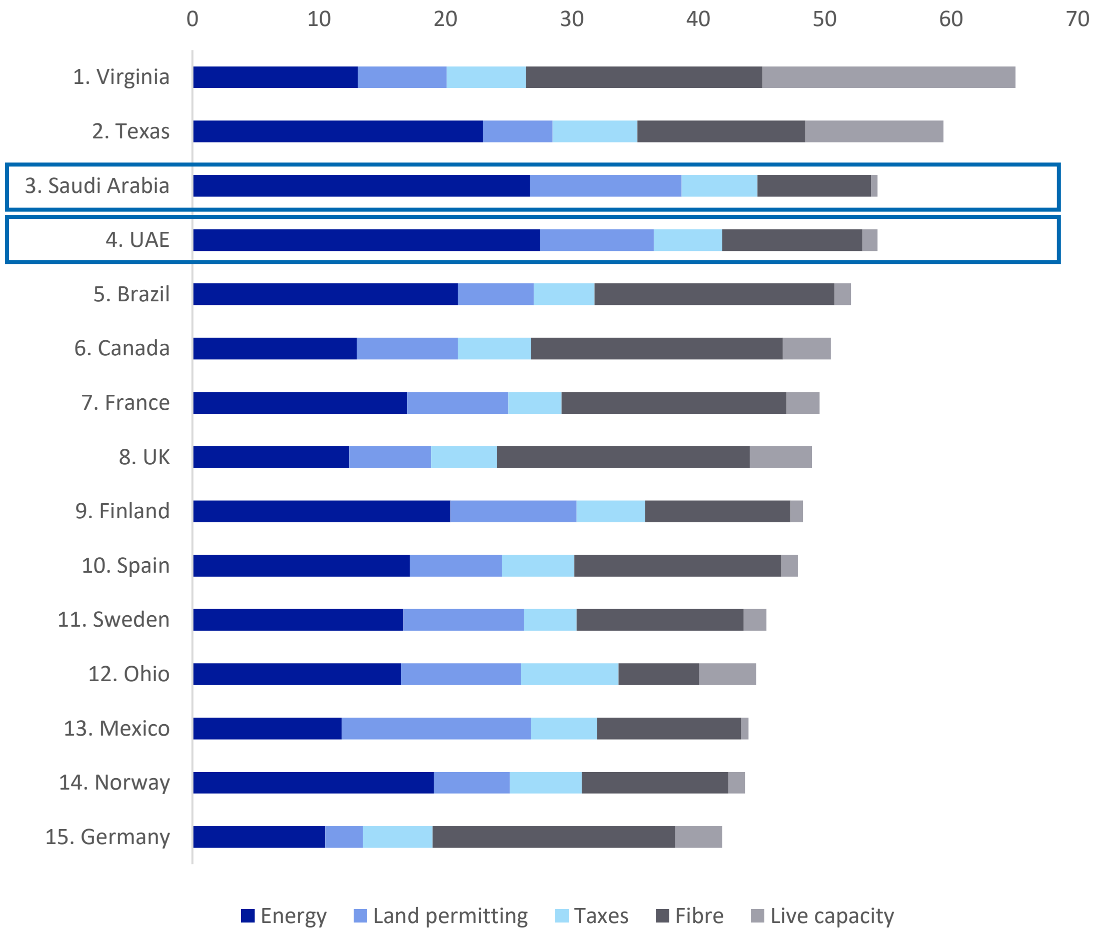
Global forecasts of live IT capacity (MW): compound annual growth rate 2025-2027



Data centre capacity and investment in UAE, Saudi Arabia



Data centre development ranking



Source: Knight Frank Data Centres Global Forecast Report 2026 (NB: Volumes include colocations, self-build, and build-to-suit capacities), Emirates NBD, BloombergNEF, Deutsche Bank Research

3. What progress has it made so far?
ii. Investment: UAE, Saudi Arabia, and now Qatar, lead domestic and international AI push

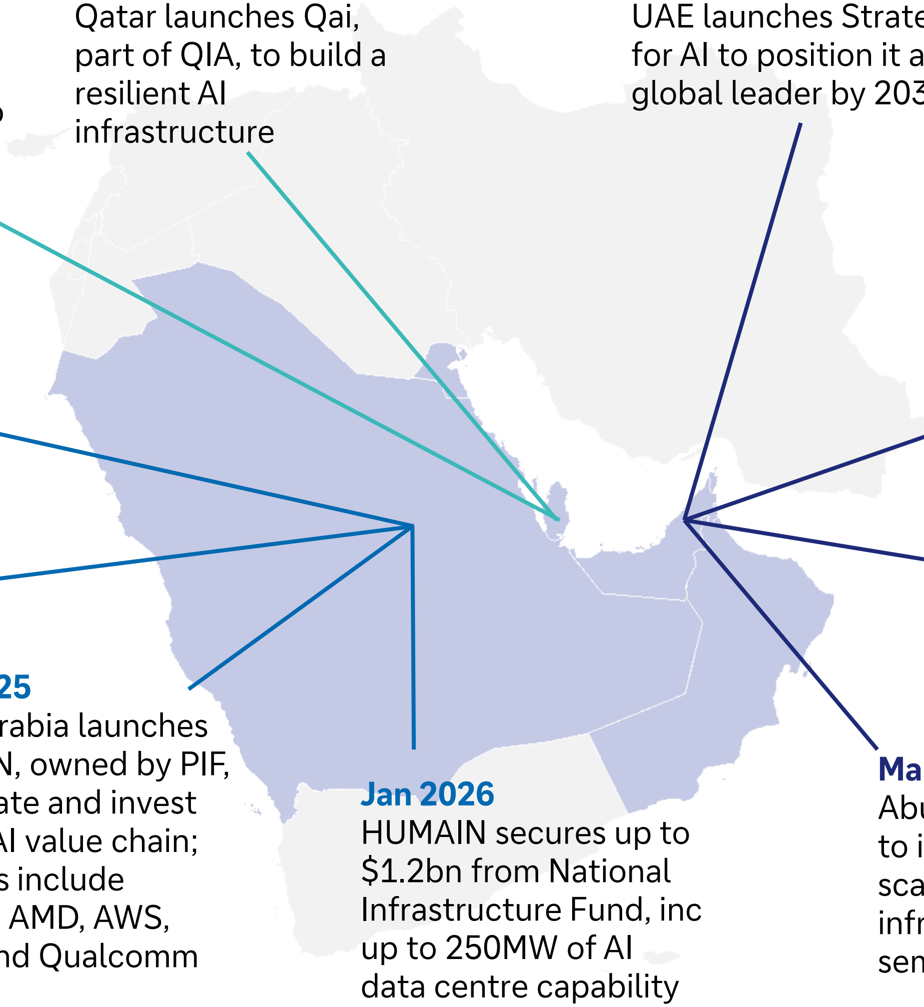
2019
Qatar launches National AI Strategy to become a knowledge-based economy

Dec 2025
Qatar launches Qai, part of QIA, to build a resilient AI infrastructure

2017
UAE launches Strategy for AI to position it as a global leader by 2031

2020
Saudi Arabia launches National Strategy for Data and AI (NSDAI) to establish it as a global leader by 2030

2018
UAE sets up G42 in Abu Dhabi to be an AI and cloud computing national champion



Jan 2024
Abu Dhabi sets up AI and Advanced Technology Council to oversee strategy and investment

2022
Aramco announces launch of Prosperity7 Ventures, $1bn VC fund to invest in tech startups globally

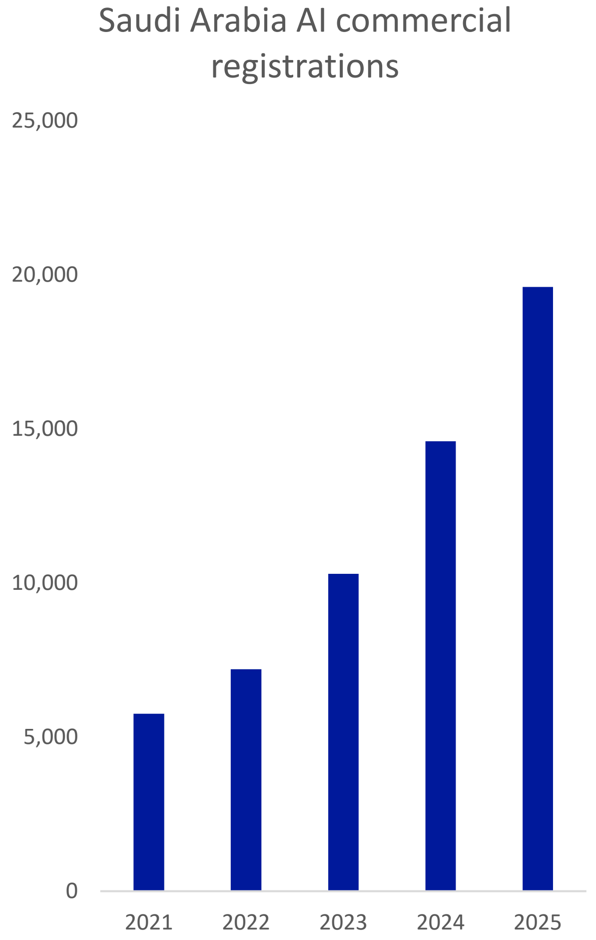
May 2025
Saudi Arabia launches HUMAIN, owned by PIF, to operate and invest across AI value chain; partners include NVIDIA, AMD, AWS, Cisco and Qualcomm

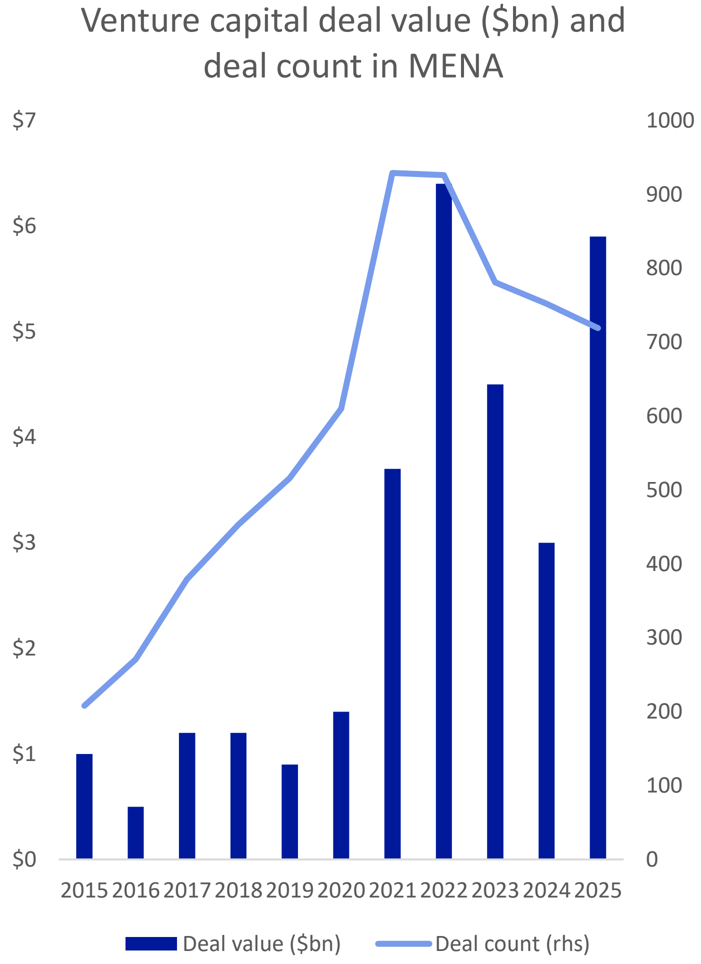
Jan 2026
HUMAIN secures up to $1.2bn from National Infrastructure Fund, inc up to 250MW of AI data centre capability

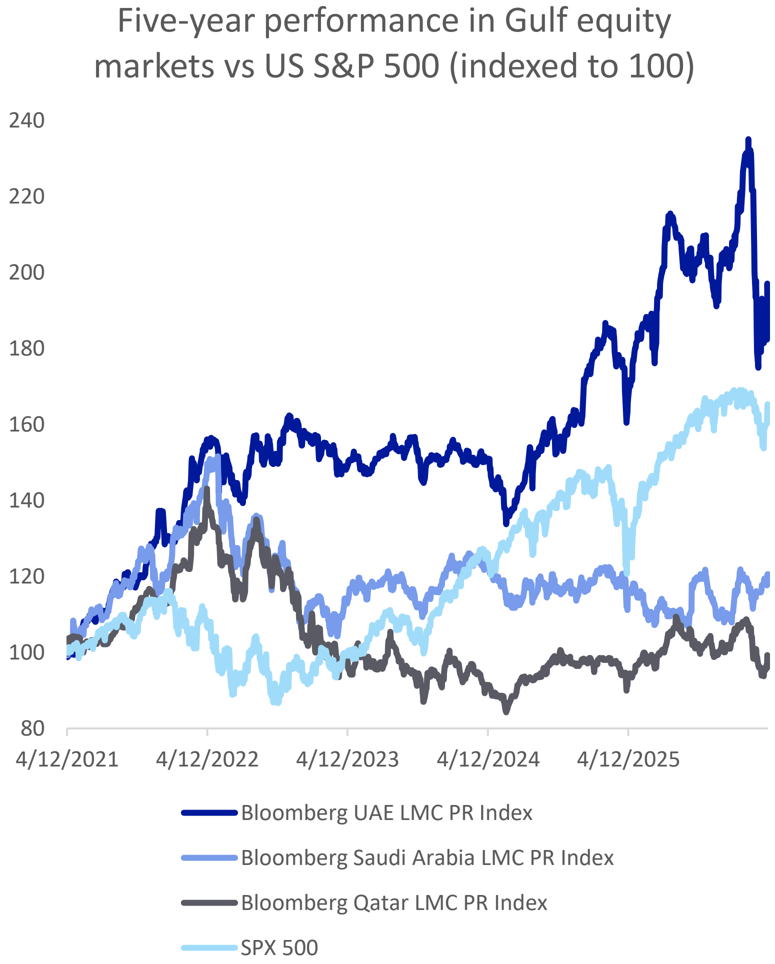
March 2024
Abu Dhabi creates MGX to invest capital at scale in AI infrastructure and semiconductors

Source: Company announcements, media reports, Deutsche Bank Research

3. What progress has it made so far?
iii. Domestic activity: surge in new company formation and investment amid mixed markets



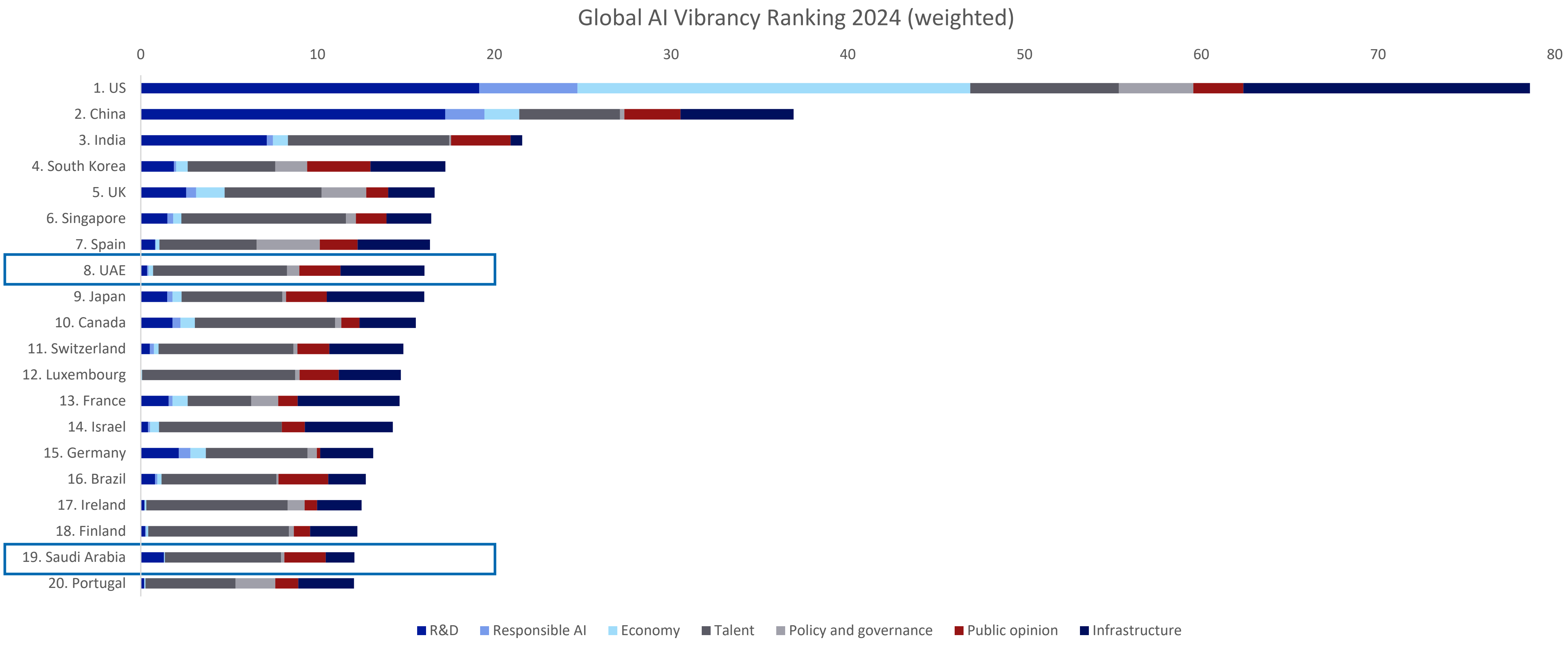




Source: Saudi Arabia Ministry of Commerce, Pitchbook, Bloomberg Finance LP, Deutsche Bank Research

3. What progress has it made so far?

As a result, UAE ranks No. 8 globally in AI vibrancy; Saudi Arabia also scores in the top 20

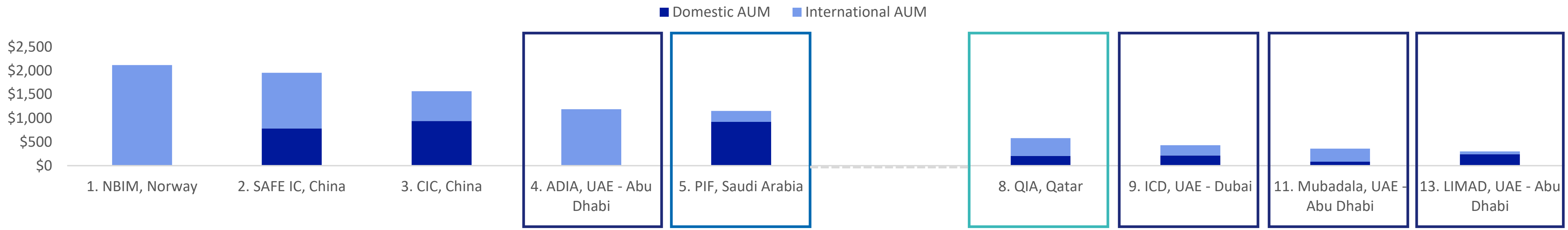


Source: Stanford University Human-Centered Artificial Intelligence Global Vibrancy Tool, Deutsche Bank Research

4. Why does this matter to the rest of the world?

i. Gulf sovereign wealth funds manage about $5tn and invested $119bn in 2025, up 43%

UAE, Saudi and Qatari sovereign wealth funds dominate top table (AuM, $bn)



Abu Dhabi Investment Authority (ADIA) (founded 1967)

Public Investment Fund (PIF) (1971)
Top SWF investor in 2025 at $36.2bn, mostly for EA Sports

HUMAIN (2025)
PIF-owned AI operator, investor

Qatar Investment Authority (QIA) (2005)

QAI (2025)
AI builder, investor

Investment Corporation of Dubai (ICD) (2006)

Mubadala (1984)
No. 2 SWF investor at $33.7bn in ’25 and No. 1 investor in AI/digital at $12.9bn

LIMAD (formerly ADQ) (2018)

MGX, Abu Dhabi (2018)
UAE international investment vehicle

G42, Abu Dhabi (2018)
AI development company

Notable investments

Anthropic

Stargate Project

OpenAI

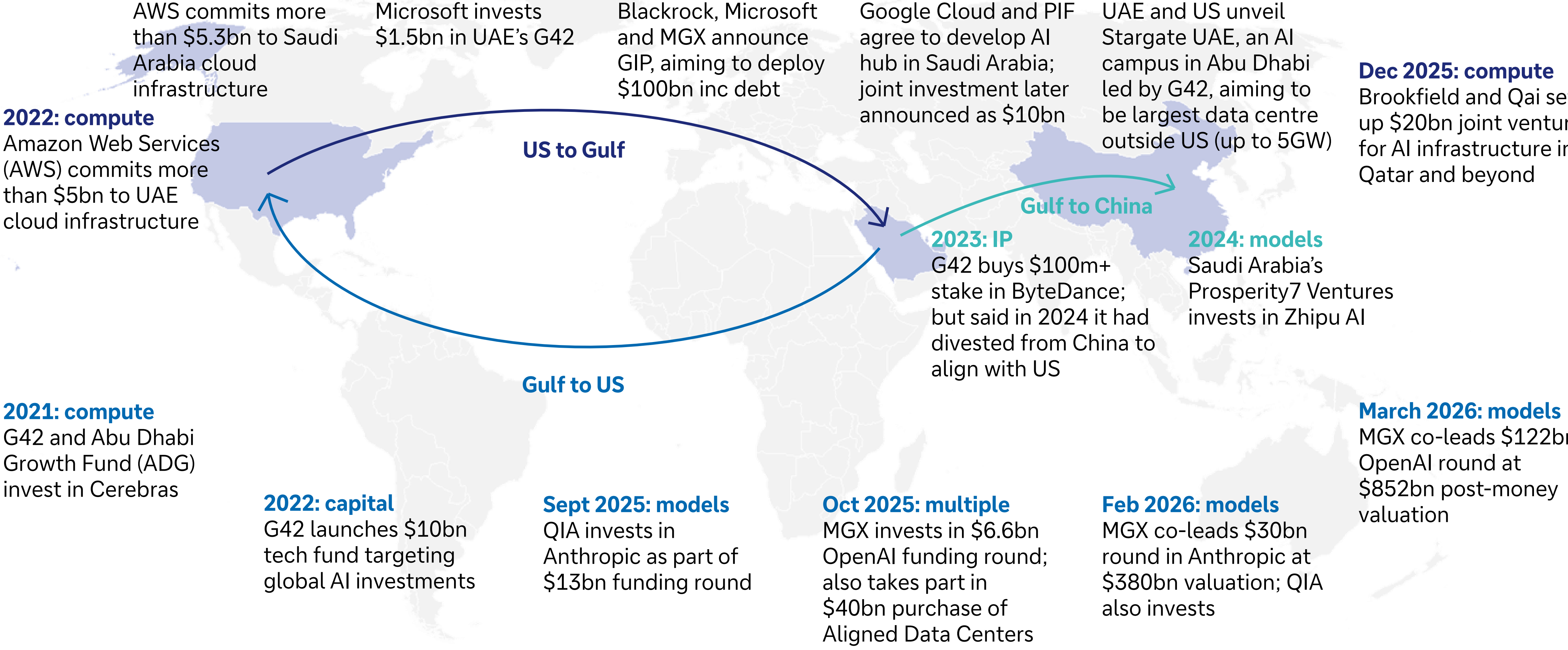
Aligned Data Centers

Microsoft

Source: Global SWF, media reports, Deutsche Bank Research; NB Gulf Seven assets are latest period USD figure if available, estimation otherwise

4. Why does this matter to the rest of the world?

ii. Investment goes both ways: Gulf countries and US focus on capital, compute and models



2022: compute
Amazon Web Services (AWS) commits more than $5bn to UAE cloud infrastructure

March 2024: compute
AWS commits more than $5.3bn to Saudi Arabia cloud infrastructure

April 2024: capital
Microsoft invests $1.5bn in UAE's G42

Sept 2024: capital
Blackrock, Microsoft and MGX announce GIP, aiming to deploy $100bn inc debt

Oct 2024: compute
Google Cloud and PIF agree to develop AI hub in Saudi Arabia; joint investment later announced as $10bn

May 2025: compute
UAE and US unveil Stargate UAE, an AI campus in Abu Dhabi led by G42, aiming to be largest data centre outside US (up to 5GW)

Dec 2025: compute
Brookfield and Qai set up $20bn joint venture for AI infrastructure in Qatar and beyond

2023: IP
G42 buys $100m+ stake in ByteDance; but said in 2024 it had divested from China to align with US

2024: models
Saudi Arabia's Prosperity7 Ventures invests in Zhipu AI

2021: compute
G42 and Abu Dhabi Growth Fund (ADG) invest in Cerebras

2022: capital
G42 launches $10bn tech fund targeting global AI investments

Sept 2025: models
QIA invests in Anthropic as part of $13bn funding round

Oct 2025: multiple
MGX invests in $6.6bn OpenAI funding round; also takes part in $40bn purchase of Aligned Data Centers

Feb 2026: models
MGX co-leads $30bn round in Anthropic at $380bn valuation; QIA also invests

March 2026: models
MGX co-leads $122bn OpenAI round at $852bn post-money valuation

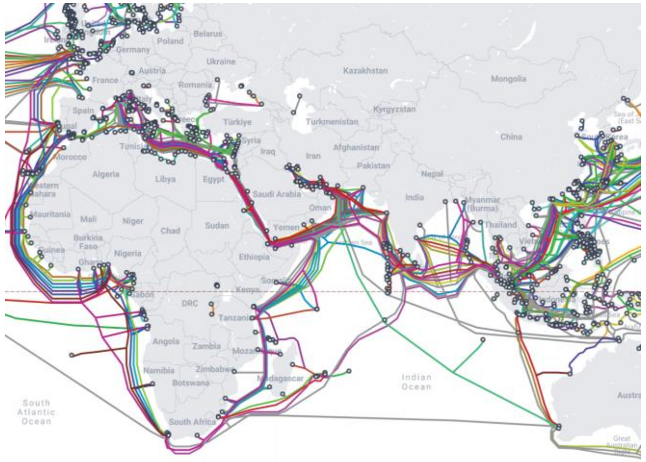
Source: Company announcements, media reports, Deutsche Bank Research

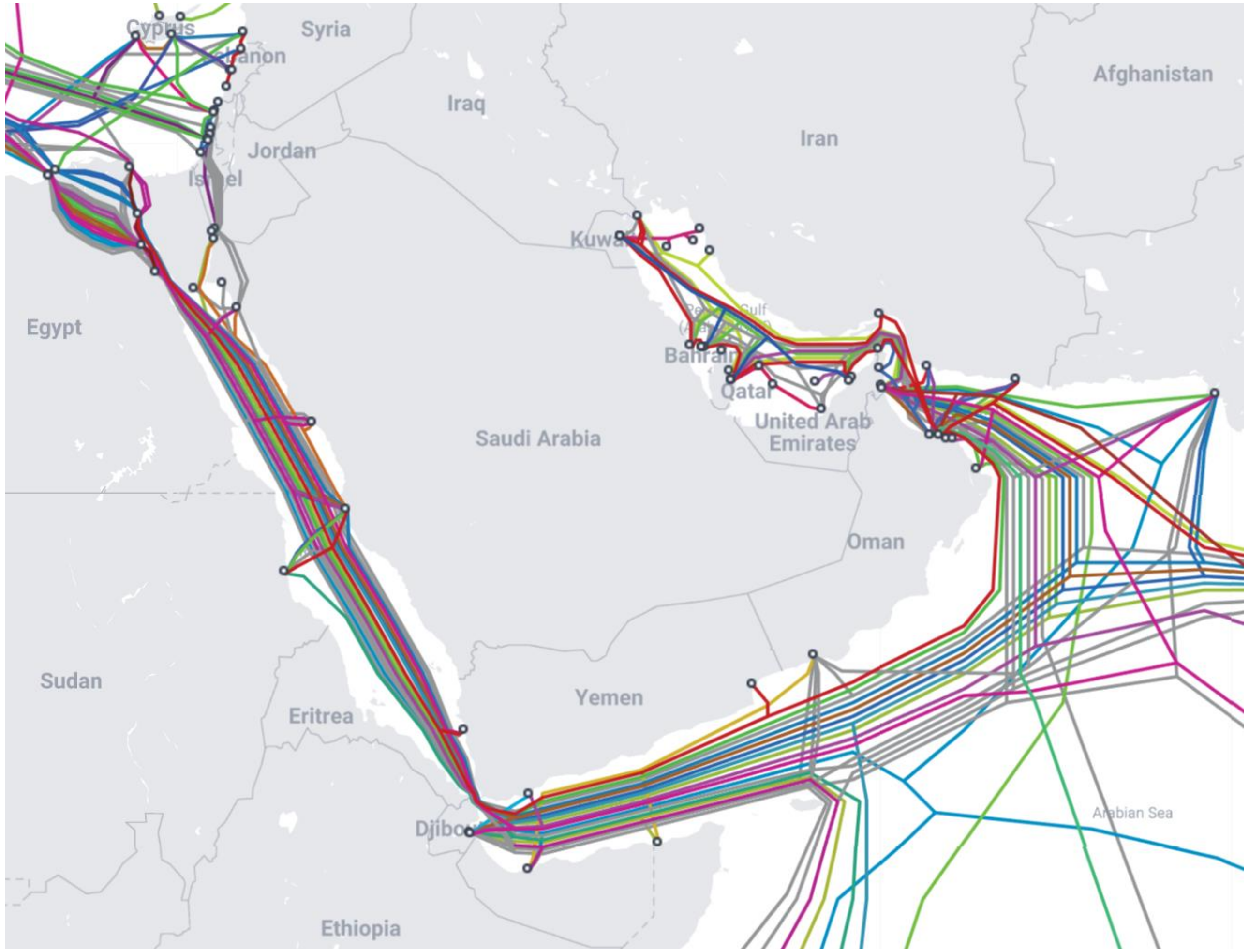
4. Why does this matter to the rest of the world?

iii. The region is at a crossroads not just for energy but also for AI demand, supply chains and data

Four examples:

1. The region is a key source of demand: IT spending in MEA was $155bn last year, 4% of the global market (source: IDC). Almost all of its hardware is imported.
2. Qatar produces about one third of global helium, key for cooling superconducting magnets and producing chips.
3. Taiwan, which produces almost all of the most advanced chips, relies on LNG for about 43% of its electricity – and one third of its LNG usually comes from Qatar.
4. About 30% of all intercontinental internet traffic passes through about 17 submarine data cables in the Gulf.





Source: *Telegeography Submarine Cable Map, Deutsche Bank Research*

5. What could be the long-term implications?

Despite short-term threats, the war could boost demand for AI investment in the longer term

1. Cybersecurity

   – Enhanced detection and response to threats, especially to critical infrastructure

Shaky ceasefire unlikely to stop cyberattacks from Iran-linked hackers for long

World Apr 8, 2026 6:14 PM EDT

2. Supply chains

   – Regionalisation of manufacturing and built-in redundancy of supply

Iran war chokes off helium supplies in threat to chipmakers and healthcare

3. Sovereign AI infrastructure and data

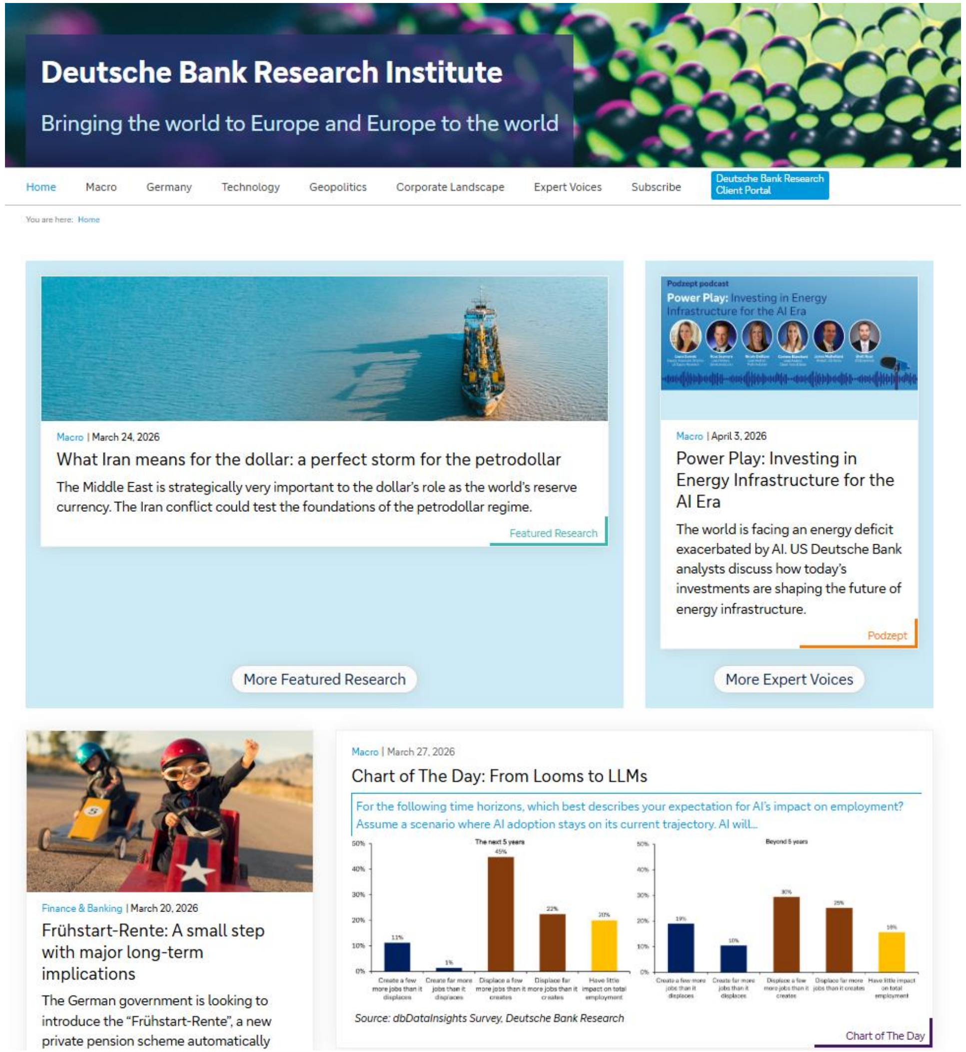
   – Investment in independent digital infrastructure, owned and operated domestically

The Iran War Will Disrupt the Digital Order and Redefine Sovereignty

S. YASH KALASH / APR 3, 2026

Source: Associated Press, Financial Times, Tech Policy Press, Deutsche Bank Research

Recent relevant research



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AI is not convinced AI is a major disinflationary force (April 3, 2026)
Power Play: Investing in Energy Infrastructure for the AI Era (April 3, 2026)
Tracking the effects of AI on the US labor market (March 10, 2026)
The AI sell-off: Over or underdone? (March 4, 2026)
What AI says about AI eating itself and the world… (Feb 16, 2026)
AI Q&A: Why have market vibes suddenly shifted on AI? (Feb 16, 2026)
Three AI themes for 2026: the honeymoon is over (Jan 20, 2026)
Would the real AI bubble please stand up? Boom or bust in 20 slides (Dec 11, 2025)
Happy birthday, ChatGPT? OpenAI faces three threats as Altman issues "code red" (Dec 3, 2025)
AI 101: Economy: Five ways AI is driving growth (Nov 18, 2025)
AI 101: Technology: hype-free guide for users (Oct 2, 2025)

Adrian Cox is a Managing Director and Thematic Strategist at Deutsche Bank Research in London. He is also Managing Director of the Deutsche Bank Research Institute. He joined Deutsche Bank in 2009 after working as an award-winning journalist and editor at Bloomberg News and the Financial Times in London, Brussels and New York. He is a graduate of Cambridge University and has an MBA from City University in London.

Appendix 1

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Deutsche Bank
Research

What Q1's emerging themes mean for 2026 and beyond

April 2026

Luke Templeman | Galina Pozdnyakova

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

Deutsche Bank
Research Institute

Contents

Thought experiment

If, at the beginning of 2026, someone asked how far the S&P 500 would fall if the Strait of Hormuz was closed, what would you have answered?

Probably more than the 6-7% peak-to-trough drop in Q1

The result reflects not just that corporate earnings forecasts remain resilient and investors are fearful of being out of the market if there is a sudden ceasefire in Iran and markets rebound. It also reflects how some of the broader themes and megatrends in markets are influencing the outcome.

Our thanks to Krystle Coe for her assistance with this presentation

1. What themes are driving markets?

The themes driving markets in Q1 – based on AI analysis of our daily market commentaries

1. Escalating geopolitical tensions in the Middle East
2. Stagflationary fears and hawkish central bank response
3. "Picks and shovels" AI investment theme
4. Equity market correction and increased volatility
5. Heightened oil price volatility and supply chain adaptation
6. Disruption in the semis and technology sectors
7. Slowing economic momentum in Europe
8. The weakening euro
9. Landmark legal rulings on social media and AI
10. Escalating geopolitical tensions in the Middle East

How much did each key theme in Q1 impact the S&P 500 (as a proxy for risk assets more broadly)?

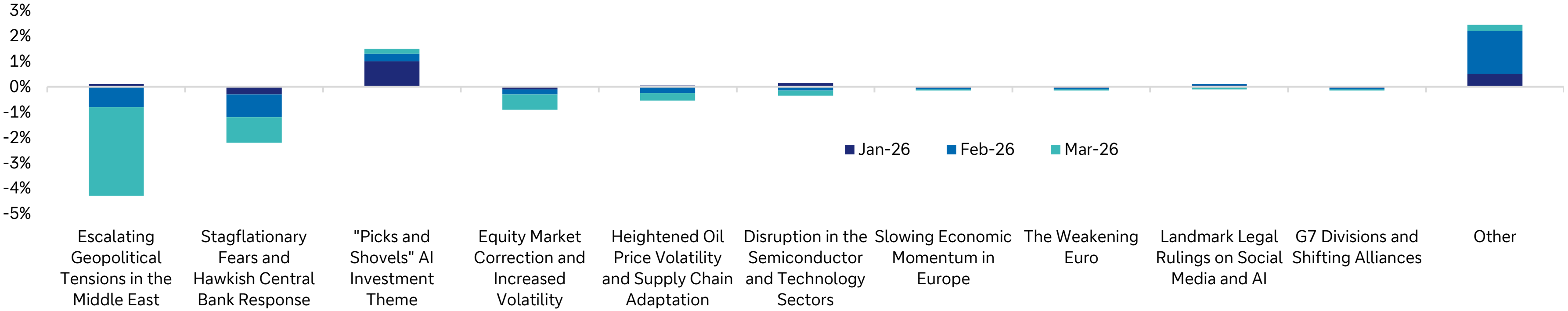
Iran conflict
It mattered … a lot … but the reality is that other factors support markets

AI disruption
The sell-off is just one of many over the last 3 years. Good tech earnings for Q1 & 2 may assuage fears

Fed policy
Central banks typically don't react to supply-side shocks, but the Fed may delay cuts

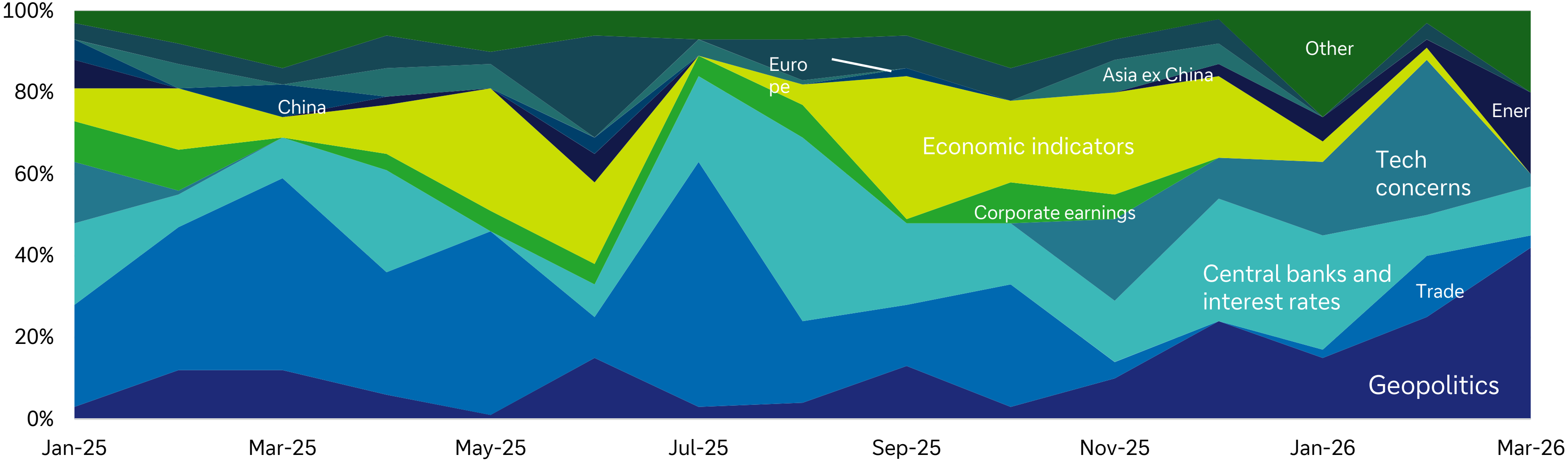
Energy markets
Investors are not pricing in a recession, but if oil remains above $90, they may suddenly do so

Impact of Q1's key themes on the S&P 500 (AI assisted by dbLumina)



Source: dbLumina, Deutsche Bank.

Our AI analysis shows the importance of each 'theme category' to markets each month



Source: dbLumina, Deutsche Bank.

The impact of today's themes 1 year ahead

We asked our AI model, dbLumina, to analyse the impact of the themes for the quarter. The 'positive' and 'negative' indicators are relative to expectations for the year at the beginning of the quarter. For example, a net negative score for the S&P 500 does not mean we expect the index to fall. Rather it means we see the rising probability that the result will be lower than previously expected.

	Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1	Middle East Conflict	Significantly Negative	Significantly Negative	Moderately Positive	Significantly Positive
2	Stagflation & Hawkish central banks	Significantly Negative	Significantly Negative	Significantly Positive	Significantly Positive
3	"Picks & Shovels" AI	Significantly Positive	Significantly Positive	Slightly Positive	Slightly Positive
4	Market Correction	Slightly Negative	Moderately Negative	Neutral	Neutral
5	Oil Price Volatility	Moderately Negative	Moderately Negative	Moderately Positive	Significantly Positive
6	Tech/Semiconductor Disruption	Slightly Negative	Moderately Negative	Neutral	Neutral
7	Slowing European Economy	Slightly Negative	Slightly Negative	Neutral	Neutral
8	Weakening Euro	Slightly Negative	Slightly Negative	Neutral	Slightly Negative
9	Landmark Legal Rulings	Slightly Negative	Moderately Negative	Neutral	Neutral
10	G7 Divisions	Slightly Negative	Slightly Negative	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	-9	-9	-1	-6

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

The impact of today's themes 3 years ahead

We are far more optimistic into the medium term. That appears to reflect the fact that the background economic and corporate fundamentals are relatively solid. Still, there are numerous trends working against global economic growth, although these factors will be specific to individual countries.

Theme	GDP Growth	S&P 500	Interest Rates[1]	Inflation[1]
1 Middle East Conflict	Slightly Negative	Slightly Negative	Slightly Positive	Slightly Positive
2 Stagflation & Hawkish Banks	Neutral	Slightly Positive	Moderately Negative	Moderately Negative
3 "Picks & Shovels" AI	Significantly Positive	Significantly Positive	Moderately Negative	Moderately Negative
4 Market Correction	Neutral	Slightly Positive	Neutral	Neutral
5 Oil Price Volatility	Slightly Negative	Neutral	Slightly Positive	Slightly Positive
6 Tech/Semiconductor Disruption	Moderately Positive	Moderately Positive	Slightly Negative	Slightly Negative
7 Slowing European Economy	Slightly Negative	Slightly Negative	Neutral	Neutral
8 Weakening Euro	Neutral	Neutral	Neutral	Neutral
9 Landmark Legal Rulings	Neutral	Slightly Positive	Neutral	Neutral
10 G7 Divisions	Slightly Negative	Slightly Negative	Neutral	Neutral

	GDP Growth	S&P 500	Interest Rates[2]	Inflation[2]
Sum of positive/negative indicators (positive is +1, negative is -1)	-1	+3	+1	+1

1) For interest rates and inflation, "positive" implies upward pressure on rates/prices
2) A score >0 for interest rates and inflation is 'good' ie. they are expected to go lower.

Emerging themes – part 1

We asked dbLumina to help us identify the key emerging themes that have the potential to most impact markets and economies in the short term

1 The 'AI hardware' squeeze and supply chain bifurcation

In Q1, there was a clear "bifurcation" between AI beneficiaries (hardware) and laggards (software), as investors grew more discerning.

12-Month Outlook: The focus will shift from the conceptual "AI disruption" to the tangible constraints and opportunities in the hardware manufacturing process.

We expect increased volatility in semiconductor stocks as investors scrutinise margin pressures, capex guidance, and the ability to maintain technological leadership.

US policy on the export of advanced chips will likely impact earnings for a concentrated but critical group of companies.

2 Sovereign fiscal fragility and the rising cost of debt

Mounting government deficits and rising debt service costs may become more prominent, particularly given the energy price shock.

The dollar's drop and then rally in Q1 partly reflected growing unease about the US fiscal and policy path. In Europe, fiscal concerns in France widened spreads in a sign of how rapid the issue can impact markets.

12-Month Outlook: We expect fiscal discipline to emerge as a continuous market narrative rather than a sporadic one.

Yield spreads may spike rapidly as markets price in a 'great divergence' in fiscal health. This may lead to near-term headwinds for equity markets in more-indebted nations and could lead to renewed domestic political tensions, particularly in Europe.

Emerging themes – part 2

A key learning from the 'emerging theme' analysis is the focus on some longer-term themes that have been pushed aside by the conflict in Iran

3 Reshoring and the proliferation of "friend-shoring"

This builds on the theme of 'middle powers' building alliances between themselves, restructuring global supply chains. In Q1, Canada was particularly active with Prime Minister Carney visiting numerous middle power countries to seek closer economic ties. Meanwhile, the EU and India progressed a trade deal and, within Europe, there was a greater push to build out domestic military capabilities.

12-Month Outlook: 'Middle power' countries are likely to increase their capital investment and M&A activity in sectors deemed strategically important, such as semiconductors, critical minerals, and defence.

This will benefit industrial, materials, and energy sectors in politically aligned countries. However, it may also lead to higher costs and inefficiencies for multinational corporations as they are forced to reconfigure their supply chains.

This may drag on margins and add low-level, persistent inflation.

4 Strain in US-consumer and labour market dynamics

There is a worrying paradox of weak consumer sentiment but a resilient (potentially fragile) economy – a trend confirmed by economic reports and surveys in Q1. In particular, the "two-speed" economy became more apparent, with reports noting job cuts in sectors like government, finance, and warehousing (healthcare remained strong). This suggests a lack of broad-based strength.

12-Month Outlook: The health of the US consumer is our most closely watched variable for the US economy and the S&P 500. Any slowdown will put the Fed in a difficult position, forcing it to choose between fighting inflation and supporting a weakening economy. A key indicator is the performance of consumer discretionary and staples stocks.

Current evolving risks

The conflict in the Middle East has acted as a catalyst, amplifying pre-existing inflationary pressures and forcing central banks into a hawkish corner, thereby creating a significant risk of a global stagflationary shock

1 The stagflationary shock: A vicious cycle

The global economy is facing a stagflationary threat

The conflict in Iran has caused a dramatic energy price shock which has occurred in an environment where inflation was already proving persistent.

Key central banks have responded with cautious or hawkish rhetoric, signalling that inflation remains a priority even at the cost of economic growth.

2 Geopolitical escalation and fragmentation

The conflict in Iran is the epicentre of global risk and has quickly metastasised with the involvement of non-state actors (Hezbollah, Houthis etc), shipping disruption, and attacks on regional neighbours.

The divisions within the G7 have become clear with European allies showing frustration at the lack of consultation. That further risks a fracturing of traditional Western alliances

3 Equity market de-risking and fragile sentiment

Global equity markets are in a correction phase with the S&P 500’s peak-to-trough drop being more severe than the average response to geopolitical shocks.

The positive narrative around the "Picks and Shovels" of AI, while still present, has been completely overshadowed

4 European economic vulnerability

Europe is particularly exposed as it faces an energy shock and a sharp economic slowdown.

The ECB is in an exceptionally difficult position, forced to maintain a hawkish stance to fight imported inflation despite a weakening domestic economy

Emerging risks underestimated by the market – part 1

While investors are currently grappling with the immediate impact of the Iran conflict, there are several second- and third-order consequences that appear to be underestimated

1 European sovereign debt concerns

The underestimated risk

The combination of a hawkish ECB, slowing growth, and the need for increased fiscal spending on defence and energy security is sowing the seeds of a renewed sovereign debt crisis in the Eurozone.

Historical lesson & analysis

The crisis of 2011-2012 showed how quickly concerns over the fiscal position of one member state can create systemic contagion. The yields on peripheral debt are a potential canary in the coal mine. A hawkish ECB, may have a difficult choice to make, particularly if one of the options involves removing the safety net that has protected the Eurozone for a decade.

The divisions in the G7 suggest that a coordinated political solution, which was crucial to resolving the last crisis, will be much harder to achieve this time.

2 The “picks and shovels" AI bubble

The underestimated risk

While the "picks and shovels" narrative appears robust, it is not immune to a bubble. The massive capital expenditure into the AI supply chain is predicated on future returns that are still largely theoretical.

Historical lesson & analysis

History is replete with infrastructure-led bubbles, from the railway booms of the 19th century to the dot-com fibre-optic rollout. In each case, the building of the infrastructure was a real economic activity, but the eventual returns on that capital were wildly overestimated, leading to a collapse.

The risk is that the market has capitalised various firms based on a gold-rush mentality without fully accounting for the risk of overcapacity or technological obsolescence.

Emerging risks underestimated by the market – part 2

Just two of the big things that markets take for granted is the leadership of Big Tech firms and the ability of companies to adapt their supply chains. But what if developing events erect permanent obstacles?

3 The unravelling of the "Big Tech" social contract

The underestimated risk

The recent court verdicts against Google and Meta are likely not isolated events but the start of a fundamental, and costly, regulatory and legal realignment for Big Tech.

Historical lesson & analysis

The recent rulings are comparable to the landmark rulings against Big Tobacco. There, it took decades for the full financial and social impact of the tobacco litigation to be fully realised. Markets appear to be viewing today's verdicts as one-off legal setbacks. However, they may mark a pivotal shift in the "social contract" for technology firms.

The legal precedent could open the floodgates to class-action lawsuits and force a complete overhaul of business models built on user engagement and data monetisation. In turn, this will lead to structurally lower growth and profitability for some of the market's largest constituents.

4 The illusion of supply chain adaptation

The underestimated risk

While markets have been calmed by signs of supply chain adaptation, such as Saudi Aramco re-routing oil, these are suboptimal and less efficient solutions that will create a drag on global productivity.

Historical lesson & analysis

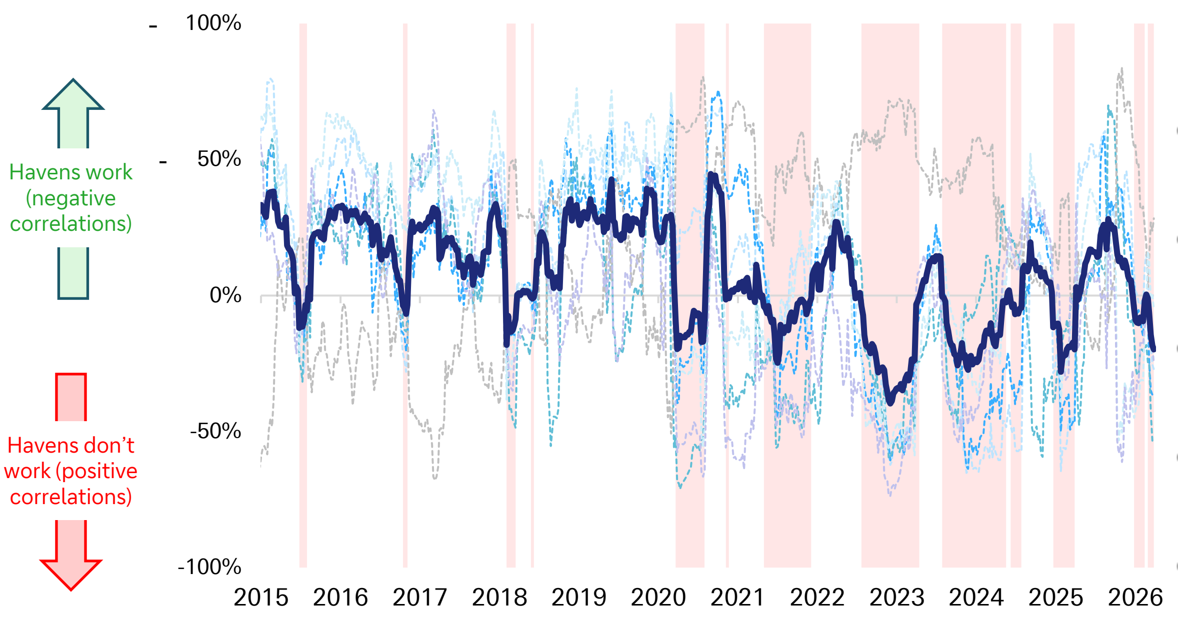
The response to covid and the Ukraine war showed that supply chains can adapt. However, that adaptation came at the cost of higher inflation and lower efficiency. The 'friend-shoring' and re-routing currently being celebrated are, in essence, a tax on the global economy.

Moving from a just-in-time, efficiency-focused model to a just-in-case, resilience-focused one has a cost that may be underestimated in long-term growth and inflation forecasts. This structural inefficiency will linger long after the immediate crisis has passed.

The mysterious case of the disappearing haven assets

Our 'haven indicator' shows that traditional havens have been largely absent in the 2020s

Our haven indicator (20-wk rolling, avg across pairwise correlations with S&P 500)



Source: Bloomberg Finance LP, Deutsche Bank.

How our haven indicator works

We take six traditional haven assets:

1. Gold
2. US dollar
3. Japanese yen
4. Swiss franc
5. 10yr treasury
6. 10yr bund

We then analyse their individual correlation with the S&P 500 (our proxy for 'risk') over time. Finally, we take an average of the six correlations.

The red shading indicates periods where the average haven **did not work** (ie. positively correlated with the S&P 500). This includes:

1. Iran war in 2026
2. US tariff crisis in 2025
3. Peak fear about interest rate rises in 2022
4. Initial covid turmoil in 2020

2. The market impact of specific themes

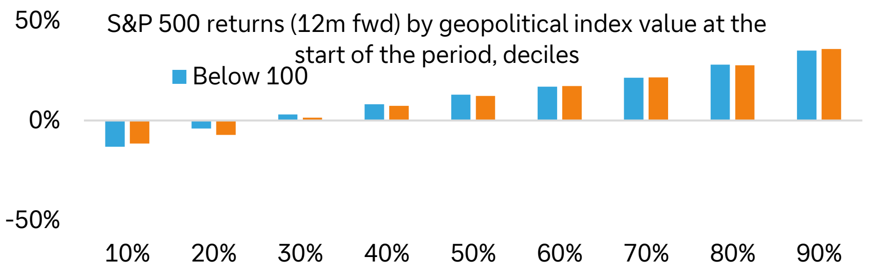
Oil shocks and markets

See our in depth reports here and here

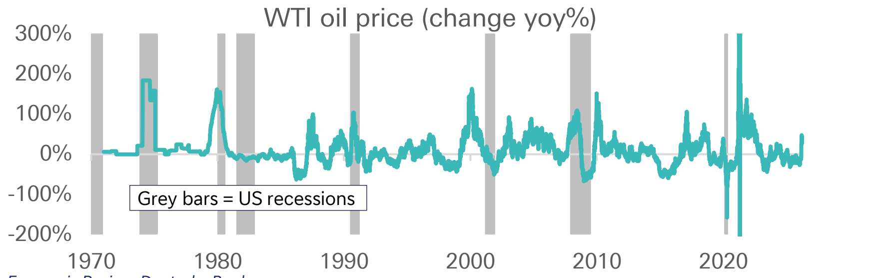
The moves in 10yr treasury yields are highly dependent on whether oil markets and thus US inflation are affected

- Linear models suggest that even very large oil price shocks may have muted effects on the US economy. But the assumption of linearity may be at odds with the historical observation that oil price shocks tend to coincide with recessions.
- Historical episodes of geopolitical tensions show little evidence of prolonged US equity drawdowns, with the median return of 9.8% in the 12 months after event starts. This holds regardless whether an event was related to the Middle East.
- The indifference to geopolitical uncertainty in the medium term is present not only in the US stocks. Among different economies since 1900, it's mostly European and Australian/Canadian stocks that experienced notably lower returns when the geopolitical risk index has been above 100 (a level it has been at for a few months now).
- Among commodities, oil is a better proxy for geopolitical risk in markets (outside covid) than gold even with the US less reliant on oil than in the 1970s. The impact on gold has actually been marginally lower when the geopolitical risk index is above 100 in periods since 2010
- Gold correlations also point to a lack of broader geopolitics as the key driver, especially recently – gold has been negatively correlated with crude and US defense stocks in the last couple of months
- What about defence stocks? – those respond to fiscal actions due to geopolitics and have been more responsive to the slides in geopolitics risk more than the upsides recently. This is to be watched given the run up in valuations

Whether geopolitical risk is high or low, S&P 500 returns after 12m tend to be similar



Recessions often coincide with sharp increases in oil prices.



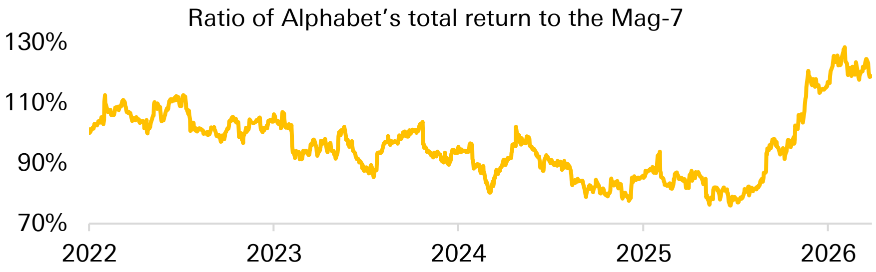
Source: Bloomberg Finance LP, Finaeon, Caldara, Dario and Matteo Iacoviello (2022), "Measuring Geopolitical Risk," American Economic Review, Deutsche Bank.

Software and technological disruption

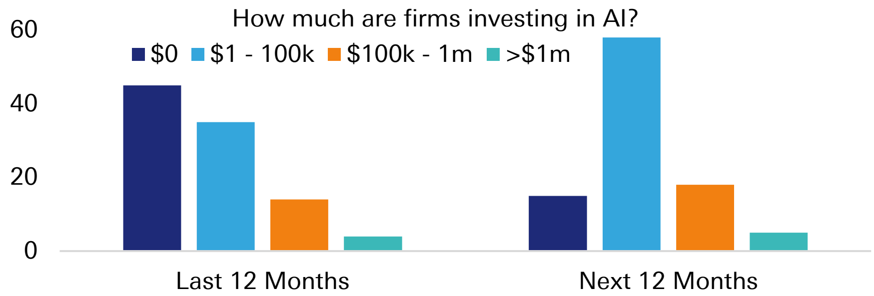
This sell-off is not the first … but the frequency of new AI tools has accelerated in recent months

- The software scare saw the largest sell-off in US software and services stocks vs the broader market since the ChatGPT release in late 2022. Notably, the divergence between "hard" and "soft" IT industry groups was unusually wide.
- Semis and tech hardware were the relative winners in tech. Notable losses occurred in a small number of management and development real estate firms
- Other parts of the AI ecosystem have been notably resilient. Meanwhile, the drop in software & services means there is a technical reinforcement to the selloff
- Amid all the AI investment themes, there is more-or-less a consensus that power demand will surge. This helps shows that concerns about other parts – demand and supply, are becoming more frequent.
- On top of the Google example, following the DeepSeek announcement, Nvidia's stock fell around 20%. Revenue estimates have doubled since then.
- There is also a lot of uncertainty about the impact on the more tangible firm outcomes, such as labour costs and customer satisfaction
- We aren't ignoring the real issues. Several factors support the "AI disruption" narrative. The issue is that the sell-off has been so indiscriminate. Thus, deleveraging and momentum selling of broader industries was a key driver.
- Lastly, public credit markets are the key to a real risk-off event. These show a relatively small exposure to technology (8% in $HY and 10% in IG) and recent spread widening is still modest. Private credit is more exposed but several factors including dry powder limit the risk of a proper turmoil as discussed here.

The Google example: Alphabet lagged after the release of ChatGPT but revenues continue to be strong which assuaged market's concerns about its LLM progress



Will software companies' clients substitute with their own AI tools? Many firms want to increase AI spend but the investment may not fully replace solutions from specialised firms.



Source: Bloomberg Finance LP, Duke University, FRB Richmond and FRB Atlanta, The CFO Survey - Q4 2025 (November 11 - December 1, 2025), Deutsche Bank.

Central bank policy amidst an oil shock

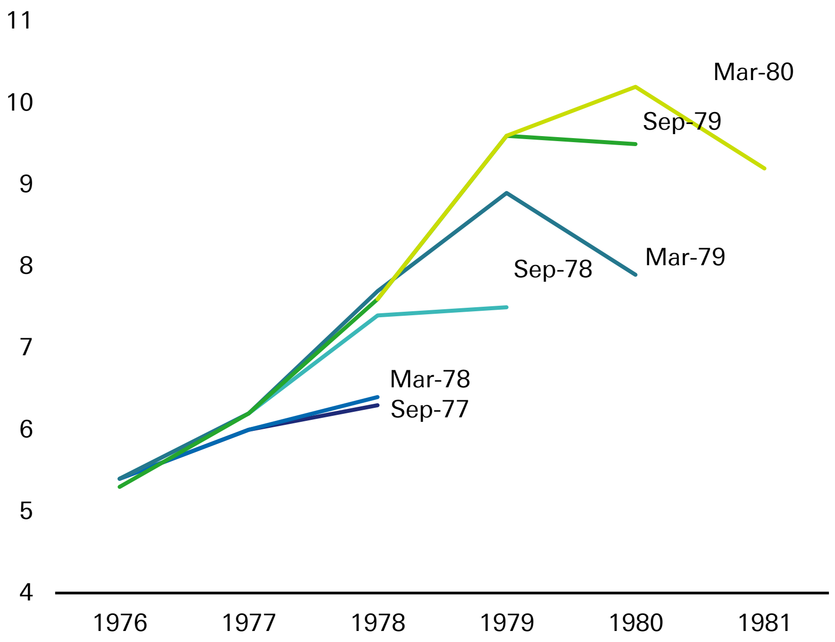
See our in-depth reports here and here

US interest rates are likely to remain on hold through to September

- Central banks face another energy shock, and were criticised last time for acting too late. This time, we can already see how they're trying to avoid a repeat, and are sounding less dovish (and sometimes hawkish) for a given level of inflation.
- Unlike in 2022, there's now an explicit acknowledgement that inflation has been above target for an extensive period. Indeed, Fed Chair Powell said in his March press conference that looking through energy shocks has "always been dependent on inflation expectations remaining well anchored", and he also mentioned the "broader context of five years now of inflation above target."
- ECB President Lagarde said the central bank will "do all that is necessary to ensure inflation is under control and French and Europeans don't suffer the same inflation increases like those we saw in 2022 and 2023."
- There is already discussion about the prospect of rate hikes pre-emptively, even before inflation has moved higher, rather than the more reactive conversation that took place in 2021-22.
- For the Fed, the latest trends suggest that a wait-and-see approach could gain broader support from more officials in the near term, absent a sudden deterioration in the labour market or a spillover of energy prices into core inflation. Our baseline expectation remains that the FOMC has likely delivered its final rate cut under Powell's leadership, and rates will remain on hold until September.

It is easy to underestimate inflation during an energy shock

The Fed's staff inflation forecasts during the 2nd oil shock in the 1970s (%)



Source: Bloomberg Finance LP, Deutsche Bank.

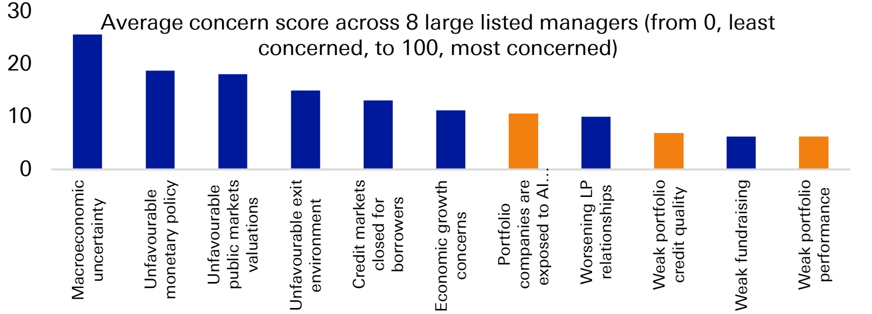
Private markets

See our in-depth reports here and here

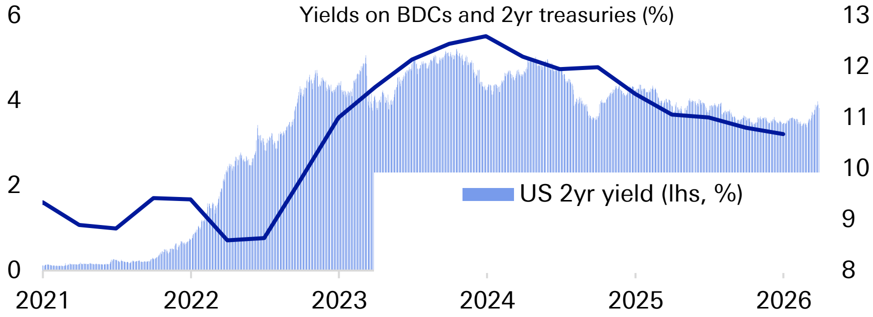
More transparency and communication are likely as private capital targets a more risk-averse and liquidity-demanding investor base to grow.

- In spite of the negative headlines and market performance, the largest managers continued with their exceptional fundraising and AUM growth. The focus is now on keeping the fundraising momentum, including among retail investors, in 2026.
- Where will the AI disruption fallout hit? There will be losers in private markets but price action in private capital-related assets that can be traded is likely misinterpreting the impact
- Any serious fallout from the actual impact of AI and credit fears on private capital is likely to be concentrated in the smaller, lower-quality corners of the industry.
- Many negative headlines have come from liquidity rather than credit risks amid withdrawal requests
- Dry powder continues to rise and is a sizable buffer to sector-specific problems.
- Outside software, exits are growing quickly – this should assuage investor concerns about private equity and free up more capital to do deals.
- There is thus some scope for a reversion in the recent selloff.
- We expect corporate buyers to step back into M&A. In fact, this may become private managers' preferred exit route given that public markets remain volatile amid geopolitical tensions.
- The biggest risk is a confidence crisis that grips wider financial markets even if it is not based on strong fundamental problems

Potential losses on investments is low down the list of concerns for private capital managers – the economy and public markets are far more impactful



Reported dividend cuts can relate to falling interest rates rather than solely outright credit distress given that BDC yields are driven by short-term rates



Source: Bloomberg Finance LP, Deutsche Bank.

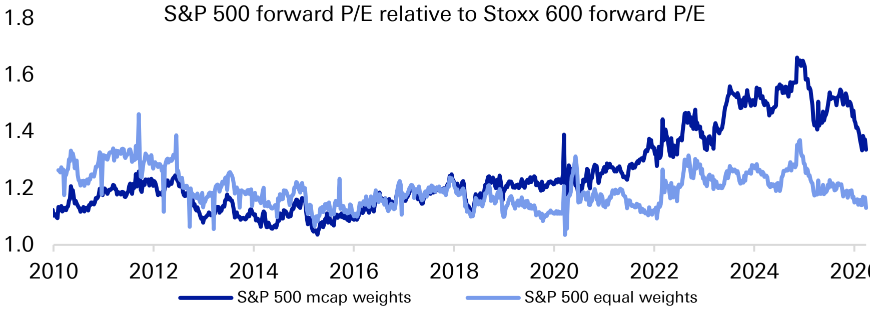
Corporate earnings – what the latest set of results tells us

See our in-depth report here

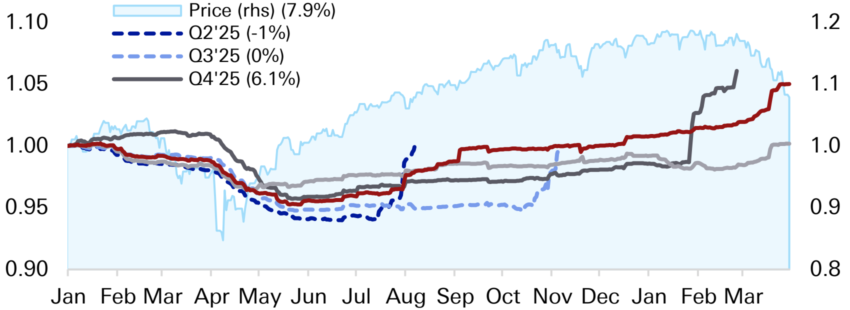
US stocks had a good earnings season with Q4 consensus EPS meaningfully moving up It was more of a mixed picture in Europe

- It's been another strong season in the US, with more dispersion in Europe. Notably, since the start of the earnings season, sectors that posted larger earnings beats have performed worse. This sets up for more gains once investor sentiment around the rotation trade and AI stabilises.
- The rotation into a wider set of stocks (despite strong earnings from tech) is a sign of persistent risk appetite. It also shows that once the dust settles on software fears, a rerating is likely. The IT sector now trades effectively on par with the market, lower than during the 2022 tech wash out in relative terms.
- There has also been a strong yield preference among investors, with dividend stocks in favour before the Iran conflict as the Nasdaq has been slipping. High-growth names including those outside TMT have also been performing strongly until March, showing the rotation has been rather exclusively out of tech as opposed to a full risk-off move.
- Zooming out, investor aversion to software means the US now significantly trails rest of the world. But perhaps more importantly, this shunning of US large cap tech pushed other DM markets to unusually high valuations. Only 4 economies out of 23 are now relatively cheaper vs own history over the last 5 years than the US.
- The equal-weighted version of the S&P 500 is now lower relative to European markets than the average since the data begins in 2010.
- Earnings will be another catalyst, in addition to valuations, that may rerate US markets v RoW higher (despite momentum headwinds).

The P/E premium of the S&P 500 over the Stoxx 600 is now also the lowest since 2022, wiping out the gains made since investors' AI enthusiasm proliferated.



S&P 500 earnings evolution since 2025



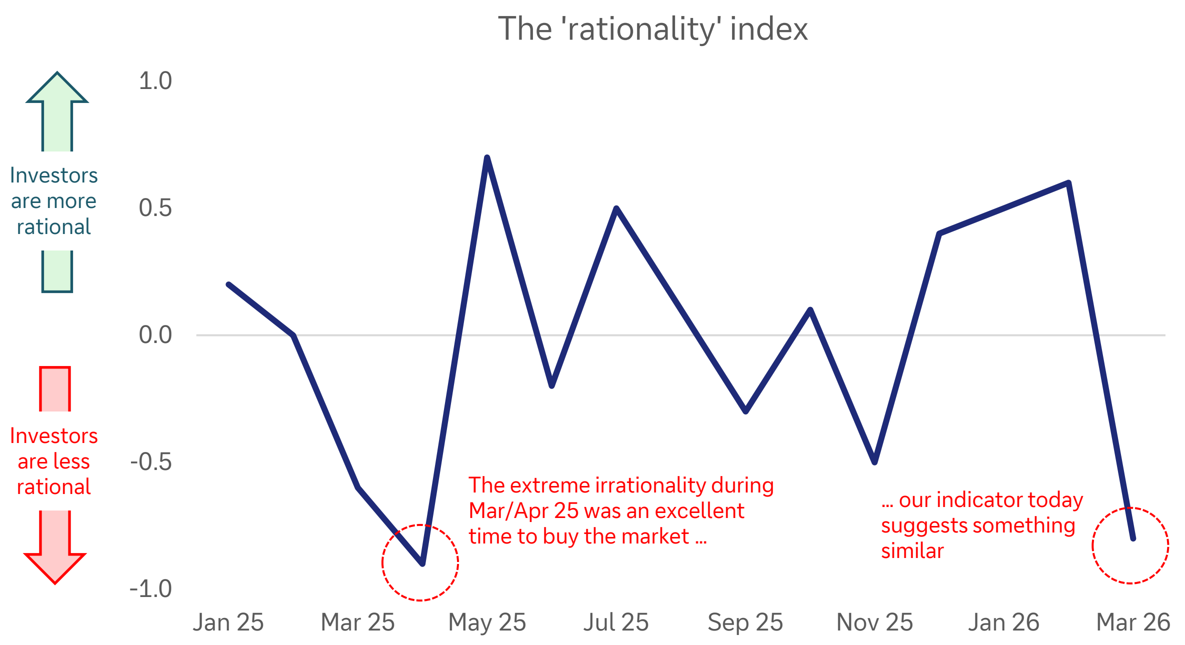
Source: Bloomberg Finance LP, Deutsche Bank.

3. Investor psychology over the last quarter

Are investors rational? Right now, our indicator says “No”

Investors appear to be good at analysing markets during normal times and terrible during periods of uncertainty

Our dbLumina gauge indicates investors have been irrational during market drops and periods of geopolitical tensions



Source: dbLumina, Deutsche Bank.

What does it mean to be ‘rational’

March 2026 Extreme Irrationality (Panic/Fear):
The escalation of the Iran conflict led to a spike in oil prices and a broad market sell-off, characterized as a "stagflationary shock". This overwhelming fear, with investors largely selling off risk assets, indicates a highly emotional and irrational response, despite brief moments of optimism.

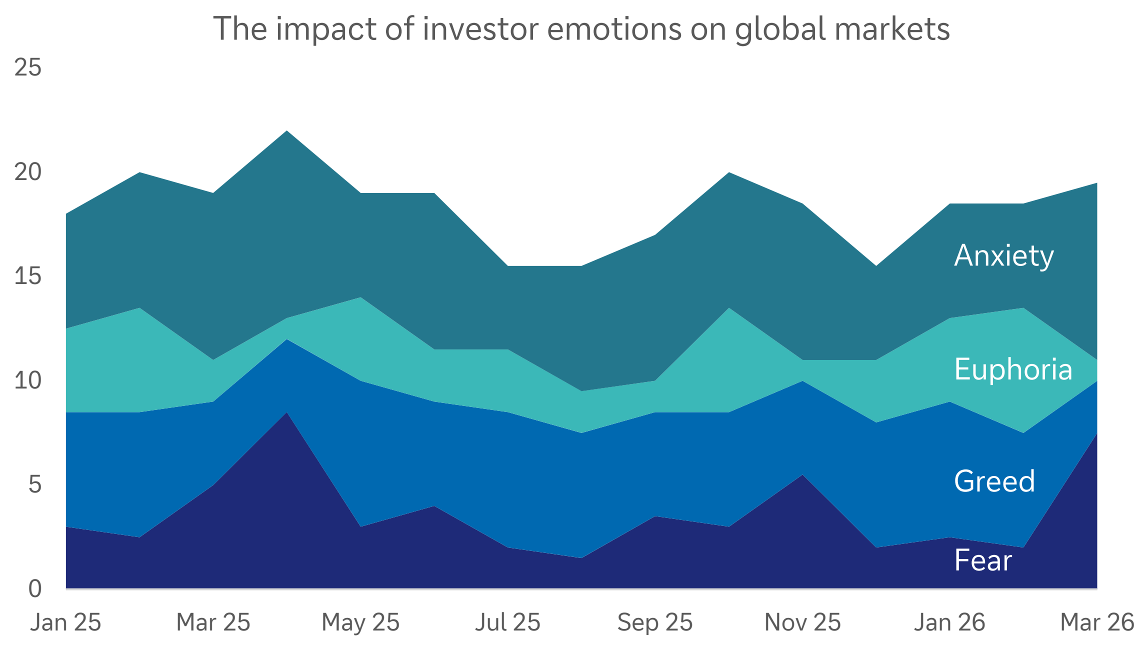
Fear Indicators:

Sharp, disproportionate reactions to threats, widespread sell-offs on announcements, panic, flight to safety beyond what might be fundamentally warranted, strong emotional language in descriptions (e.g., "roiled," "plunge," "blistering sell-off," "panic selling,“).

The impact of four market-moving emotions: fear, greed, euphoria and anxiety

dbLumina tells us that anxiety and greed can go hand-in-hand even as fear and euphoria wax and wane.

Anxiety and fear ruled Q1



Source: dbLumina, Deutsche Bank.

Status-quo bias is strong. That is creating more market volatility

Investors have been reluctant this decade to consider the upside and downside before the event. So, when something 'bad' happens, market drops can be more aggressive than expected.

Some examples:

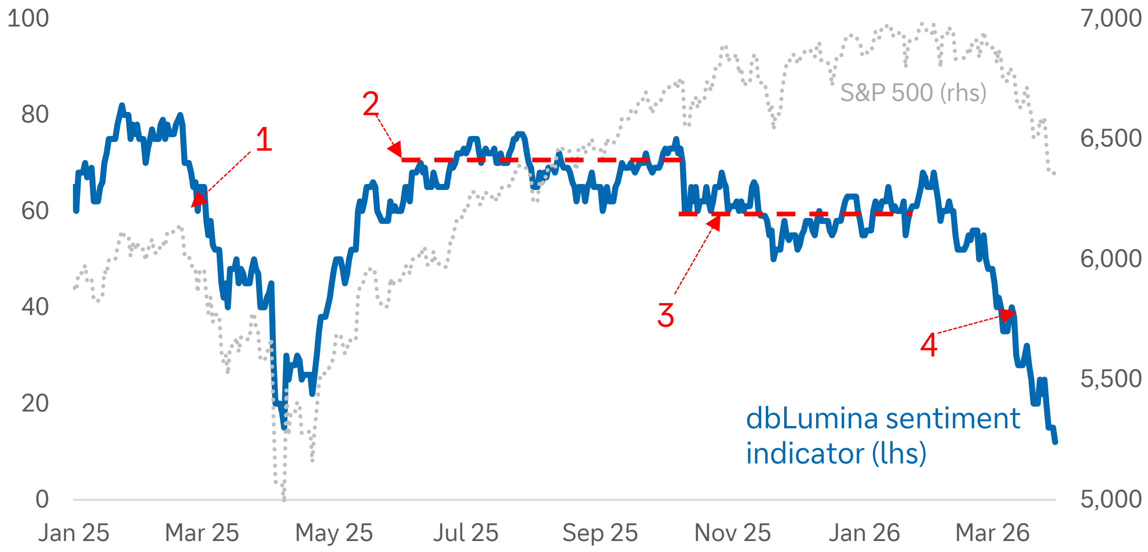
1. Demography and fiscal largesse is slow moving and predictable
2. Rush to price in the downsides of tech disruption
3. Fear of considering sovereign risk
4. Fear of underperforming peers is greater than the desire to beat them
5. Unwillingness to price in the upside of iterative technology improvements
6. Narrow vision of what globalisation is and the fear that it has peaked

We created our own gauge of sentiment. It uses AI to interpret macro, economic, and market signals.

This is our response to the fact that many confidence indices today fail to reflect reality. That is because they can fail to take into account the technological and social megatrends that now affect markets.

It is too early to say our gauge has predictive qualities but …

dbLumina sentiment indicator v S&P 500



Source: Bloomberg Finance LP, dbLumina, Deutsche Bank.

1. Indicator drops 20 points in Feb-25 before the most dramatic stock market declines relating to tariffs.

2. Consistent strong sentiment as markets recovered from the tariff-induced sell-off.

3. Lower sentiment as markets top-out on valuations and software/PC nerves grow. Yet, no catalyst for a large drop.

4. Our sentiment indicator commenced a steep drop on 3-Feb. It fell 20 points before the war in Iran kicked off.

4. How Q1s themes fit in with our megatrend framework

How Q1's themes fit inside our megatrend model

The most important megatrends signals in Q1

Technology
- The drop in the Nasdaq and low productivity

Sovereign debt & the gov. role
- Rising interest costs and fiscal policy uncertainty hurt, partially offset by robust treasuries demand

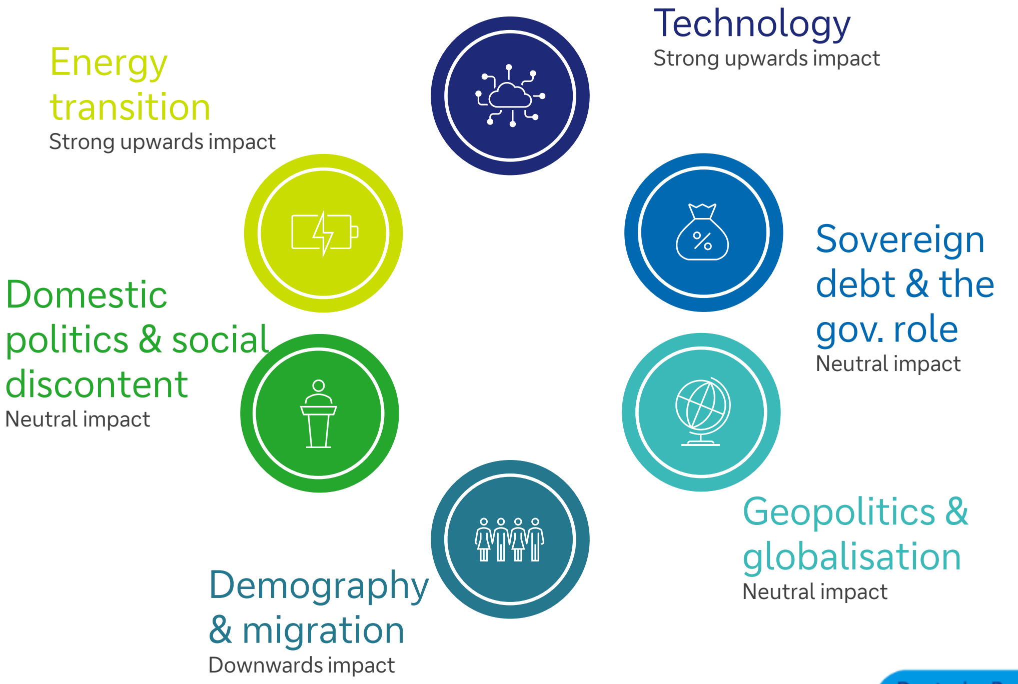
Geopolitics & globalisation
- The deteriorating geopolitical backdrop is balanced somewhat by robust global activity

Energy transition
- Rising prices of key metals

Domestic politics & social discontent
- Affordability and consumer sentiment remain a drag

Demography & migration
- A slow-burn in the working-age population

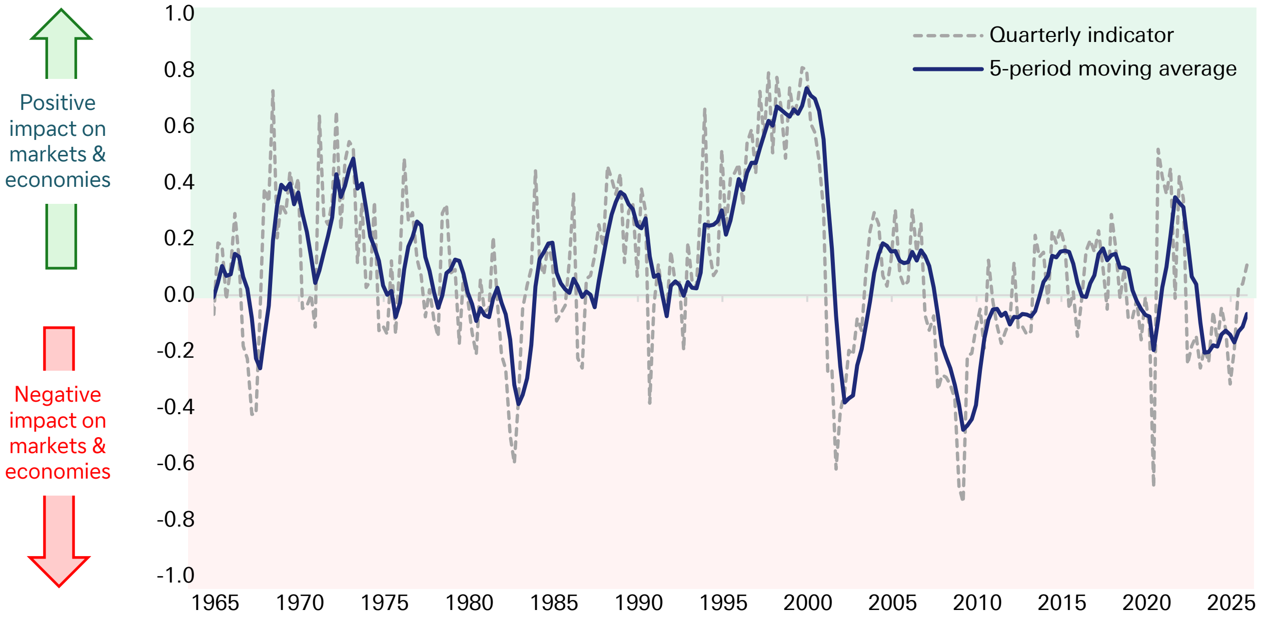
The overall impact of the ~60 signals on our megatrend model is positive for markets and economies. Our indicator has been improving in recent quarters:



Our megatrend tracker incorporates ~60 indicators

This has risen over the last 12 months, indicating that the broader megatrends are improving and so should have a positive impact on markets and economies

An aggregate of 6 megatrends



Source: Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

How the indicator works

We start with the following six megatrends:

- Technology
- Demography and migration
- Sovereign debt and deficits
- Geopolitics and globalisation
- Domestic politics and social discontent
- Energy transition

We then analyse ~60 economic, market, and social indicators that relate to each of the megatrends and analyse their impact on four key variables:

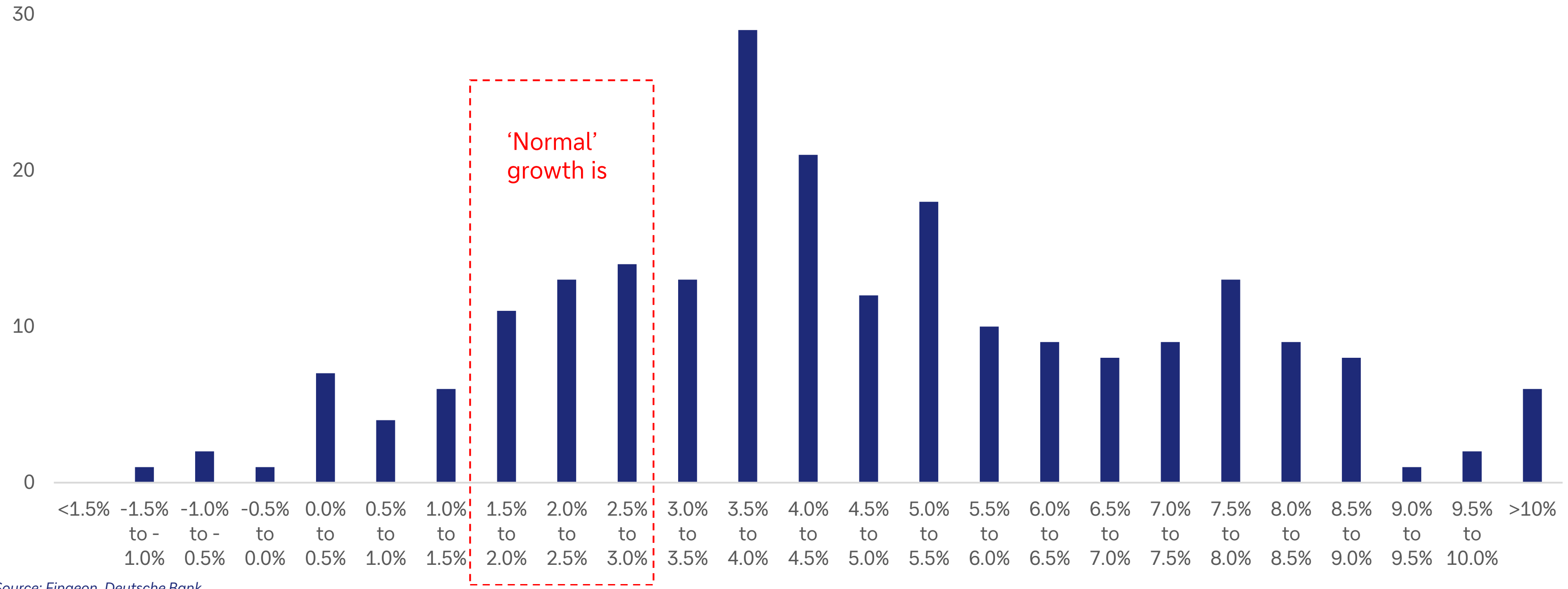
- GDP
- Equity prices
- Treasuries
- Inflation

When we aggregate the results, it gives us an indication of how the six megatrends have affected economies and markets over time.

History shows that assuming 'normal' economic growth of around 2.5% into the medium term is very risky – in fact 74% of years have seen growth over 3% pa (on a rolling 10yr basis).

This is particularly the case when a world-changing technology is involved

Frequency of US GDP growth 1790-present (rolling 10yr, annualised)



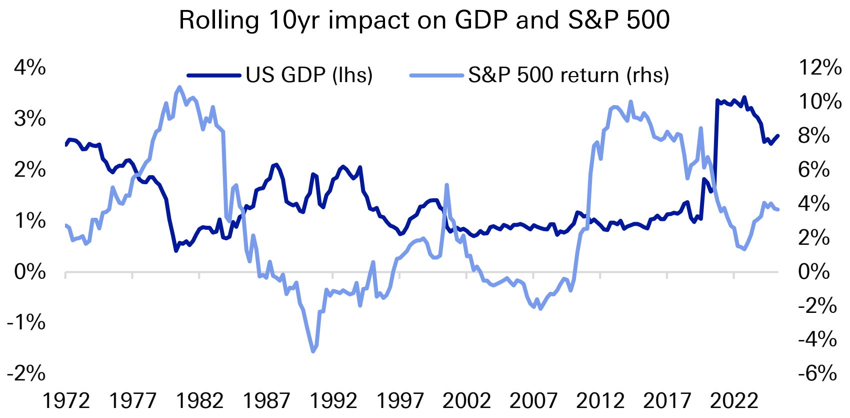
Source: Finaeon, Deutsche Bank.

5. Individual megatrends: The Q1 impact

Technology

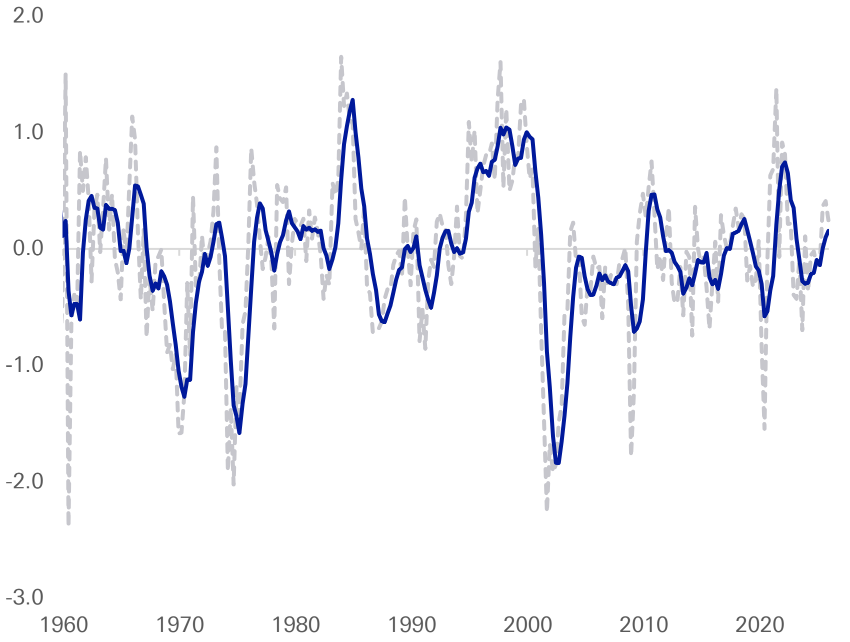
Rising tech spending lacks productivity growth for now

The -13% QoQ fall in tech stocks will be a drag on our indicator in Q1 that has been uplifted by increased tech spending in GDP accounts in recent quarters. Despite the increases in tech investment, business productivity remains somewhat tepid.



Economy-wide technology growth indicator

Higher = more spending, higher productivity

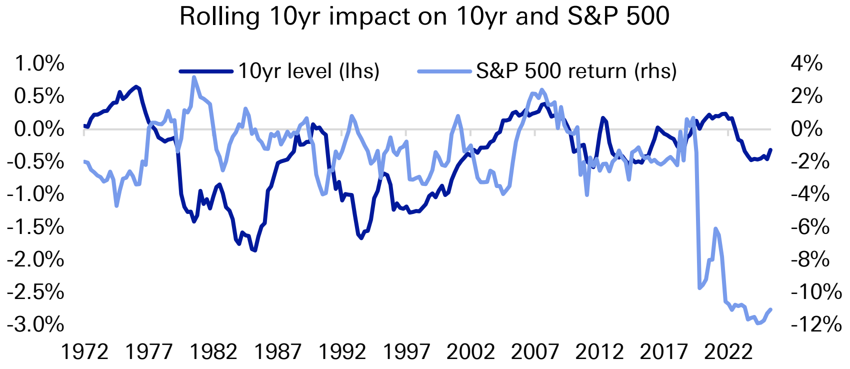


Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Sovereign debt and the government's economic role

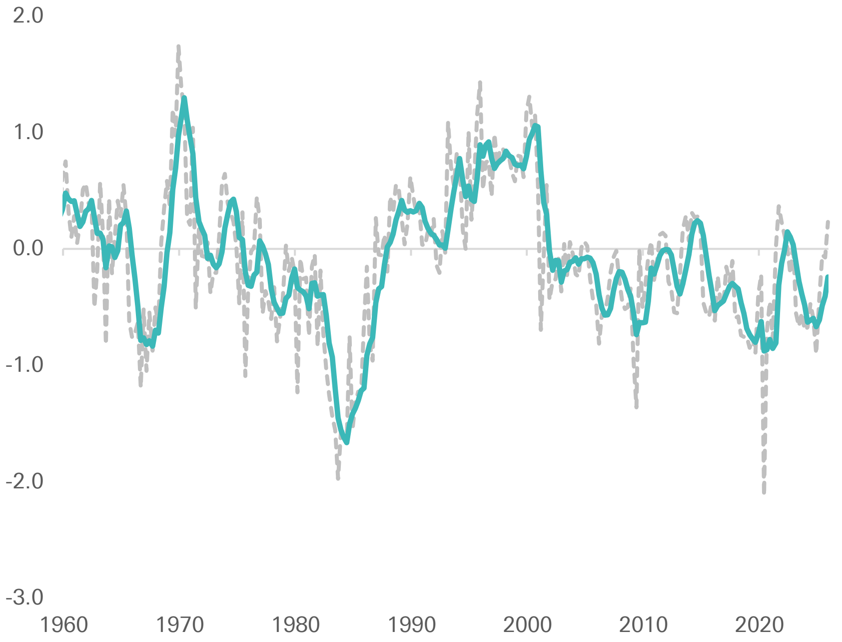
Stock gains have become dependent on fiscal expansion this decade

Our indicator remains below average for the last 10 years, amidst high debt-to-GDP, rising interest costs and elevated fiscal policy uncertainty. But a narrowing deficit, the still robust demand for treasuries and comparatively contained yields have been propping up our composite in recent quarters.



US sovereign debt and the government's role indicator

Higher = improving fiscal position, lower govt spending



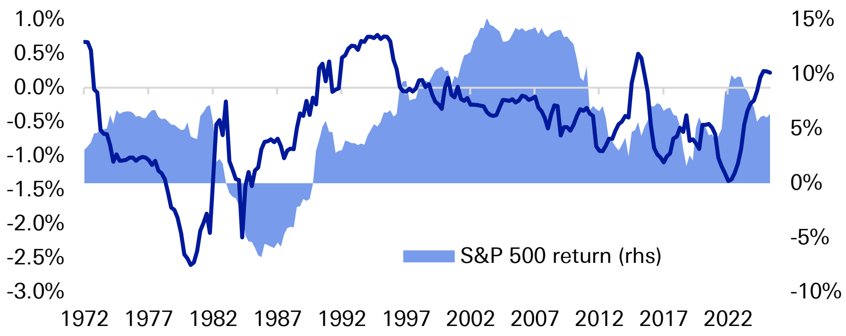
Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Geopolitics & globalisation

Our gauge is at a 10-year neutral level as trade and activity hold up despite deteriorating policy

Deteriorating geopolitical backdrop and policy uncertainty have been outweighed by still robust trade and global activity backdrop, especially since the post-covid disruptions in recent quarters. Overall, however, our index has shown significant deterioration since 2010 and relative to 1990s-2000s prints.

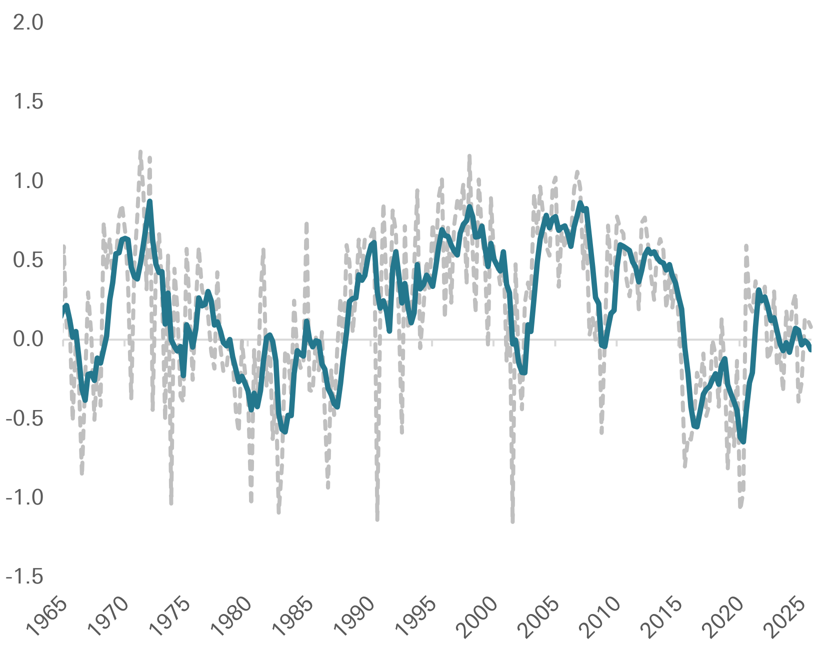
Rolling 10yr impact on CPI and S&P 500



Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Geopolitics & globalisation index

Higher = more stable geopolitical backdrop, more trade



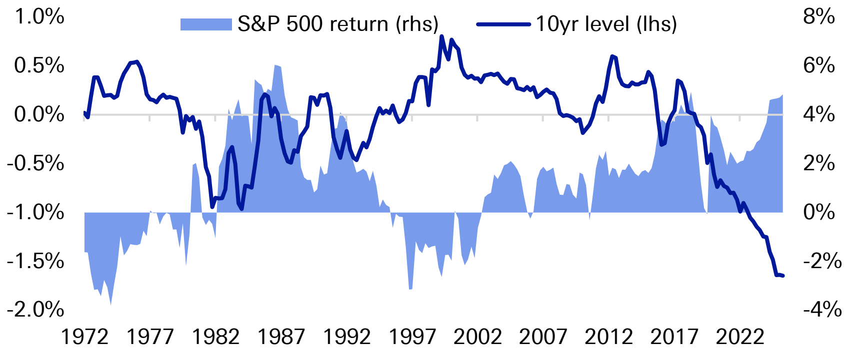
Deutsche Bank Research Institute

The energy transition

A net rise in factors helping the transition has been associated with lower yields and higher stock returns

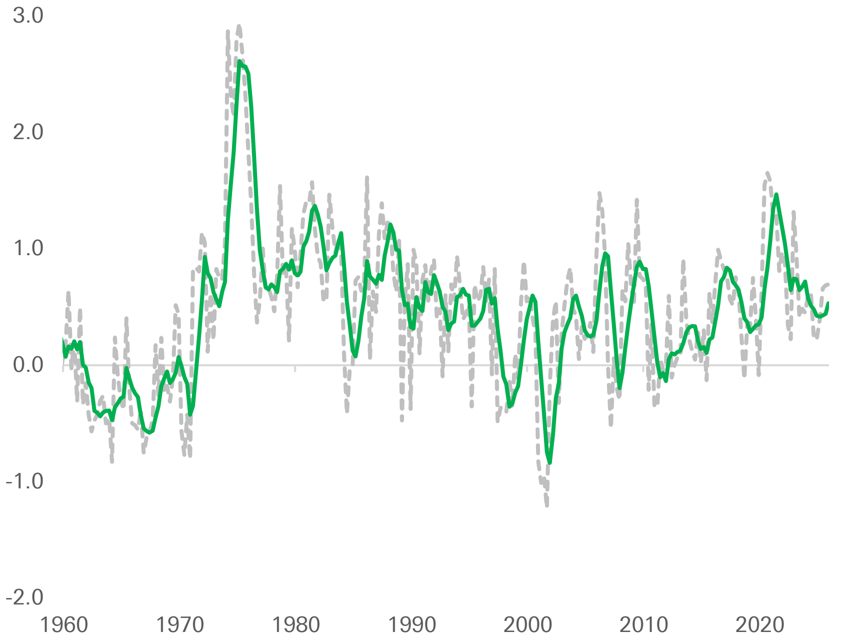
Rising share of renewables and falling CO2 relative to GDP have been the upward drivers of our index, offsetting the impact of higher policy uncertainty and growth in absolute fossil fuels production. Metals key to the transition have also been rallying.

Rolling 10yr impact on 10yr and S&P 500



Energy transition index

Higher = more renewables, greater transition incentives



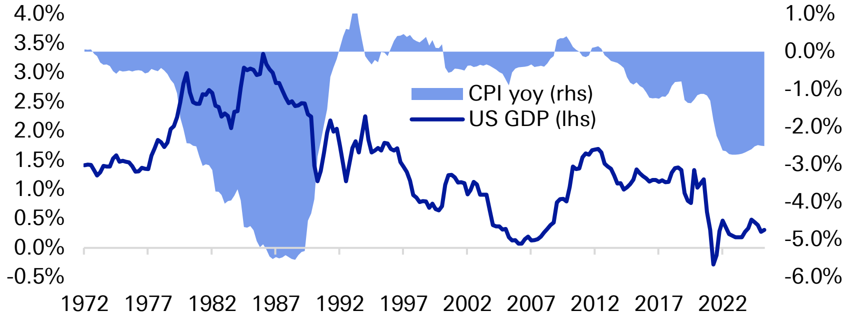
Source: Bloomberg Finance LP, Finaeon, Haver Analytics, EIA, Deutsche Bank.

Domestic politics & social tensions

The US economy has become gradually less exposed to some of the key factors affecting consumer mood

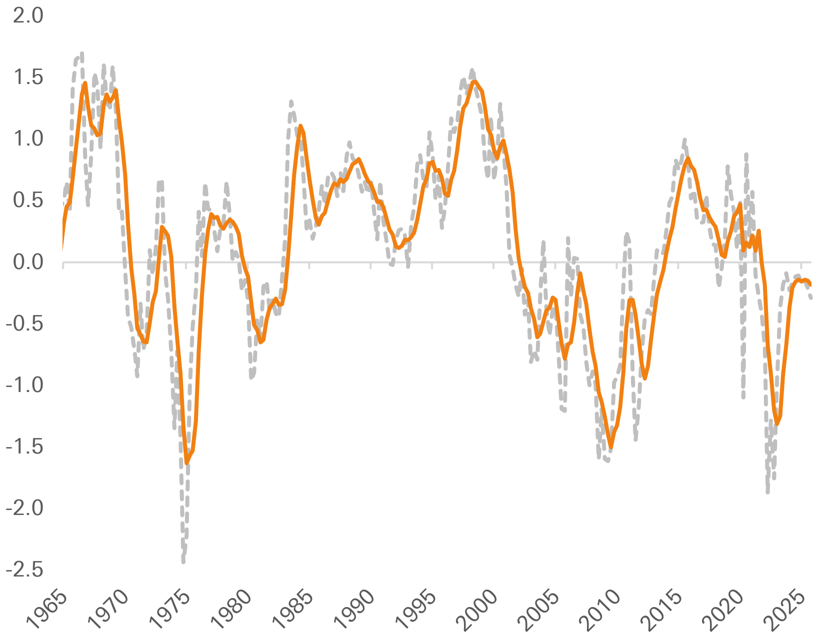
Affordability and consumer sentiment remain a drag on our indicator, with some relative deceleration in labour market indicators also starting to weigh in. While off the lows seen in early 2020s, our gauge is yet to signal material improvement and a few indicators may deteriorate soon, including inflation.

Rolling 10yr impact on GDP and CPI



US domestic politics and social tensions index

Higher = better sentiment, higher affordability, better labour market

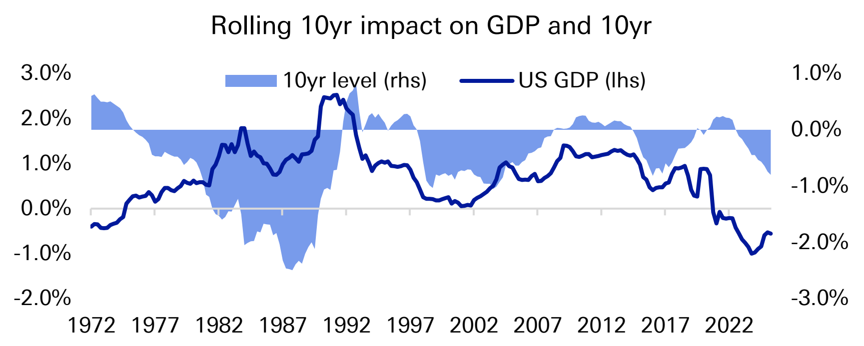


Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Demographic shifts

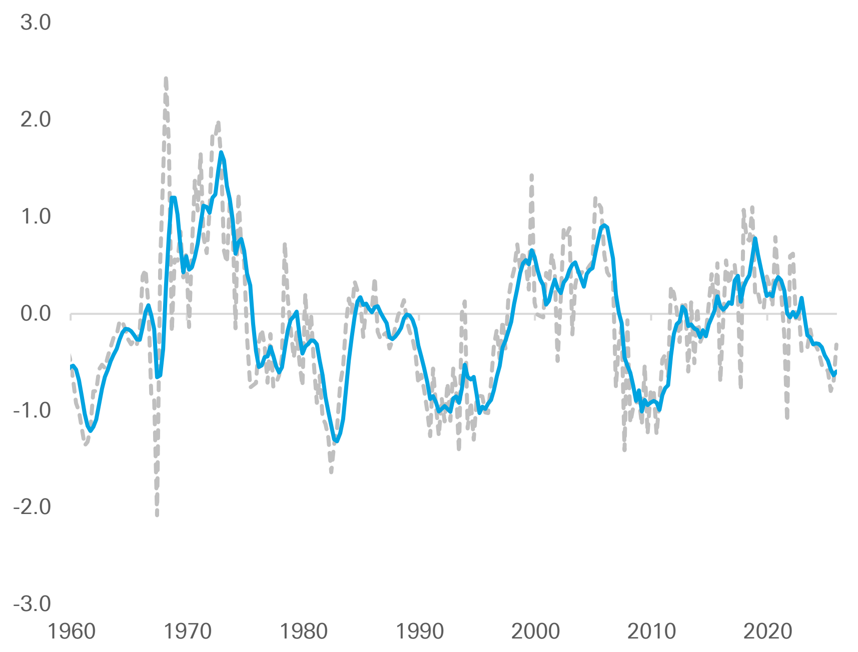
A long-term fiscal headwind, but adaptation in some countries will likely mitigate some of the downside

Deteriorating trends in working-age population ratio, births and immigration have been exacerbated by negative housing market dynamics recently, keeping our index c -0.5 standard deviations below the 10-yr mean. In the 1980s, the relationship between GDP (higher), CPI (lower) and better demographics was especially pronounced. This effect is much weaker now.



US demographic shifts index

Higher = younger working age population, more households and immigration



Source: Bloomberg Finance LP, Finaeon, Haver Analytics, Deutsche Bank.

Appendix 1

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Iran ceasefire: an end in sight?

April 10, 2026

Authors

Helen Belopolsky
Global Head of Geopolitical Research

Context

— On 7 April, the US and Iran agreed to a two-week ceasefire with Iran agreeing to re-open the Strait of Hormuz. The US has halted Trump's threat of an imminent large-scale military action. Both sides have agreed that the ceasefire period will be used for intensive diplomacy aimed at a longer-term solution.

— The situation remains extremely fragile as the two sides postures continue to widely diverge on Iran's nuclear programme, the pace of sanctions relief, potential reparations, Iranian support for its proxies, and long-term control of the Strait of Hormuz. While the ceasefire marks a temporary de-escalation, the crisis remains unresolved, with risk of renewed conflict depending on the outcome of ongoing negotiations.

What happens next?

— A "deal," which has been, and continues to be, our baseline scenario allows all parties to walk away "relative" victors. The US can credibly claim to have significantly degraded Iran's ability to project power beyond its borders, particularly through sustained strikes on missile infrastructure, command networks and elements of its regional proxy architecture. At the same time, Iran can argue that it has demonstrated resilience and deterrent credibility. Israel can point to successfully degrading the missile and military infrastructure of one of its greatest existential threats.

— Despite the precarious nature of the ceasefire, we assess that the probability of a longer-term solution is strong given the longer this crisis lasts, the greater the incremental gains for the US will encounter growing diminishing returns. We have already witnessed increasing domestic and international pressure on Trump, which we expect will only increase as the economic fallout of crisis persists and expands.

Victory Without Peace?

— However this ends, it will be a **gamechanger for the Middle East**. The long-standing Middle East geopolitical equilibrium has been shattered. A new, highly unpredictable reality will emerge, leaving a multitude of critical questions about the shape of the post-conflict landscape.

— We judge that **Iran's calculus** that it could endure more pain than its adversaries' political systems could tolerate have played out in its favour despite the significant costs. Iran will likely walk away from this conflict with enhanced leverage over the Strait of Hormuz, potential sanctions relief, and with greater political leverage having withstood the US and Israeli attacks. It will, however, need to cautiously rebuild relations with the Gulf.

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- **The Gulf has been paying the price** of the US/Israeli action having faced hundreds of missiles and drones over the past month. The Gulf is no longer buffered from conflict. Consequently, its relationship with the US will inevitably be impacted. We expect to see a diversification in Gulf relations in the aftermath of the acute crisis.
- **Israel and Lebanon :** Israel's military operations against Hezbollah could continue unabated despite the ceasefire with Iran. There appear to be differentiated opinions on whether Lebanon was meant to be included in the ceasefire. Nonetheless, we expect Israel to continue to make inroads in Lebanon and beyond to seize the moment to expand operations to target adversaries.
- From a geostrategic perspective, both **China and Russia are benefiting** in positioning themselves as defenders of international law and the post-WWII order. They are also likely benefiting from President Trump's hawkish foreign policies in recent weeks, which is creating significant volatility for allies and adversaries alike. They may be able to make inroads with regional allies in the aftermath of the crisis.

Appendix 1

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Q1 Tech Performance Review

April 1, 2026

Marion Laboure | Camilla Siazon

Deutsche Bank Research Institute

IMPORTANT RESEARCH DISCLOSURES LOCATED IN APPENDIX 1.

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Oracle Plans Thousands of Job Cuts in Face of AI Cash Crunch

Nvidia's Jensen Huang predicts $1tn in AI chip revenue over 2 years

The $1.6 Trillion Meltdown That Swept Through Software Stocks

China Is Embracing OpenClaw, a New A.I. Agent, and the Government Is Wary

Source: Deutsche Bank Research, Bloomberg Finance LP, Reuters, Yahoo Finance, White House.

SUMMARY: Will the AI disruption narrative resume in Q2?

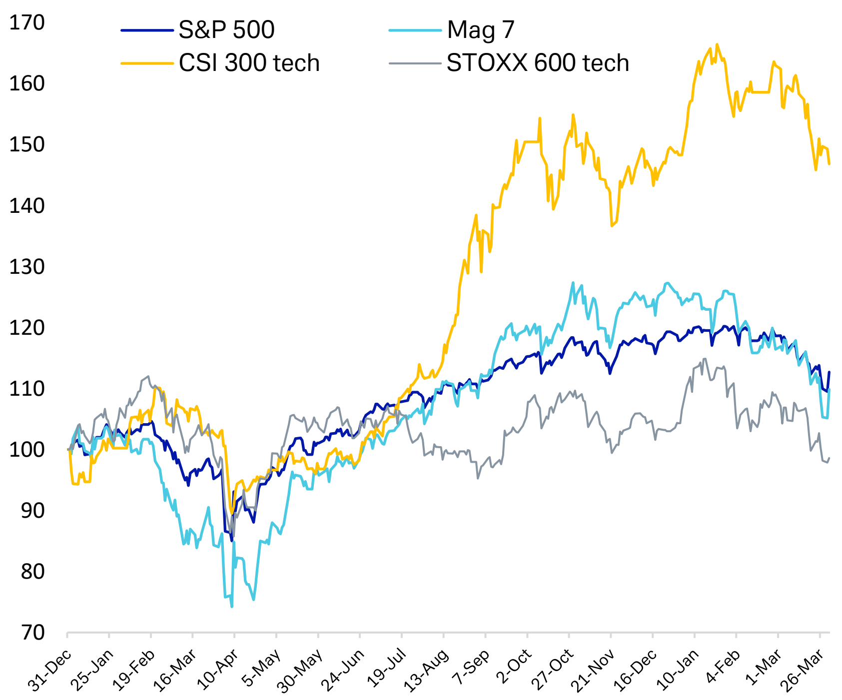
Tech equities declined in Q1, initially driven by concerns around AI disruption, before a broader macro sell-off—triggered by escalating tensions in the Middle East—accelerated the downturn. The bulk of the decline occurred in Feb, when new AI tool releases and a widely circulated research note on potential US employment risks intensified fears and prompted a rotation away from tech. As a result, the Mag 7 (-12%) and the tech-heavy Nasdaq Composite (-7%) both ended the quarter lower.

The sector faced two key challenges: first, rising doubts about the sustainability of AI-driven demand; and second, concerns that AI disruption could weigh on software revenues and the US labour market. Software stocks were particularly affected, with the S&P 500 software segment declining -24%—its worst performance since the peak of the global financial crisis. More broadly, several of the Magnificent 7 companies ended the quarter in bear market territory relative to their October 2025 peaks.

The sell-off was also global. The STOXX 600 Technology index fell -5.5%, while the Hang Seng Tech (-16%) and CSI 300 Technology (-9%) saw sharper declines. Our European analysts believe peak AI disruption concerns may be behind us, whereas their US colleagues' analysis of positioning in the sector indicates it is still short.

Looking ahead to Q2, much will depend on developments in the Middle East. If tensions ease, the key question is whether the AI disruption narrative re-emerges after its February peak—and whether it continues to weigh not only on likely losers, such as software and IT services, but also on beneficiaries of AI spending, including chipmakers and hyperscalers.

Global tech equities have sold off along with broader macro

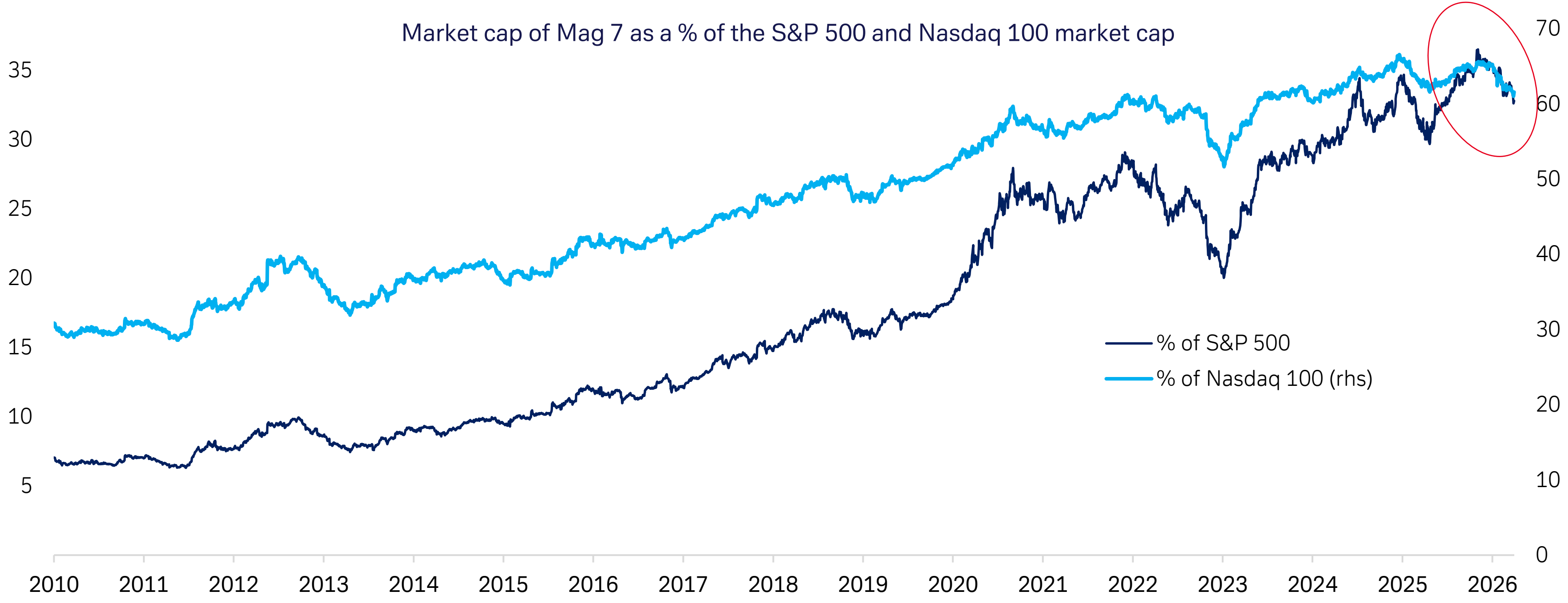


Sources: Deutsche Bank Research, Bloomberg Finance LP. Updated April 1 2026.

1. MAGNIFICENT 7

Even before the escalation in the Middle East, a rotation away from tech and large caps was underway, as doubts over earnings sustainability began to build. This shift accelerated in February, when fears that AI could drive higher US unemployment—alongside the release of new AI tools—intensified the sell-off.

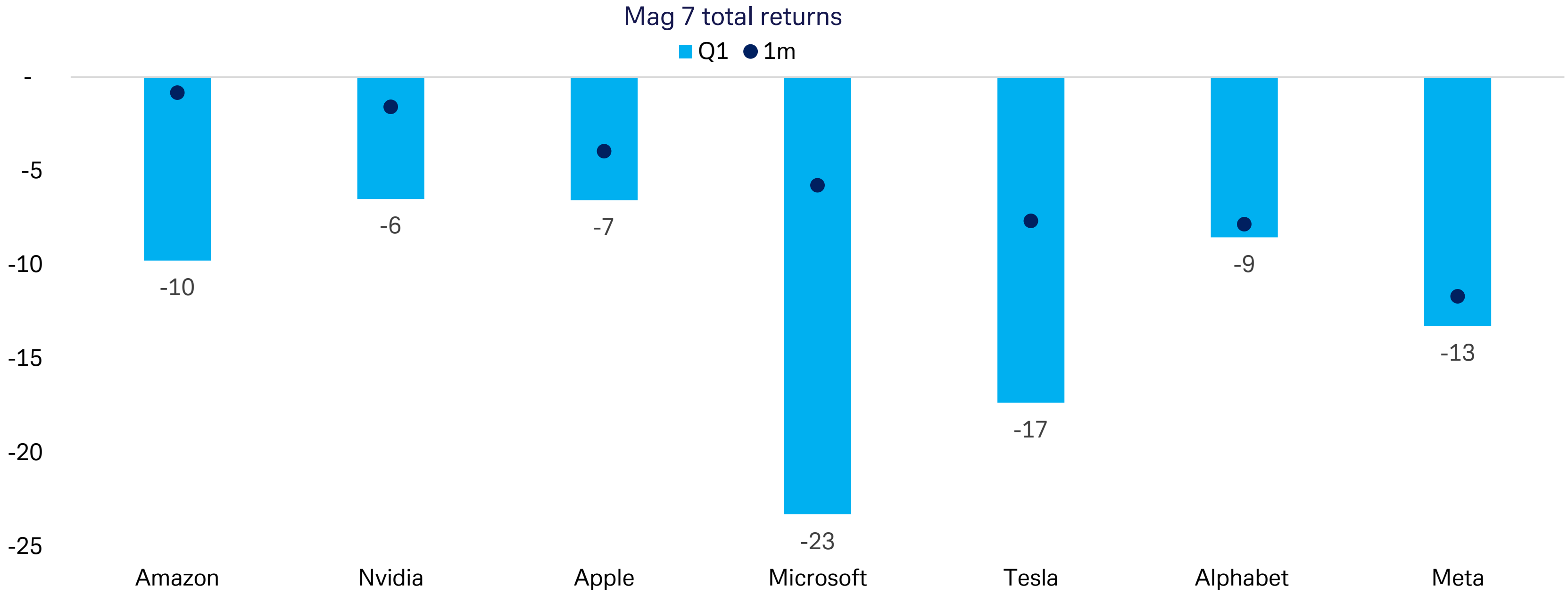
Market cap of Mag 7 as a % of the S&P 500 and Nasdaq 100 market cap



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated March 31 2026.

1. MAGNIFICENT 7

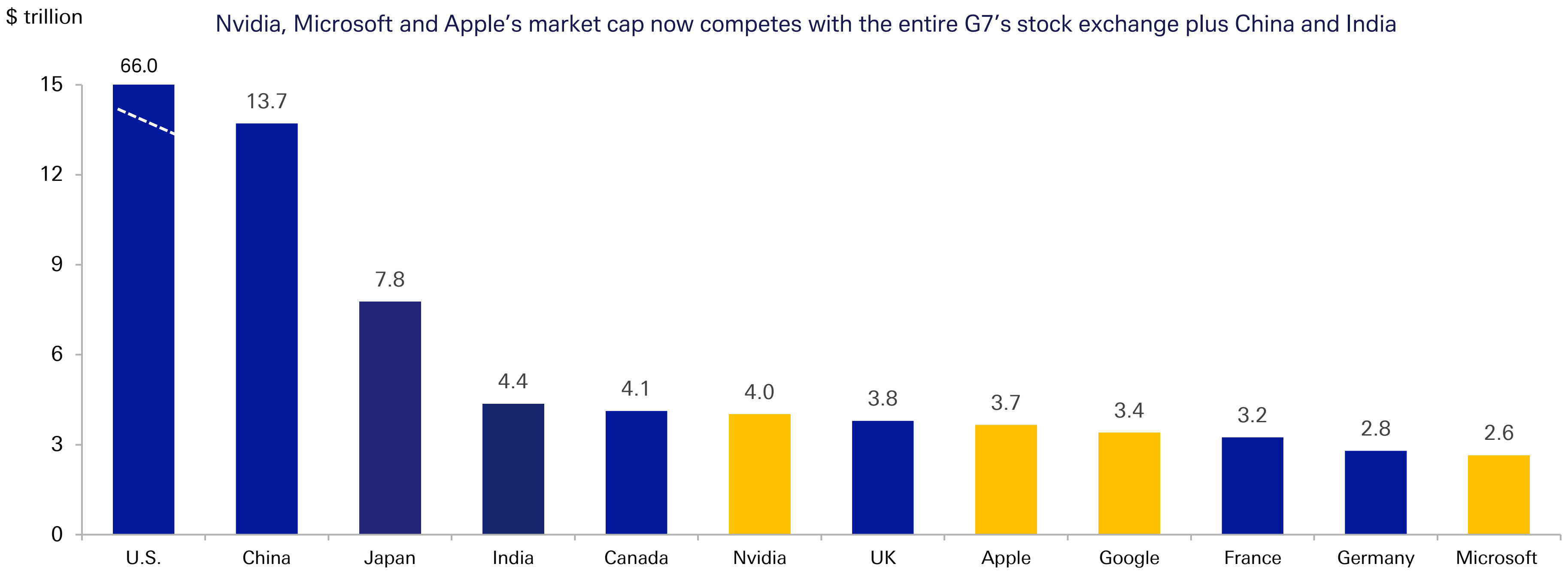
All Magnificent 7 companies underperformed the S&P 500 (-4.35%) this quarter. Among the laggards, Nvidia (-6.5% in Q1) recorded its worst quarterly performance since January 2025—when DeepSeek launched—as underwhelming earnings reinforced growing scepticism around the AI trade. Microsoft (-23%) also posted its worst quarterly performance since 2008.



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated April 1 2026.

1.1 MAGNIFICENT 7

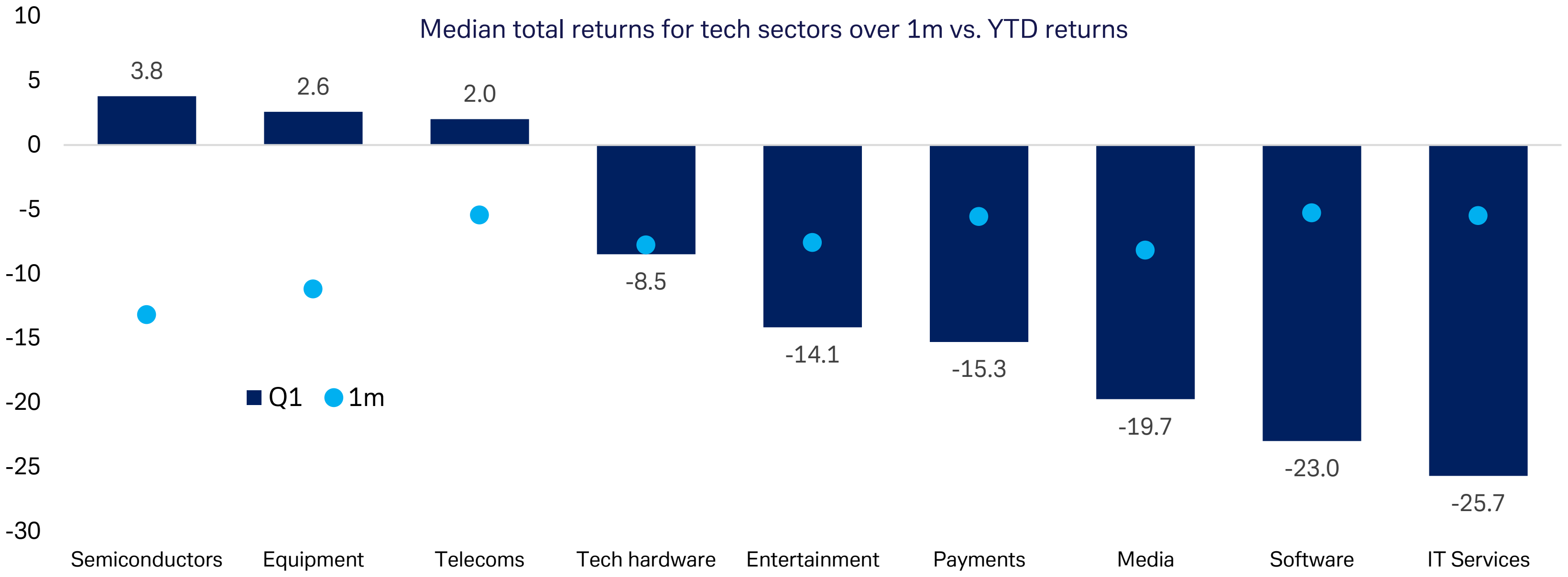
Despite the ongoing sell-off, Big Tech continues to compete with economies for influence. Nvidia continues to maintain a market cap of over $4trn, while in January, Google also crossed the $4trn threshold for a few days after it announced a multi-year deal to power the AI technology for Apple's iPhones.



Source: Deutsche Bank Research, Bloomberg Finance LP. Market cap in US$ of G7 plus China and India's entire listed equity market on Bloomberg. Updated March 31 2026

2. GLOBAL INDUSTRY BREAKDOWN

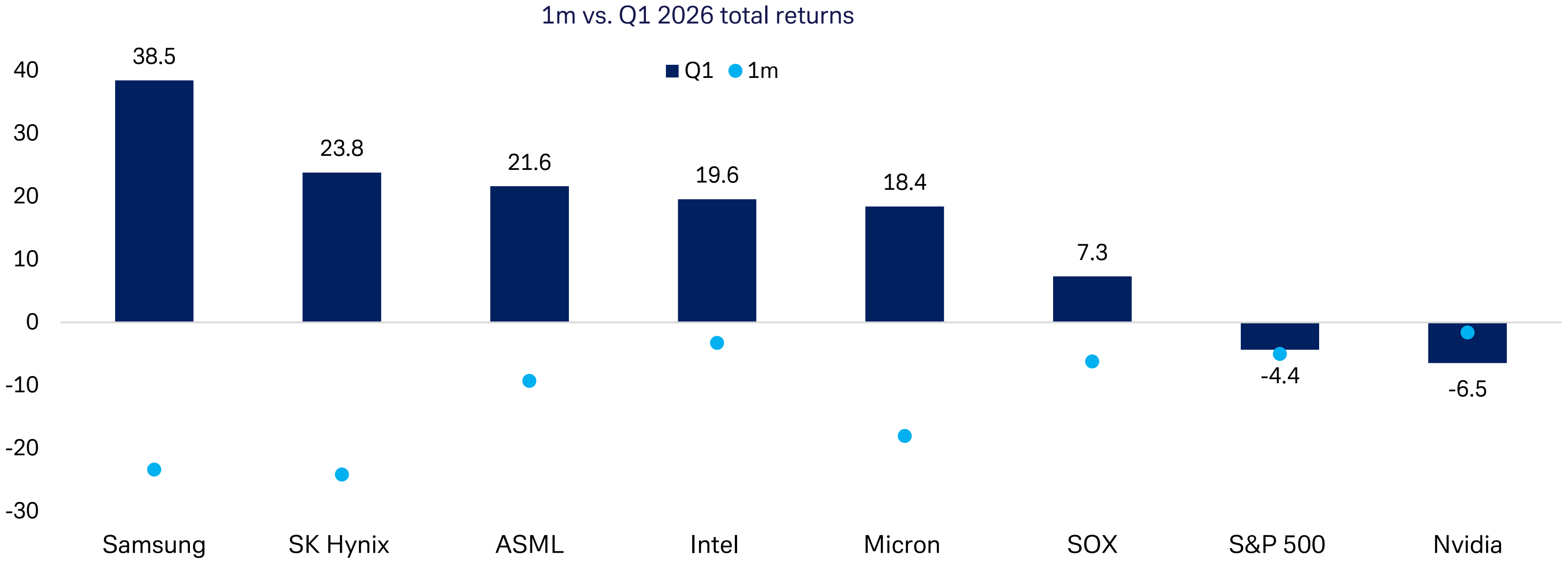
Software and IT services categories suffered the largest losses in Q1, although they recovered slightly in March. Out of those companies in Q1, Oracle US (-24%) and Oracle Japan (-36%), EPAM Systems Inc (-34%) and Tata Consultancy (-29%) were among the worst performing, as fears around AI disruption to the global software business model dominated the narrative. Until March, global semiconductor companies fared relatively well with Asian companies like Hanmi Semiconductor (+88%) and eMemory Technology Inc (+49%) leading.



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated March 2026.

2.2 SEMICONDUCTORS

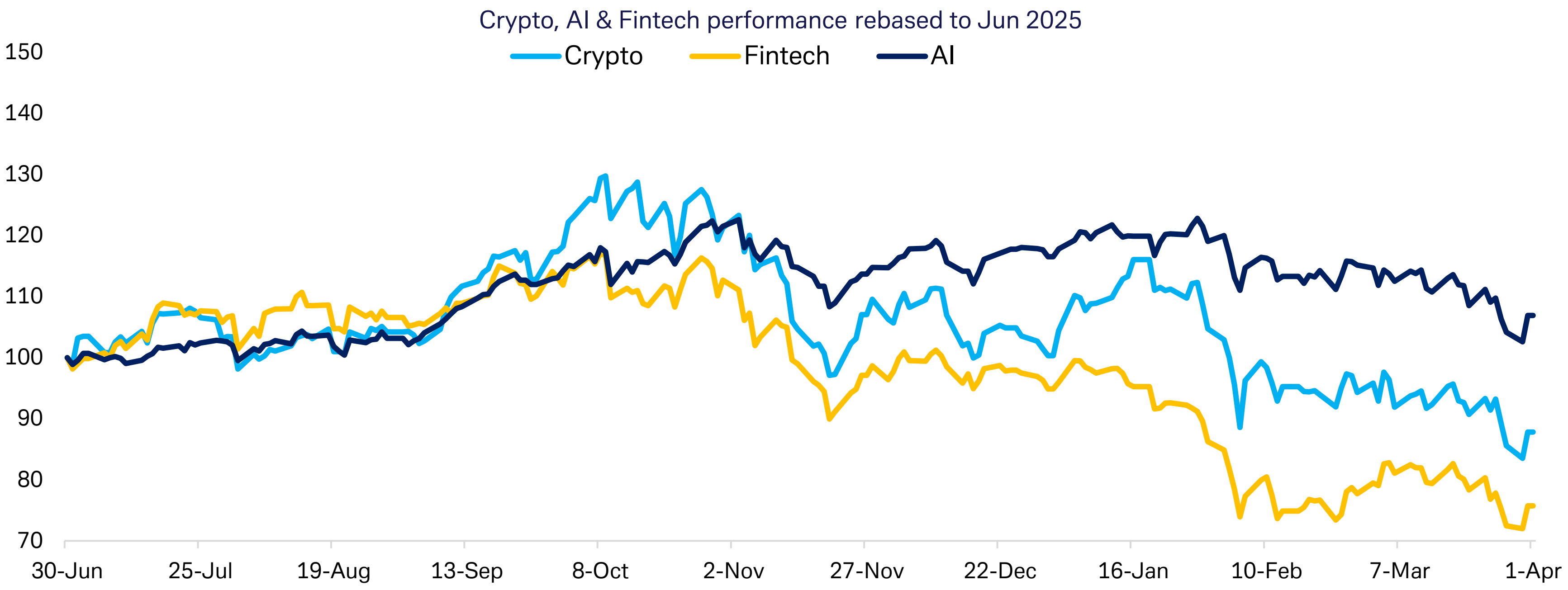
In March, chipmakers performances were hit by Google's TurboQuant launch—an algorithm suggesting AI data centres may require less memory for compute. This triggered a sharp sell-off in memory manufacturers such as Micron (-18% in March) and Samsung (-23% in March), although both remain up year-to-date, supported by the global memory chip shortage. More broadly, uncertainty around US export controls, growing scepticism toward the AI trade, and geopolitical risks also weighed on the sector.



Source: Deutsche Bank Research, Bloomberg Finance LP. Updated March 2026.

2.3 CRYPTO, AI & FINTECH

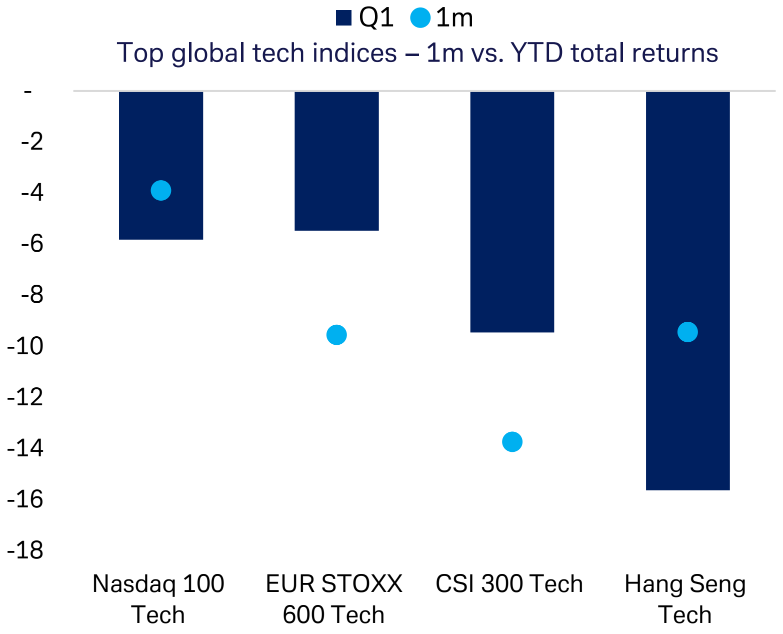
Bitcoin has fallen for five consecutive months until March, making it one of the clearest laggards of early 2026 and weighing on crypto-ETFs. The crypto fell -22% to $68,194 in Q1. However, as headlines have shifted towards the Middle East conflict, crypto, fintech and AI ETFs have managed to recover some of their declines.



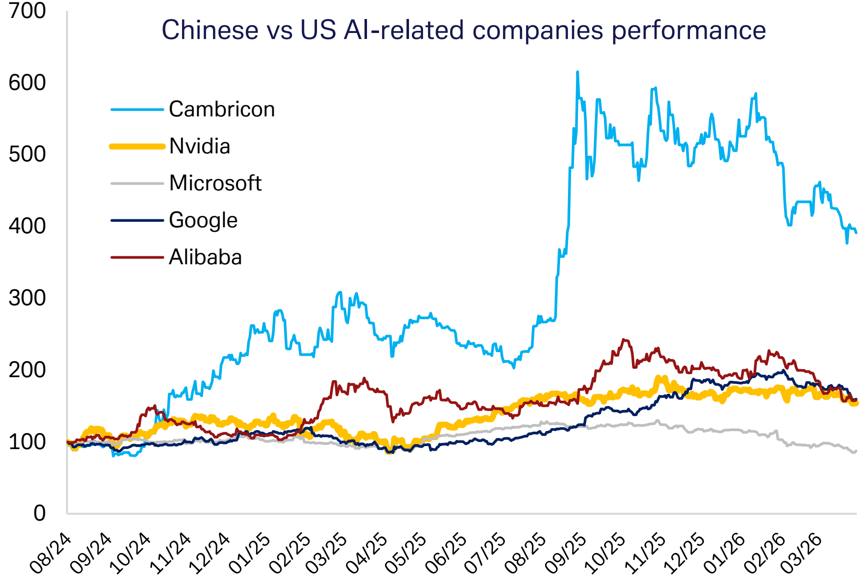
Source: Deutsche Bank Research, Bloomberg Finance LP. Data updated March. The following are based on AIQ ETF, BLOK US Equity ETF, and ARKF US Equity ETF.

3. REGIONAL BREAKDOWN

Globally, tech equities have suffered under the broader macro dip. Notably, our European equity analysts have upgraded tech from underweight to neutral and turned overweight software within tech, believing that the AI disruption worries have peaked (see here). In the US, investor positioning towards tech is still short.



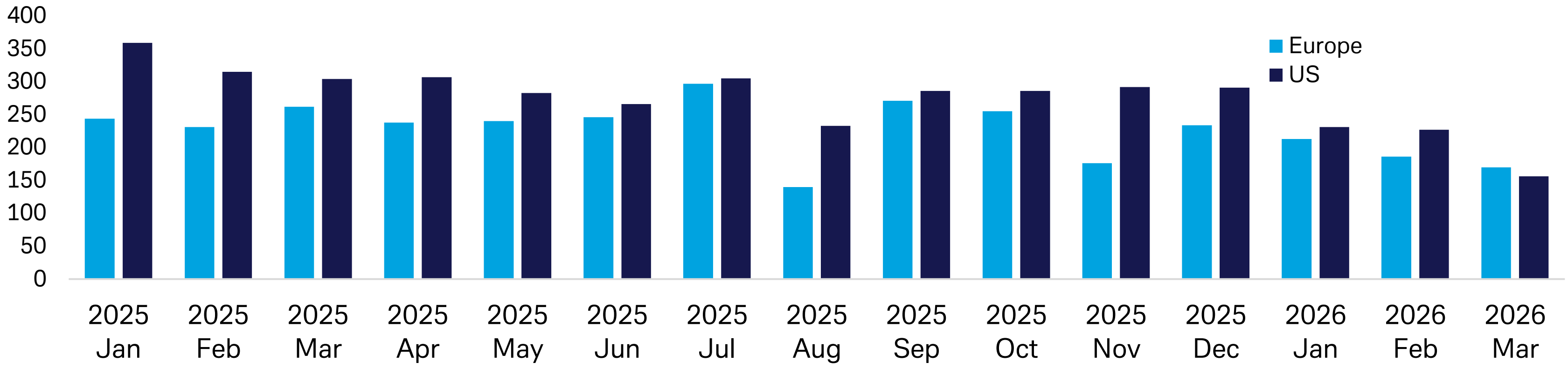
Source: Deutsche Bank Research, Bloomberg Finance LP. Updated March 31 2026.



Source: Deutsche Bank Research, Bloomberg Finance LP. Indexed to 100. Updated March 31 2026.

4. M&A

US and Europe tech M&A, no. of deals announced per month



Source: Deutsche Bank Research, SDC data up to March 31. Values include Completed, Ontended, Pending Regulatory, Partially Completed, Pending, Unconditional

AUTHORS

Marion Laboure is a Senior Economist at Deutsche Bank in London and a Lecturer at Harvard University.

She brings extensive experience across the private sector, public policy, and monetary policy, having worked at the European Commission, the IMF, the Luxembourg Central Bank, and Barclays.

She has received multiple distinction, including first prize from the American Society of Actuaries, Revue Banque as a rising star in finance, selection as one of the 45 standout women in fintech, and being named as a "cryptocurrency mastermind" by Business Insider.

She is the author of Democratising Finance, published by Harvard University Press.

Camilla Siazon is an Associate on Deutsche Bank's Macro and Thematic team in London. She covers digital assets, geopolitics, and global macroeconomic trends.

She previously worked at geopolitical advisory firms Eurasia Group and Greenmantle, where she focused on political and economic risk analysis.

She holds a dual MA/MSc in International History from Columbia University and the London School of Economics, as well as a BA in History from Barnard College, Columbia University.

Appendix 1

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Deutsche Bank
Research Institute

AI is not convinced AI is a major disinflationary force

April 9, 2026

Authors

Matthew Luzzetti, Ph.D.
Chief US Economist

Brett Ryan
Senior US Economist

Justin Weidner
Economist

Amy Yang
Economist

Introduction

In a recent note, we discussed how AI assesses the potential negative effects of AI on the unemployment rate (see "Will AI drive unemployment higher this year? AI thinks not"). While several AI tools agree that the impact is likely to be limited over the next year -- all three tools we referenced assign a 5% probability AI will lift the unemployment rate by 0.5pp or more over the next 12 months -- there was disagreement further out. The three tools assigned a 10-30% probability of more substantial disruption at the 5-year horizon.

In this note we take the same approach to assess the impact of AI on inflation. In contrast to the unemployment rate, where there appears to be considerable uncertainty about the size of the impact, our sense from client conversations is that there is a strong consensus that AI represents a substantial disinflationary force that will also exert meaningful downward pressure on interest rates over the coming years.

Does AI agree with this consensus? Surprisingly not.

Why AI is expected to be a significant disinflationary force

The case for AI to exert a meaningful drag on inflation is reasonably straightforward. AI is expected to substitute cheap capital / technology for more-expensive labor, driving input costs lower. By leading to sustainably higher productivity growth, it will push unit labor costs substantially lower, weighing on price pressures. To the extent that AI raises the unemployment rate, it will also reduce inflation through increasing labor-market slack, reducing wage growth, and weakening aggregate demand. Finally, AI represents an important equalizing force for entrepreneurs, ushering in countless small businesses with lower barriers to entry that will compete and drive margins lower. This is our understanding of the consensus view on why inflation and rates will be sustainably lower due to AI.

But there are countervailing forces to this narrative. In the near term at least, AI has triggered a substantial positive aggregate demand shock, as investment has surged in anything AI-related, including software, chips, computers and peripheral equipment, and data centers. These investments are competing with alternative demands for resources. In addition, AI is likely to represent an ongoing source of demand that could create bottlenecks and price pressures in key points of the supply chain -- the spillover to higher electricity prices is most obvious. Finally, while the initial impact of AI-driven higher unemployment would be weaker demand and softer inflation pressures, we expect policy levers would be triggered to ultimately limit the rise in unemployment and put a floor under aggregate demand. Fiscal policy is best placed to serve this redistribution role -- indeed stronger productivity implies faster GDP growth which must also imply stronger aggregate income growth, making redistribution possible -- but monetary policy could also be part of the initial response.

So, while we find the disinflationary consensus to be compelling, there are forces that could at least limit the magnitude of these forces. What does AI say?

What dbLumina says

We begin by asking DB's proprietary AI tool, dbLumina, how it assesses the probabilities around inflation effects from AI over two different horizons -- the next 12 months and five years ahead. We use the following prompt:

I'm interested in studying the different probabilities of outcomes for AI's impact on US inflation. I have in mind: the probabilities it raises inflation by 0.1pp or more, has minimal impact on inflation, slightly reduces inflation and meaningfully reduces inflation. Please consider two different horizons: 1 year and 5 years. Define slightly reduces as -0.1pp to -0.4pp and meaningfully reduces as -0.5pp and more at both horizons. Please summarize the results in a table. Provide the reasoning for your findings as well as citations to academic papers in the table.

Over the next year, dbLumina sees risks as tilted slightly towards higher inflation from AI. There are roughly equal probabilities assigned to the outcomes where AI has a minimal impact on inflation (35%) and lifts inflation by any amount (40%). In contrast, AI lowering inflation over the next year has a total probability of only 25% (5% assigned to AI lowering inflation by 0.5pp or more over the next 12 months and 20% probability of a drag of -0.1pp to -0.4pp). dbLumina's rationale is outlined within the table in the final column. As we noted in the prior section, dbLumina cites the positive impact on aggregate demand through AI-related investment as well as a potential boost to energy prices as rationalizing risks of higher inflation.

Looking further out, the risks are better balanced. Five years ahead, dbLumina now sees a roughly 50% probability that AI lowers inflation by any amount; however, the overall distribution is roughly flat. While the modal outcome is a slight reduction in inflation (-0.1 to -0.4pp), dbLumina assesses it as more likely that AI lifts inflation by any amount over the five-year horizon (25%) than meaningfully reduces inflation over that period (20%). That skew does not fit the consensus view on AI's disinflationary impact.

Figure 1: Probability of AI's net impact on US inflation

Scenario (Net Impact on Inflation)	Quantitative Definition (Both Horizons)	Probability (Next 12 Months)	Probability (Next 5 Years)	Reasoning and Citations
1. Raises Inflation (Demand-side effects, investment boom, or bottlenecks dominate)	+0.1 pp or more	**40%**	**25%**	The initial phase of AI adoption is likely to be characterized by strong investment and demand-side pressures, potentially leading to a modest increase in inflation. The International Monetary Fund (IMF) also projects a near-term rise in inflation due to higher investment and aggregate demand . Over a longer horizon, the demand-pull effects of sustained AI-driven growth could still lead to higher inflation, as suggested by some long-term scenarios in a BIS working paper where demand effects eventually dominate.
2. Minimal Impact (Supply and demand effects largely offset; slow adoption)	-0.09 pp to +0.09 pp	**35%**	**25%**	The impact of AI on inflation in the short term could be balanced, with productivity gains partially offsetting increased demand. The precise timing and magnitude of these opposing forces are uncertain, leading to a significant probability of a minimal net effect . The long-term impact of AI on inflation remains highly uncertain, with competing forces of productivity gains and increased demand potentially neutralizing each other.
3. Slightly Reduces Inflation (Supply-side efficiency gains, cost reductions, or weak demand)	-0.1 pp to -0.4 pp	**20%**	**30%**	Early productivity gains from AI could begin to put downward pressure on prices, but this effect is expected to be limited in the first year as adoption is still in its early stages. As AI becomes more widely adopted, its productivity-enhancing and cost-reducing effects are expected to become more significant, leading to a moderate reduction in inflation.
4. Meaningfully Reduces Inflation (Widespread cost-cutting, extreme productivity boom, or demand shortfall)	-0.5 pp or less	**5%**	**20%**	A significant reduction in inflation within a year is considered unlikely as the widespread adoption and deep integration of AI required to achieve such an outcome takes time.

Source: Deutsche Bank Research

This result is somewhat surprising, as it goes against the strong consensus that AI is likely to represent a transformative disinflationary force. Given the tension, it is important to investigate whether any difference in assessments is unique to dbLumina or is represented across a broader selection of AI tools.

What do other AIs say?

To test the robustness of our results in the previous section, we provide the same prompt to ChatGPT 5.2 and Claude Opus 4.6. Though there are some differences in probabilities, the main point of tension found in dbLumina's analysis — that the risks to inflation are likely to be more balanced than commonly assumed — is also present across other AI tools.

For example, the alternative AI tools view the most likely outcome as minimal impact over the next year (50-60% probability across AI tools). All tools see a meaningful reduction in inflation at the one-year horizon as a tail risk (5%) and see increases in inflation as more likely than a meaningful reduction. dbLumina is the most skewed at 40% vs. 5%, but ChatGPT (20% vs. 5%) and Claude (25% vs. 5%) agree with the direction of the skew. Across tools, the role of the positive aggregate demand shock from AI-related investment features consistently, as does the potential boost to electricity prices.

Figure 2: Summary of AI views on the inflation effects of AI

1-year horizon	Increases Inflation (+0.1 pp or more)	Minimal impact (-0.09 pp to +0.09 pp)	Slightly reduces (-0.1 pp to -0.4 pp)	Meaningfully reduces (-0.5 pp or less)
dbLumina	40%	35%	20%	5%
ChatGPT 5.4	20%	60%	15%	5%
Claude Opus 4.6	25%	50%	20%	5%
Average	**28%**	**48%**	**18%**	**5%**

5-year horizon	Increases Inflation (+0.1 pp or more)	Minimal impact (-0.09 pp to +0.09 pp)	Slightly reduces (-0.1 pp to -0.4 pp)	Meaningfully reduces (-0.5 pp or less)
dbLumina	25%	25%	30%	20%
ChatGPT 5.4	15%	25%	40%	20%
Claude Opus 4.6	20%	25%	35%	20%
Average	**20%**	**25%**	**35%**	**20%**

Source: ChatGPT 5.2, Claude Opus 4.6, Deutsche Bank Research

There is general agreement on the likely impact of AI on inflation over the longer (5 year) timeframe. ChatGPT and Claude on balance see somewhat greater probabilities of lower inflation outcomes. Claude sees slightly reducing inflation as the most likely outcome (35%), which along with meaningfully reduces (20%) suggests a 35pp skew towards lower versus higher inflation. ChatGPT shows an even more notable skew towards lower inflation, with 60% applied to this outcome (40% for slight decline and 20% for meaningful decline) versus 15% for higher inflation scenarios. Across the tools, AI sees 50-60% probability of lower inflation over the 5-year horizon versus 15-25% probability it will lift inflation over that horizon. That said, across all the tools a positive impact on inflation is seen as roughly as likely as a meaningful reduction in inflation. Indeed, the average probability of each outcome is 20% across the tools.

We provide detailed reasoning in an appendix.

Conclusion

AI does not necessarily agree with the strong consensus that AI is likely to exert a meaningful drag on inflation pressures over the coming years. How should we interpret this finding?

One option is to discount the result. Surely AI must be disinflationary, the consensus believes, so AI is simply missing something. While perhaps appealing, it strikes us that this approach is fundamentally flawed — if AI is categorically wrong about this question, perhaps we should rethink our assessment of how transformative it is likely to be for complex knowledge work like forecasting, at least in its current form.

A second (and opposing) possibility is that the consensus is somewhat off base in assuming AI will be an unequivocal disinflationary force. Perhaps it is not accounting for the inevitable countervailing forces that will keep inflation supported against the coming drag. Factors supporting demand, at least in the near term, are notable.

A middle ground is that AI is taking a sensible approach by assigning relatively flat probabilities across outcomes in a highly uncertain environment with longer time horizons. Having been trained on a corpus of text from economists, AI is simply acting as the proverbially two-handed economist, hedging its views against an unknowable backdrop.

As our colleague Jim Reid noted in his recent piece (see "Innovation, jobs and inflation: Lessons from 250 years of disruption"), the long history of inflation alongside transformative technologies suggests that some combination of the latter two explanations may be appropriate. History has shown that the force of the policy response will be at least as critical as the technology in determining the broader impact on inflation.

At this point, AI appears to largely agree.

Appendix

ChatGPT's Reasoning:

1. *Why the 1-year horizon is centered on "minimal impact"*

Over a 1-year window, AI is unlikely to dominate US inflation dynamics. Inflation at that horizon is usually driven by monetary policy, wage growth, housing services, commodity prices, supply chains, and fiscal conditions. Even if AI adoption is rapid in some occupations, aggregate price indexes tend to move slowly because:

- General-purpose technologies often have delayed macro effects due to complementary investments and reorganization.
- Early adoption is concentrated in specific tasks and firms rather than economy-wide.
- Measured productivity gains often lag technical capability.
- Consumer price pass-through from lower business costs is partial and uneven.

This logic supports assigning the highest short-run probability to outcome B.

2. *Why 5 years shifts toward disinflation*

Over 5 years, it becomes more plausible that AI lowers inflation modestly. The main channels are:

- **Productivity:** AI can reduce labor input per unit of output in coding, customer support, back-office work, design, and analytics.
- **Logistics and allocation:** Better forecasting and optimization can lower inventories, waste, and delays.
- **Labor market effects:** If AI slows wage growth in exposed tasks or increases effective labor supply, unit labor costs may ease.

- **Competition:** Lower entry barriers and faster imitation may reduce pricing power in some sectors.

These channels support assigning the greatest 5-year mass to outcome C, with a meaningful tail probability on D.

3. *Why a strongly inflationary AI scenario remains possible*

AI could still raise inflation if demand effects exceed supply effects. For example:

- AI investment booms may raise demand for semiconductors, energy, cloud infrastructure, and specialized talent.
- Higher asset prices and stronger business confidence could stimulate aggregate demand.
- If AI increases market concentration or markups, firms may not fully pass cost savings to consumers.
- Adjustment costs may temporarily depress measured productivity.

This keeps a non-trivial probability on A at both horizons, though less so after 5 years.

4. *Why very large disinflation is not the base case*

A reduction of inflation by 0.5pp or more would require broad, fast, and pass-through-heavy productivity gains. That is possible, but not the central scenario, because:

- Diffusion across regulated and service-heavy sectors is often slow.
- Many gains may show up in quality improvements rather than lower measured prices.
- Some firms may retain gains as profits.
- Complementary bottlenecks, especially power and compute, could offset part of the supply benefit.

Claude's Reasoning:

1-Year Horizon (2025–2026)

At the 1-year horizon, the dominant expectation is that AI will have **minimal direct impact on aggregate US inflation** (Scenario B, 50%). The reasoning is as follows:

- **Adoption lags:** While ChatGPT and similar tools have achieved rapid user adoption, enterprise-level integration into production processes — the kind that affects unit costs and prices — takes considerably longer. Brynjolfsson, Rock, & Syverson (2021) document a "productivity J-curve" where new general-purpose technologies initially *reduce* measured productivity as firms invest in complementary intangible capital (reorganization, training, workflow redesign) before benefits emerge.
- **Small macro footprint:** AI-related spending, while growing rapidly, remains a small share of GDP. Total US AI investment (hardware, software, services) is estimated at roughly $150–200 billion annually — significant but under 1% of GDP. Its price-level effects are correspondingly small relative to forces like monetary policy, fiscal policy, and global commodity markets.

- **Inflationary risk (Scenario A, 25%):** The probability of a slight inflationary impact is non-trivial because of the surge in AI-related capital expenditure. Data-center construction, semiconductor demand, and energy consumption for AI workloads are all increasing rapidly. The IEA (2024) projects data-center electricity consumption could double by 2026. This demand-pull inflation in energy and investment goods could slightly raise headline inflation, even if AI's net long-run effect is deflationary. Additionally, some firms are raising prices for AI-enhanced products and services.
- **Limited deflationary potential (Scenarios C & D, 20% & 5%):** Some early-adopting sectors (customer service, coding, content creation, basic legal/financial analysis) may already see modest cost reductions. However, these effects are too small and too narrowly distributed to materially affect the CPI within one year.

5-Year Horizon (2025–2030)

At the 5-year horizon, the probability distribution shifts meaningfully toward deflationary outcomes, with **Scenario C (slight reduction, 35%)** becoming the modal outcome. The reasoning:

- **Productivity gains materialize:** By 2030, AI will have had sufficient time for broad enterprise adoption. Based on historical GPT diffusion patterns (David, 1990; Brynjolfsson et al., 2021), 5–7 years is typically when measurable productivity gains begin appearing in macro data. Baily, Brynjolfsson, & Korinek (2023) project AI could add 0.4–0.9pp to annual labor productivity growth, which would translate to meaningful downward pressure on unit labor costs and thus prices.
- **Services inflation moderation:** Services constitute roughly 60% of the CPI basket and have been the primary driver of post-pandemic inflation persistence. AI is disproportionately suited to automating or augmenting service-sector tasks (Eloundou et al., 2023), suggesting it could address the stickiest component of inflation. Healthcare administration, financial services, legal services, education, and government services are all areas where AI could reduce costs.
- **Supply chain and logistics optimization:** AI-driven improvements in demand forecasting, inventory management, routing, and supply chain coordination could reduce goods-sector costs, compounding the services-side effects.
- **Scenario D tail risk (20%):** The probability of meaningful deflation (−0.5pp or more) is substantial at the 5-year horizon. This reflects the possibility of faster-than-expected AI capability improvements (e.g., toward more autonomous AI agents) and rapid adoption. If AI achieves the upper end of productivity estimates from Korinek & Suh (2024), the deflationary impact could be significant. However, Acemoglu (2024) provides a more skeptical baseline, arguing that AI's impact will be concentrated in a limited set of tasks and that general equilibrium effects (e.g., Baumol's cost disease in non-automated sectors) will limit aggregate impact.
- **Residual inflationary risk (Scenario A, 20%):** Even at the 5-year horizon, AI could prove inflationary if: (a) AI-driven productivity gains primarily accrue to capital owners, boosting wealth and demand without proportionate supply increases; (b) AI-driven energy demand significantly tightens electricity markets; (c) AI enables firms to engage in more effective price discrimination and dynamic pricing, extracting more consumer surplus; or (d) geopolitical competition over AI leads to industrial policy and subsidies that are inflationary.

Key Uncertainties and Caveats

- **Technology trajectory:** The pace of AI capability improvement is deeply uncertain. A slowdown in scaling laws would favor Scenarios A/B; breakthroughs in reasoning, planning, or autonomous agents would favor C/D.
- **Policy response:** The Federal Reserve will adjust monetary policy in response to AI's effects on inflation, potentially dampening both inflationary and deflationary extremes. Regulatory choices (AI safety requirements, licensing, labor protections) will also affect adoption speed and cost.
- **Pass-through dynamics:** Even if AI reduces production costs, the extent to which these savings are passed through to consumers (vs. captured as higher margins) depends on competitive structure. Highly concentrated industries may retain savings.
- **Labor market adjustment:** If AI displaces workers faster than new jobs are created, reduced consumer spending could create deflationary demand-side effects beyond the supply-side productivity channel.

Conversely, if labor markets remain tight due to demographic trends, AI-driven productivity gains may be necessary just to prevent inflation from rising.

- **These probabilities represent informed subjective estimates** based on available research as of mid-2025. They are not derived from a formal probabilistic model and carry substantial uncertainty.

Appendix 1

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What Iran means for the dollar: a perfect storm for the petrodollar

March 24, 2026

Authors

Mallika Sachdeva
Strategist

The long-term legacy of the Iran conflict for the dollar could be the way it tests the foundations of the petrodollar regime. If fault lines are further exposed, there could be significant downstream effects to the dollar's use in global trade and savings, and the dollar's role as the world's reserve currency.

The world saves in dollars in large part because it pays in dollars. The dollar's dominance in cross-border trade is arguably built on the petrodollar: globally traded oil is priced and invoiced in USD. This arrangement can be traced to a deal struck in 1974 where Saudi Arabia agreed to price oil in USD and invest surpluses in USD assets, in exchange for US security guarantees. Because oil is a core input to global manufacturing and transport, there is a natural incentive for global value chains to dollarize, and global surpluses to accumulate in USD.

The foundations of the petrodollar regime have been under pressure even before this conflict. Most Middle East oil is now sold to Asia not the US; sanctioned oil from Russia and Iran has already been trading off dollar rails; Saudi Arabia has been localizing defence, and experimenting with forms of non-dollar payment infrastructure such as Project mBridge.

The current conflict may expose further fault lines, by challenging the US security umbrella for Gulf infrastructure and the maritime security for global trade in oil. Damage to Gulf economies could encourage an unwind in their foreign asset savings held largely in dollars. In this context, reports that the passage for ships through the Strait of Hormuz may be granted in exchange for oil payments in yuan should be closely followed. The conflict could be remembered as a key catalyst for erosion in petrodollar dominance, and the beginnings of the petroyuan.

A bigger risk could come if the world begins to move away from globally traded oil and gas itself, to more resilient sources of energy including domestically available fuels, renewable energy, and nuclear power. The energy choices of the Global South, Europe and North Asia will be key to track. A move away from oil could be as powerful as the pressure to price it in other currencies.

A world that becomes more self-sufficient in defence and energy would also be a world that holds less USD reserves. The huge strategic importance of the Middle East to the dollar's role as the world's reserve currency should not be underestimated. The current conflict may be the perfect storm for the petrodollar.

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What Iran means for the dollar: a perfect storm for the petrodollar

The dollar is the world's reserve currency for a very simple reason: the world pays for global goods and services in dollars and is willing to save resulting surpluses in dollar assets. From 1945-1971, the dollar was "backed" by gold – namely, global central banks were able to exchange $35 for 1oz at the Fed. This was the foundation of the international monetary system known as Bretton Woods. In 1971, the US broke the dollar's link to gold. Since then, the dollar has been in a purely fiat regime – one that is backed by the sovereign credit worthiness of the US and willingness of the world to save in its debt.

Enduring support for the fiat dollar arguably comes from the dominance of the dollar in the pricing of cross-border trade. Globally traded goods and services are largely priced in USD, with payments exchanged over US controlled payment rails. Global surpluses are thus built in USD and mostly invested back into US assets. Corporates are incentivized to save and borrow in the currency of their payables and receivables, banking systems are dollarized, and central banks save in dollars to act as effective lenders of last resort. This drives demand for USD reserves. See more on this link here.

A crucial anchor to this system is the petrodollar: the fact that most globally traded oil is priced and invoiced in dollars. Because oil is so central to global manufacturing processes – from petrochemicals, fertilizers and transport, to running factories and offices – companies are incentivized to price end products in dollars as a natural currency hedge to a key cost. This is a big reason global goods and services are priced and traded in US dollars.

The reason oil is priced in dollars can be traced back to the 1974 petrodollar arrangement between the US and Saudi Arabia. In simple terms, Saudi Arabia agreed to price its oil exports in USD and invest oil surpluses into US Treasuries. In exchange, the US provided security guarantees and military protection. The rest of the GCC followed. Structural foreign demand for US debt lowered the funding costs of the US government, acting as an indirect means of payment for the security umbrella extended to the region. To the extent that this arrangement supported broader global invoicing and saving in USD, it has been crucial to the USD's role as the world's reserve currency.

The current conflict has arguably shaken some core foundations of the petrodollar regime: the security-for-oil-pricing arrangement. US military assets and bases in the Gulf have come under attack in the war. Oil infrastructure in the Gulf has also been hit. And the US ability to provide the maritime security to ensure the global flow of oil has been challenged with the closure of Hormuz. The US security umbrella has been fundamentally tested.

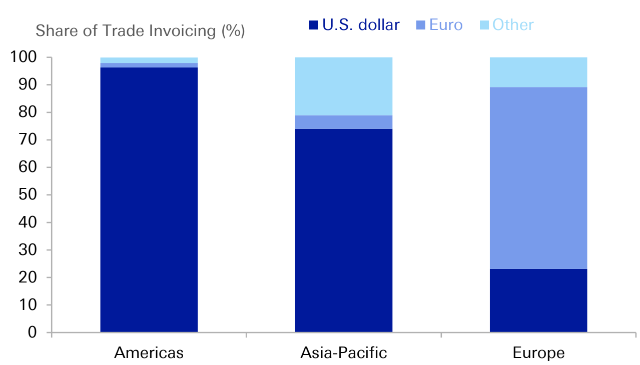
The legacy of this conflict for the dollar could be the ways in which it tests the foundations of the petrodollar regime. In the long-run, if the world uses less oil, the Gulf draws more deeply on existing dollar savings, if the Gulf moves closer to Asia in its trade and investment relationships, and eventually prices less oil in dollars – there could be significant downstream effects to the dollar's usage in global trade and savings.

The system was already under threat: many changes have been afoot

A lot had already begun to change before the US-Israel-Iran war broke out. We highlight a few key shifts that were underway:

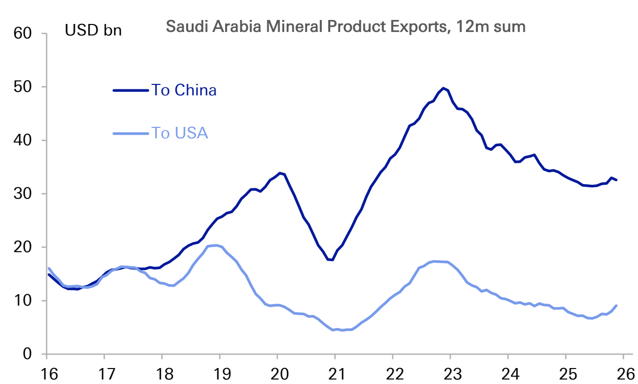
1. **The US was no longer the biggest buyer of Middle East oil.** With the shale revolution making the US energy independent, Saudi Arabia had been selling more than four times as much oil to China as to US. 85% of Middle East crude oil goes to Asia. This already introduced a fundamental instability with reports that China was keen for more oil to be invoiced in yuan.
2. **Saudi Arabia was already looking to localize more of its own defence.** Under Vision 2030, Saudi Arabia had been targeting an increase in the domestic content of military spending to 50%, looking to reduce dependence on foreign imported arms.
3. **Saudi Arabia had joined Project mBridge and signed FX swap lines with China.** Project mBridge is an initiative involving the PBOC, HKMA, Bank of Thailand, and the central banks of the UAE and Saudi Arabia. It uses blockchain technology to facilitate payments in the central bank digital currencies of each country. Crucially it does not depend on USD correspondent banking or SWIFT, and is at the minimum viable stage. The rails to transact outside of the dollar have already been built.
4. **Sanctions on Russia and Iran meant significant oil trade was already taking place outside of dollar rails**. Sales of Russian and Iranian oil have been priced and transacted in a range of local currencies from ruble, yuan, rupee with non-dollar payment infrastructure having been used.

Figure 1: The dominance of the dollar in cross-border trade has a lot to do with the petrodollar



Source : Deutsche Bank, Federal Reserve

Figure 2: There were already signs of instability to petrodollar foundations: Saudi Arabia sells four times as much oil to China now as to the US



Source : Deutsche Bank, CEIC

The conflict has introduced new sources of instability for the petrodollar

First, the US security umbrella is being tested with Gulf economies facing attacks on US bases, oil fields and infrastructure in a war that began with US military action in the region. Aside from the Gulf, pain has arguably been disproportionately felt by traditional US allies: Europe, Japan and Korea that are large energy importers exposed to the Straits.

Second, direct diplomacy rather that US-led maritime security has been the conduit for oil to pass through the Strait of Hormuz. Select tankers heading for China, India and Japan have been allowed passage. Bilateral relations have served countries.

Third, and crucially there have been reports that Iran is negotiating with eight countries to allow passage of ships through the Strait of Hormuz if oil payments are paid in yuan (The Telegraph, CNN, Asia Times, Chosun). This adds further credence to the enormous power of oil pricing. The conflict could be the catalyst for erosion in petrodollar dominance and the beginnings of the petroyuan.

To be sure, there are numerous mitigants to consider. While the US is no longer the biggest global buyer of oil, it may well be on track to be the world's biggest supplier. As we argued in our earlier piece, if the US controlled the entire Western Hemisphere's oil – either directly or by proxy – it would have more reserves than OPEC, putting it in a better position to set the terms of global oil trade.

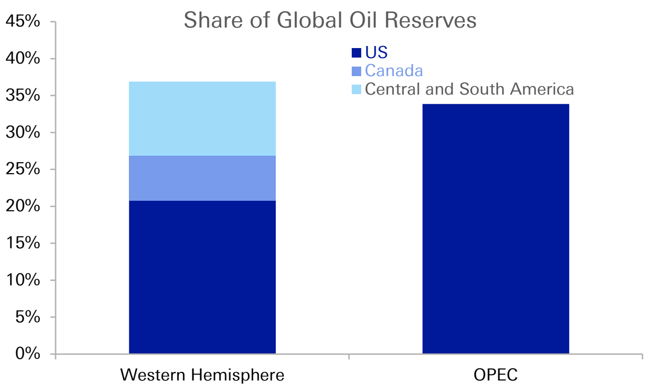
In a best-case scenario, the dollar could remain the dominant currency for oil trade if the US controls a majority of the world's supply, although further development of reserves would be required to produce surpluses sufficient to be the world's biggest *tradeable* supplier.

In a worst case scenario, one may see a fracturing of global oil pricing along trade routes and corridors. One might imagine Middle East oil passing through the Straits of Hormuz to Asia being priced in yuan, while oil from the Western Hemisphere sold across the Atlantic and Pacific to historical US allies may be priced in USD.

Another important mitigant is that Gulf economies are very invested in the USD. The GCC runs USD pegs that are backed by enormous USD savings. To maintain pegs, it makes sense for them to add weight with USD receivables. Any sign of moving away from USD revenues could invite self-fulfilling attacks on pegs.

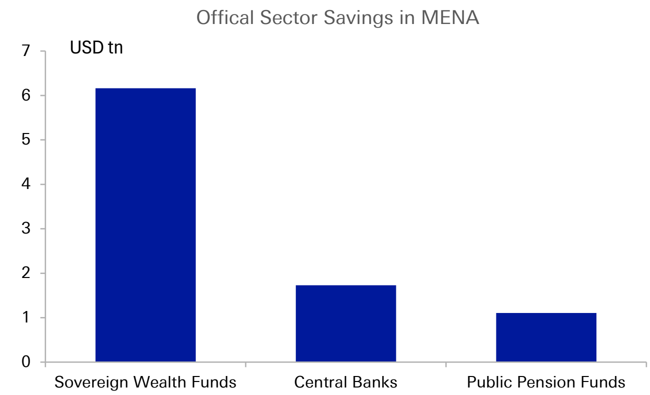
However, the Gulf may need to unwind dollar savings due to this crisis itself. The war has already led to damage of oil and civil infrastructure in Gulf economies. Escalation or a prolonged conflict could worsen this (see here for what's at stake for Gulf economies and here for our take on escalation risks). Hydrocarbon economies could suffer both from near-term hits to physical production and reduced long-term demand for fossil fuels (more below). Moreover, services sectors such as finance, tourism, aviation may face scarring if there remains an overhang of security risk. All of this could mean Gulf economies need to redeploy more of their externally held dollar savings domestically to support local economies and potentially reorient diversification programs to a different regional climate. To be sure, reserves are plentiful: the MENA region has roughly USD2tn in central bank managed reserves, and around USD6tn in sovereign wealth funds, according to Global SWF. But if USD savings are being drawn down for domestic investments, this could reduce the hurdle to other currency changes.

Figure 3: If the US controlled Western Hemispheric oil, it would have more reserves than all of OPEC



Source : Deutsche Bank, Energy Institute

Figure 4: A majority of the Gulf region's savings are held in sovereign wealth funds



Source : Deutsche Bank, Global SWF

Perhaps the biggest long-term risk comes not just from the Gulf moving away from dollar pricing, but if the world begins to move away from oil itself to which we turn next.

Bigger energy shifts could be in the offing: the world may diversify away from traded fossil fuels to domestic energy

There are many parallels between now and the 1970s. Most obviously: we have now seen a second major oil and gas shock this decade. In the 1970s, the 1973 oil embargo was followed by the 1979 Iranian Revolution, while this time, Russia's invasion of Ukraine (2022) has been followed by the ongoing US-Israel-Iran war. Global prices may remain structurally higher beyond the end of the current conflict if production facilities in the Gulf are materially damaged. And the historic weaponization of the Straits could introduce more risk premium into energy that moves on ships. Increasing self-sufficiency and domestic resilience would make sense even if oil prices came down.

The 1973 oil embargo by Arab nations on Western economies motivated big energy shifts in efficiency, diversification, and reserve building. It accelerated the development of alternative oil sources in Canada, Gulf of Mexico, Alaska, and the North Sea that helped the OECD diversify away from Middle East oil. It also led to the creation of Strategic Petroleum Reserves, and created the political will for initial investments in renewable and nuclear power.

The US may see the least change. Unlike the 1970s, the US is now energy independent in oil & gas. US industry will not be hurt immediately in the way that economies not receiving physical oil will be, and US oil producers will benefit. While there could be a consumption shock from higher pump prices, the US arguably retains the option of an export ban to manage domestic prices. WTI-Brent-Oman spreads already reflect the very different prices being paid around the world. The US growth impact is thus the most ambiguous and likely the least negative (see here and here). This should mean the US remains relatively more invested in fossil fuels.

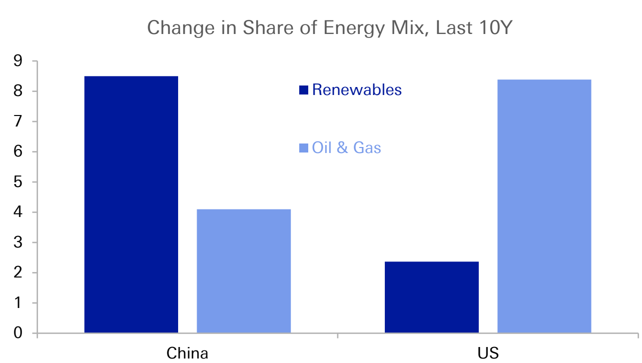
But energy-dependent regions like Europe, Asia and many parts of the Global South – that are facing multiple threats to global trade in fossil fuels – arguably have three key options:

- **The first option is to source more fossil fuels domestically**. Regions that have the ability to explore and develop more of their own oil and gas may be more motivated to do so, from the UK to Brazil, while Europe and parts of Asia may also turn back to domestic sources of coal. Even if the world does not move away from fossil fuels, global trade in it could well decline, and this is what matters for FX.
- **The second option is to double down on renewable energy**. Renewables are far more appealing on relative cost today than they were in the 1970s, due to the enormous industrial build out in China. China produces 80% of world's solar panels, 70% of the wind turbines, and 70% of lithium batteries (Science). China's overcapacity may well have been strategic, with China now extremely well placed to supply into new demand. Many Global South economies could accelerate this shift. But ultimately, this second option also comes with new dependencies on a mostly single industrial actor.
- **Genuine energy independence may only be created by a third option: nuclear power.** One could perhaps expect a significant focus on this in Europe, Japan (see here), and South Korea – traditional US allies. Just like their defence buildouts, nuclear power will have very long multi-year lead times but could have very significant FX implications.

To be sure, there will be limits to how much the world can fully move away from oil. Many industries rely on the fractionalization of oil for manufacturing. And as we have argued in our earlier work, oil is crucial to militaries: while passenger cars can run on electricity - military jets, tanks and navy ships are likely to be powered by oil for a long time. Nevertheless, there is scope for greater parts of the energy mix to transition to more reliable, secure, and potentially cheaper sources of energy. If the world shifts away from globally traded oil and gas, to more domestic sources of fuel, renewable or nuclear power, the most obvious long-term impact would be reduced oil and gas deficits in Europe and North Asia, and reduced energy surpluses in the Middle East. Reduced global oil trade would also create more room for pricing of goods and services to shift away from the USD.

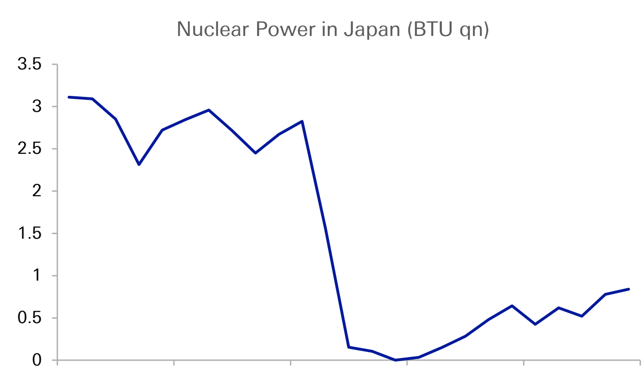
The move away from oil could be as powerful as the pressure to price it in other currencies. The petrodollar system could be at risk from both the 'petro' and the 'dollar' legs.

Figure 5: China has been leaning more towards renewable energy, while the US has been doubling down on fossil fuels



Source : Deutsche Bank, CEIC, IEA

Figure 6: Japan and Europe could ramp back up on nuclear power to build more energy independence



Source : Deutsche Bank, CEIC

Conclusion: What does this all mean for the dollar?

In the near term, we have argued that US energy independence might be worth some safe-haven premium on the dollar. The US is the only major economic region in the world that is both energy independent and is itself far from a battleground. However, there have been important offsetting factors from growing US fiscal risk around greater military spending, to the unwind of UST holdings in Asia and the Middle East to defend currencies (see pieces here and here) which mean the dollar has not strengthened much in this crisis even in the short term.

The more interesting and lasting impact of this crisis on the dollar could be on the long-term foundations, namely the willingness of the world to price traded goods and services in dollars and to save resulting surpluses in dollar assets.

The foundations of the petrodollar regime were under pressure even before this conflict. There were already signs of instability to the long-standing arrangement to price GCC oil in dollars in exchange for security: most Middle East oil is sold to Asia now not the US; sanctioned oil has already been trading off dollar rails; Saudi Arabia has been localizing defence and experimenting with non-dollar payment rails alongside other Global South central banks.

The current conflict may have exposed further fault lines, by challenging the US security umbrella for Gulf infrastructure, the maritime security for global trade in oil, and encouraging a potential unwind in Gulf dollar savings. In this context, reports that the passage of ships through the Strait of Hormuz may be granted in exchange for oil payments in yuan should be closely followed. The conflict could be the catalyst for erosion in petrodollar dominance and the beginnings of the petroyuan.

A bigger risk perhaps comes if the world begins to move away from globally traded oil and gas itself, to more resilient sources of energy from domestically available fuels, renewable energy, and nuclear power. The energy choices of the Global South, Europe and North Asia will be particularly crucial to track. The move away from oil could be as powerful as the pressure to price it in other currencies.

A world that is looking to become more self-sufficient in defence and energy would also likely be a world that will hold less USD reserves. If economic security is now a key part of national security, building greater energy resilience should be seen as complementary to the increased focus on defence spending. Reserves hitherto held in foreign assets will increasingly be needed to build domestic capacity that supports strategic autonomy (see here and here). Future surpluses could themselves reduce and be accumulated less in dollars. These are long-term dollar negatives.

The huge strategic importance of the Middle East to the dollar's role as the world's reserve currency should not be underestimated. The current conflict may be the perfect storm for the petrodollar.

Thematic FX Framework Pieces

This note is part of a broader series of thematic FX notes that explores the evolving nexus between payments, energy and defence as key pillars to the dollar's long-term status. Other FX pieces that can be read alongside this include:

FX Special Report: What Venezuela has to do with the dollar

FX Special Report: What do stablecoins mean for dollar dominance?

FX Special Report: Who can come home?

FX Special Report: The geopolitics of reserves: shifting security alliances will matter for the USD

EM Special Publication: Geoeconomics 101: implications for the dollar, emerging markets and beyond

FX Blog: On defense spending, bond markets, home bias & the yen

EM Blog: Iran - a further impetus away from recycling savings to the US?

Appendix 1

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Power Play: Investing in Energy Infrastructure for the AI Era

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This report includes contributions from the following Deutsche Bank research analysts: Brett Ryan (econ), Ross Seymore (semiconductors), Benjamin Soff (data center REITs), Michael Linenberg (airlines), Edison Yu (autos and space), Scott Deuschle (aerospace and defense), Nicole DeBlase (multi-industry and electrical equipment), Corinne Blanchard (clean tech), and Chris Robertson (LNG infrastructure and maritime transport).

The world is facing a significant energy deficit. This was true before AI and the war in Iran exacerbated the shortfall. The kinds of investments underway today will shape the future of the energy landscape and its ability to support an evolving tech-forward economy. Businesses are responding through legacy and innovative ways both to alleviate current constraints and to scale up for the long run.

To address immediate constraint issues, companies are sidestepping grid backlogs by offering readily dispatchable power sources and modular units. As the grid expands to meet growing demand, multi-industry and electrical equipment companies are enabling utilities to expand generation, transmission, and delivery. Renewables cannot be overlooked as a scaling solution as the AI era exponentially increases the demand for energy. Solar in particular is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems, these technologies can provide the kind of around-the-clock, predictable power that AI infrastructure requires. Commercial nuclear is another important component of the longer-term energy supply makeup, with the near-term focus on extending the lifetime of existing plants while looking ahead to longer-term expansion. All that said, even as the grid scales significantly and the mix of energy supply diversifies, natural gas is likely to remain a primary and backup source of power.

Authors

Laura Daniels
Deputy Associate Director
of Equity Research

Josh Ray
Associate Director
of Equity Research

Matthew Barnard
Head of US Company Research

Deutsche Bank
Research Institute

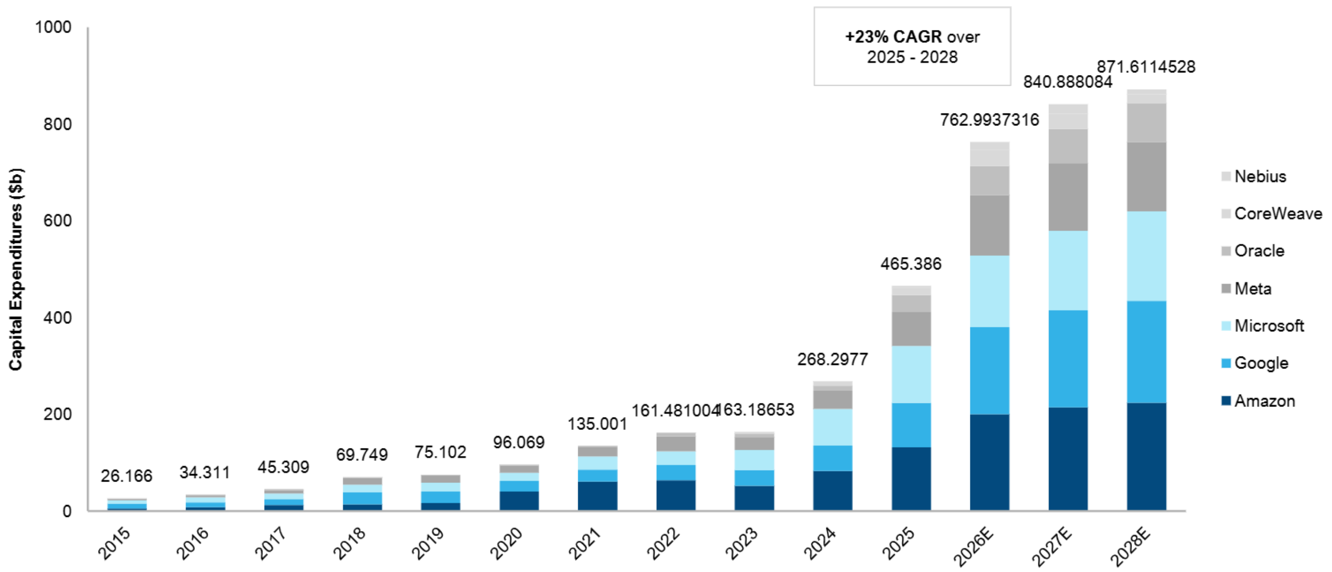
The Macro and Demand Landscape

The grid is under significant pressure. US energy generation hit a record 4.43 terawatt hours in 2025, according to the US Energy Information Administration; generation will likely break that record again this and next year due primarily to large data centers. Energy demand driven by AI data centers globally is expected to quadruple over the next four years, according to the International Energy Agency. To get a sense of how great this demand is and will be in the coming years, we can look to cloud-related investments, growth in data centers and AI workloads, and energy prices.

One of the most telling data points has been cloud-related investments. Our Semiconductors Research Team has tracked a decade of data-center-related investments via capital expenditures (capex) at several major cloud service providers to finance the setup of physical data centers as well as the electronic equipment needed to fill them. The data shows that the current buildout is unlike any other. Capex has surged over the last three years, rising from $161 billion in 2022 to $465 billion in 2025. This reflects a compound annual growth rate of +42%, a jump up from comparable three-year rates of +29% (2019-2022) and +30% (2016-2019).

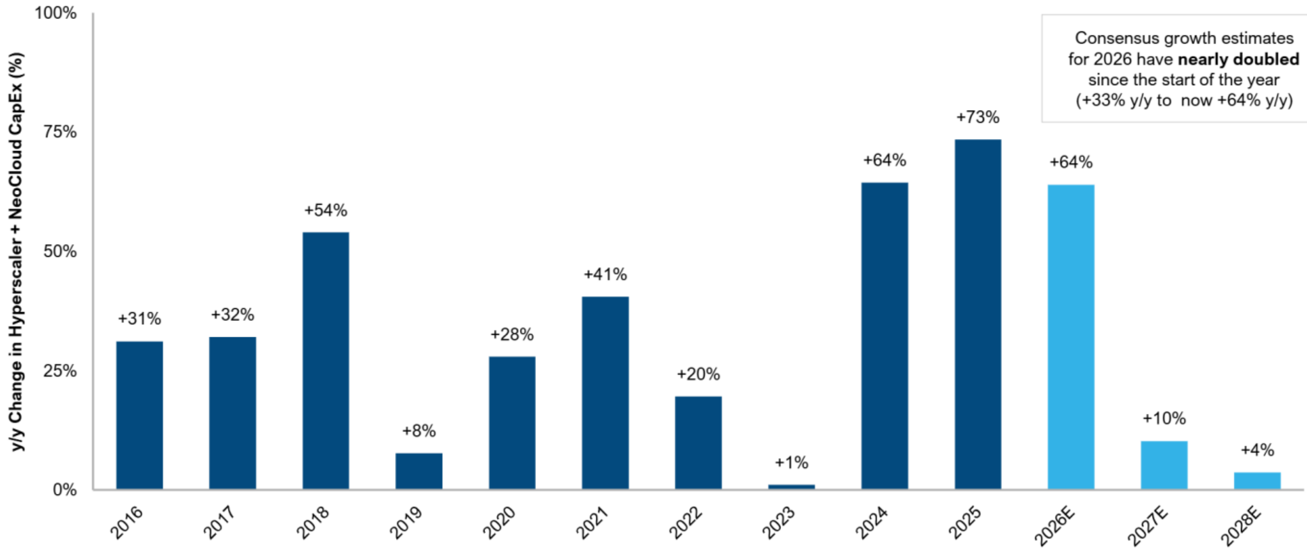
Spending is expected to step up even further to $762 billion (+64% year-over-year) in 2026. If that happens, this would mark the third year in a row that cloud-related capital investments have risen more than 60% year over year. Prevailing estimates also expect growth to slow after this growth spurt, but as we saw throughout 2025, consensus tends to underestimate the amount of spending in latter years, so future spending could well exceed today's expectations.

Figure 1: Aggregate Cloud Capex (2015 - 2028E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.

Figure 2: Annual Growth in Aggregate Cloud Capex (2016 - 2018E)



Source: Deutsche Bank Research, Bloomberg Finance LP. Note: Dataset available upon request. Forecast data above is based on median Bloomberg consensus estimates. Information is presented on an annual basis given the lack of quarterly detail by company and dispersion in consensus estimates going forward.

Data center AI workloads paint a similar AI-driven energy demand picture. Data centers represent about 5% of total US electricity usage today, according to estimates from BloombergNEF and the Department of Energy. This will likely expand with the adoption of AI, as well as with continued growth in the digital economy.

Based on midrange estimates from McKinsey, total global data center capacity demand is set to increase from 82 gigawatts in 2025 to 220 gigawatts in 2030, amounting to a 22% annual growth rate. This estimate sees generative AI workloads growing at 29% on average, plus 11% from non-AI workloads. Generative AI workloads could grow to 35% of total global data center activity by 2028, up from a mid-single-digit percentage as of 2023, according to estimates from Boston Consulting Group.

AI workloads require greater power density (essentially the amount of electricity a system needs per square foot) and more intensive liquid cooling, driving up energy consumption. Data center companies like Digital Realty (DLR) and Equinix (EQIX), for example, are already seeing these trends play out in their businesses. Digital Realty recently said it is rolling out higher-density deployments with pre-installed liquid cooling to meet AI demand. Equinix has noted that 60% of its larger deals in the last quarter of 2025 were driven by AI, up from 50% earlier last year. The power density for these deals, according to Equinix, is 33% higher than for non-AI deals.

These factors exacerbate concerns around US electricity prices and grid capacity, which were already mounting even before the war in Iran elicited a spike in global energy prices. After an extended period of stable growth in the US – over the last decade, electricity output grew at about a 0.7% annualized rate – data center electricity demand estimates are raising concerns about a knock-on impact to household utility bills as well as grid capacity. The energy problem is just as acute for central bankers. Traditionally, Federal Reserve policymakers would tend to look through the temporary inflationary impact of an energy shock (see US Economic Perspectives: Impact of oil price shocks a "frack"-tion of what they once were). However, the backdrop to the latest energy shock is one in which inflation has persistently exceeded the Federal Reserve's 2% goalpost for the better part of the last five years. This raises the risk that inflation expectations could become unanchored. In short, the energy supply outlook carries implications across a breadth of economic issues, from interest rates to technology stocks.

These concerns could result in policy favorable to companies increasing energy supply, as energy prices will likely be top of mind for voters ahead of the midterm elections this November. As we recently outlined, historically, the relationship between gasoline prices and consumer energy spending suggests that a 1 cent rise in gasoline prices (on a sustained basis) increases spending on energy goods and services. This thus reduces income that can be spent on non-energy items – by $1.15 billion, amounting to a sizable "energy tax." Mapping the impact of potential energy price shocks to consumers, we find that at a sustained $100 per barrel oil price, the projected tax benefits to consumers from last year's tax legislation (the "One Big Beautiful Bill Act") would still exceed the drag from the energy tax increase. However, at $150 per barrel, the implied energy tax increase would more than wipe out the tax cuts to consumers from that legislation and present a more serious threat to the outlook for consumer spending.

The sum total of this is that supply is already struggling to keep up. Wait times to connect a data center to the grid are exceeding four years (based on comments from Google representatives and other industry reports). Efforts are underway to increase supply, including expanding grid capacity, as well as AI players preparing to build their own sources of power that operate "behind-the-meter" (developed on the customer's side and not delivered through the public grid). Capex into investor-owned electric utilities has been steadily increasing and is projected to climb to nearly $250 billion in 2029 (according to the Edison Electric Institute). The White House's introduction in early March of a Ratepayer Protection Pledge urging AI companies to build or buy additive energy further highlights the race to bolster energy supply to meet AI needs. While the exposure across regions will vary, the conflict in Iran will add another layer of constraint to energy supply, as 20% of global oil and 20% of LNG pass through the Strait of Hormuz. Companies are looking to fill the gaps in energy supply over a spectrum of problem sets solving for immediate constraints, for scale, and for consistency.

Solving for Constraint: Readily Deployable Power

The most direct response to AI-driven power constraints is coming from companies that enable data center operators to bring electricity online without waiting for grid expansion. These companies are delivering dispatchable on-site or near-site power. While there are existing manufacturers providing these kinds of products, there are an increasing number of companies adapting existing technology to enter this AI-adjacent space. FTAI Aviation (FTAI), which traditionally operates in the airline industry, and BorgWarner (BWA), an auto industry supplier, are examples of this kind of shift.

FTAI Aviation launched FTAI Power in December 2025 with the aim of converting the company's CFM56 engines into gas turbines that can deliver energy to data centers. The engine is prevalent, with enough in operation globally that one takes off every two seconds somewhere in the world. FTAI Power will offer its aeroderivative turbine "FTAI Mod-1," a mobile, trailer-mounted conversion unit that can be delivered and installed in under two weeks. As a 25-megawatt unit, it is smaller, which allows for flexibility in stacking and growing data centers. Deliveries will likely begin around December 2026, with approximately 100 units in production per year starting in 2027.

BorgWarner is taking a similar approach. The company is an auto industry supplier that designs and manufactures powertrain systems but will partner with Turbocell (a subsidiary of the AI-optimized data center infrastructure company Endeavour) to co-develop an integrated, modular power generation system. This system would provide primary and backup power for AI data centers and would serve as a "bridge" solution, meaning data center operaters can install it quickly for power now but can keep it on to provide additional power even as longer-term energy infrastructure is built up.

Modular power generation systems like this one are designed with the high and often fluctuating power demands of AI computing in mind. Modular products also comprise standardized, interchangeable units that are more flexible in applications than the custom builds commonly found in legacy systems. These modular units, as well as the integrated battery systems and electronic components, can be scaled up or down based on specific data center needs.

Beyond quick-install energy solutions, companies further up the supply chain are adapting to the energy demanded by growth in AI. Woodward, for example, is an aerospace-industrial hybrid company whose power generation products have historically been focused on the industrial gas turbine market, but it is now broadening into supplying components for reciprocating engines and marine engines being repurposed as behind-the-meter power solutions.

Solving for Scale: Grid Expansion and Infrastructure Development

Meeting AI-driven electricity demand forecasts will require system-wide capacity and infrastructure expansion. Investments are happening simultaneously but across pathways that range from near-term grid expansion and enablement platforms, near- and longer-term renewables, and long-duration commercial nuclear supply. In terms of grid expansion, efforts span both generation capacity and transmission and distribution (T&D) infrastructure to accommodate higher loads and load density, more stringent reliability requirements, increasingly complex grid architectures, and longer-duration power needs associated with AI workloads. GE Vernova (GEV), for example, manufactures combined-cycle and peaker gas turbines (currently the most common source of baseload power in the US) and also provides high-voltage T&D equipment including transformers, high-voltage direct current (HVDC) solutions, circuit breakers, switchgear, power-sending devices, and grid automation systems. Some of its products also supply the renewable and commercial nuclear sources of power.

As existing grid infrastructure faces a critical deficit, it is impossible to ignore renewables as a scaling solution. Although US onshore wind investment faces challenges due to unfavorable policy headwinds, solar, alongside other renewables, is among the fastest-to-deploy sources of new generation. When paired with battery energy storage systems (BESS), these technologies can meet the 24/7 power needs of AI-focused data centers. There are indications that as energy demand driven by AI increases, so too do investments in renewable energy sources. A recent study from the University of Zurich, for example, found that solar and BESS have been the primary drivers of increased energy investment in regions with pre-existing data center capacity (based on ChatGPT infrastructure in the US). Increased investments in gas play a role too, but these have been offset by declines in grid-connected gas plants. These developments are currently based on investment decisions, but policy drivers could add further impetus should the environmental toll stemming from AI usage take an increasingly prominent place in political discourse.

Solar energy is well suited to the energy demand landscape due to the technology's fast scalability, low to zero marginal costs, and favorable levelized cost of energy (LCOE). As a result, the sector is seeing an increasing number of power purchase agreements (PPAs) with large technology players, including Amazon (AMZN) and Google (GOOGL). These arrangements are also conducive to a buildout of development projects that are supporting equipment suppliers.

Clearway Energy (CWEN), for example, operates energy generation assets in the US, with a specific focus on clean tech. It is positioning itself to serve the growing demand for power in the US driven by hyperscalers and data centers. The company has announced 2 gigawatts of PPAs for these projects, with possibilities for additional volumes. Additionally, it has outlined 8-13 gigawatts of capacity via co-located data center complexes, utilizing a mix of renewables and storage in addition to gas.

The scale-up of renewables also depends on component and equipment suppliers. First Solar (FSLR), for example, is a US-based solar technology company that designs and manufactures advanced thin-film modules (a kind of solar photovoltaic panel) for utility-scale projects. Nextpower (NXT), meanwhile, is an example of a clean tech equipment and systems provider, selling trackers (the mechanical systems that hold and rotate panels to follow the sun), software, electrical systems, and the foundations upon which solar technology is mounted. Solar electrical equipment also includes inverters, which convert solar-generated direct current electricity into distributable alternating current electricity delivered to both residential and commercial customers.

Commercial nuclear is another important component of longer-term energy supply, with small modular reactors (SMRs) a key part of the development pipeline. Businesses are focused on extending the lifespan of current reactors while also preparing for the possible buildout of new ones. In the near term, the priority is the maintenance, repair, and life extension of existing nuclear infrastructure. There are 54 nuclear power plants in the United States (amounting to 94 nuclear reactors), according to 2024 data from the US Energy Information Administration. These produce a net generating capacity of nearly 97 gigawatts and make up about 20% of US energy usage.

Maintenance of these plants requires the replacement and upgrading of critical reactor components, modernization of control systems, and other projects that allow current reactors to operate safely and consistently for longer periods. Suppliers such as BWX Technologies (BWXT) and Curtiss-Wright (CW) provide operating components and systems for nuclear plants, including reactor hardware, fuel-related technologies, pumps, valves, and control systems.

At the same time, the industry is preparing for potential new build activity, particularly around advanced reactor designs and SMRs. Development work spans reactor components, fuel technologies, and digital and mechanical systems designed for newer reactor platforms. BWX Technologies' work on advanced fuel and reactor components and Curtiss-Wright's involvement in systems and controls applicable to newer reactor designs are examples of how suppliers are positioning for a more modular and scalable nuclear build-out over time.

Solving for Consistency

At the core of scaling AI infrastructure is the need for stable, reliable, and low-cost power. While wind and solar power generation will undoubtably be part of the mix, natural gas can be expected to continue to play a sizable role as both a primary source of power and a backup source of power to intermittent power sources such as wind. Even with significant renewable penetration, the grid will still require backup generation capacity. Natural gas, in our view, will remain the likely fuel selected for this purpose, due to its global availability (via pipeline or liquefied natural gas imports), energy density, price, and ability to be transported and stored on site for behind-the-grid power. Natural gas also avoids many of the political hurdles that constrain other energy sources. It does not face the regulatory and political hurdles of traditional nuclear and SMRs, nor is it falling out of favor in Western markets to the same degree as coal. As such, it is likely that natural gas will continue to be in demand for both baseload and backup power generation.

The Bottom Line

Projected energy demand tied to the growing AI ecosystem and the broader digital economy is rising faster than grid capacity and interconnection timelines can adjust. Businesses are responding to address immediate constraint issues, questions of scale, and the need for consistent primary and backup generation. This will shape not only the trajectory of the AI buildout, but have reverberating impacts across the broader economy, its competitiveness, and critical infrastructure.

Appendix 1

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